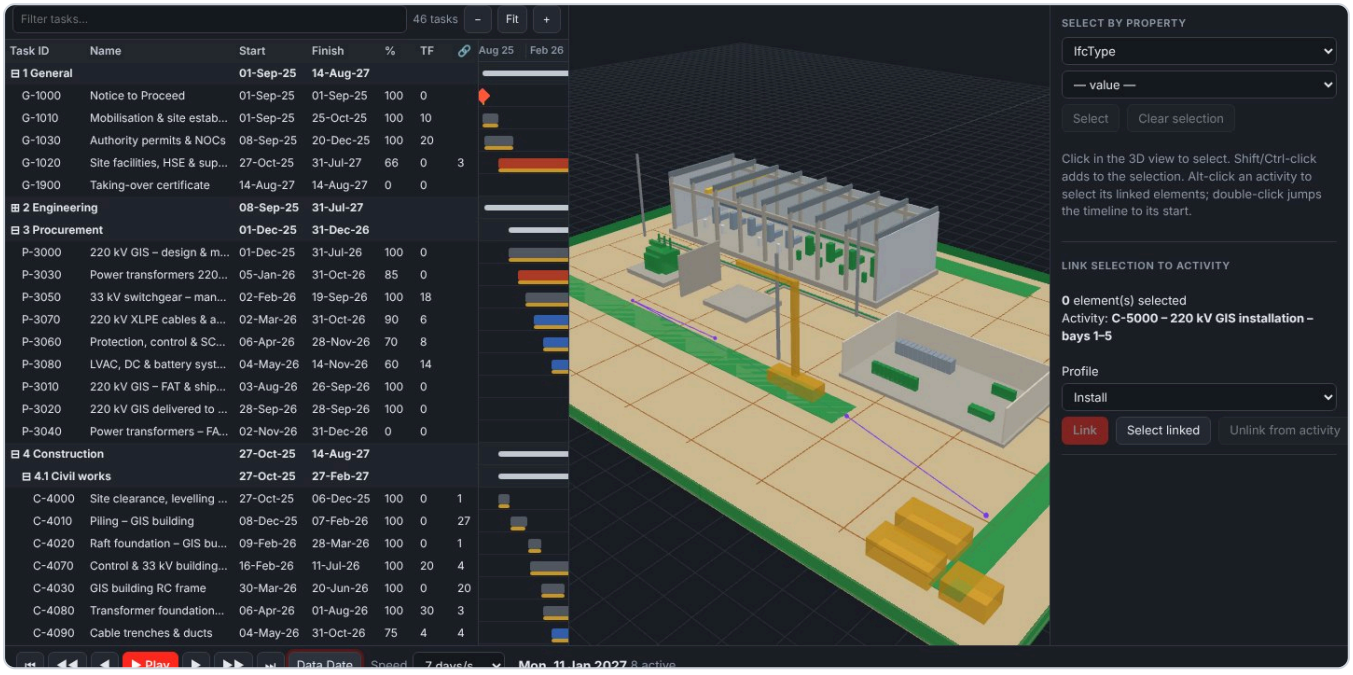
Stratum 4D
by Miqyas Labs
PLAN | SIMULATE | ANALYSE | DELIVER

User Guide

Release 0.3 · Interface layouts, 4D simulation, project database, Primavera P6 connections, dashboards and emailed reports



Worked example: 220/33 kV GIS Substation, Abu Dhabi (fictional sample project)

Miqyas Labs
ENGINEERING INTELLIGENCE

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User guide by Junaid Quamar
October 2026

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About the author

Junaid Quamar

Senior Planning Engineer

Stratum 4D was conceived, specified, designed and directed by Junaid Quamar. The feature set, its priorities and the planning workflows it supports come from his project-controls practice in Primavera P6 planning and cost engineering for HV substation, transmission and rail electrification projects.

That background shows throughout the tool: the P6 and MS Project layouts planners already know, a CPM engine that follows P6's scheduling rules, two-way P6 connections, EPC-style weighted progress reporting with manual weightage, and the 220 kV GIS substation used as the worked example in this guide.

This guide was written and prepared by Junaid Quamar.

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Product	Stratum 4D by Miqyas Labs · release 0.3 · October 2026

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1. About this guide

This guide covers Stratum 4D release 0.3: the three interface layouts (P6, MS Project and 4D), scheduling, 4D simulation, the project database, Primavera P6 connections, and the dashboard and report module with emailed reports. Every screenshot comes from the application itself, running as a Stratum Server, with one example project loaded throughout.

The example project

The example is a **220/33 kV GIS substation in Abu Dhabi**, delivered as an EPC contract. It ships with Stratum as a sample (*File → Sample: 220 kV GIS Substation (UAE)*, or the button on an empty screen), so you can follow every step yourself. The project is fictional. Names, quantities and values are illustrative, and the Islamic public holidays in its calendar are approximate because they follow the moon.

Item	Example project
Project	GIS-220 · 220/33 kV GIS Substation, Abu Dhabi (sample project)
Scope	220 kV GIS hall with 9 bays and 10 t EOT crane; two 220/33 kV power transformers with firewall and oil pits; control & 33 kV building (33 kV switchgear, protection, control & SCADA, LVAC, DC); cable trenches and 220 kV XLPE cables; earthing grid and lightning masts; roads, drainage, boundary wall and gatehouse
Contract structure (WBS level 1)	General · Engineering · Procurement · Construction (Civil works, Electromechanical works, Testing & commissioning)
Schedule	46 activities, 62 relationships (FS, SS and FF with lags), activity codes *Discipline* and *Area*, user field *Package*
Calendar	UAE 6-day week, Monday–Saturday, 07:00–16:00 with a one-hour break; UAE public holidays 2025–2027
Resources and cost	15 resources in AED: labour (design engineers, civil crew, electrical technicians, OEM specialists, testing engineers), plant (200 t mobile crane, piling rig) and procured equipment (GIS, transformers, switchgear, panels, cables)
Data date	19-Oct-26
3D model	202 elements carrying the property `Pset_Stratum.ActivityID`, so one rule links the whole model

Status at the data date

Engineering is complete apart from the as-built documentation. The 220 kV GIS arrived on site on 28-Sep-26, three weeks after its baseline date, and installation of bays 1–5 has started. The power transformers are 85% through manufacture and running 21 days late, which puts transformer erection, pre-commissioning and energisation on the critical path. The forecast taking-over date is 14-Aug-27 against a baseline of 31-Jul-27.

Milestone	Name	Current / forecast	Baseline	Status
G-1000	Notice to Proceed	01-Sep-25	01-Sep-25	Achieved
P-3020	220 kV GIS delivered to site	28-Sep-26	07-Sep-26	Achieved

Milestone	Name	Current / forecast	Baseline	Status
C-6010	220 kV energisation	31-May-27	17-May-27	Forecast

> Critical path after rescheduling (F9): `G-1020` → `G-1900` → `P-3030` → `P-3040` → `C-4900` → `C-5030` → `C-5040` → `C-6000` → `C-6010` → `C-6020`. CPM finish: 12-Aug-27.

2. Installing Stratum

Stratum runs three ways. The engine and the screens are the same in all of them; what changes is where projects are stored and which services are available.

Way to run	Where projects live	P6 connections	Emailed reports	Typical use
Desktop app (Windows, macOS, Linux)	A local SQLite database in your user folder	Yes	Yes, while the app is open	One planner, offline work
Browser	The browser's own storage	No (use XER files)	No	Quick look, demos
Stratum Server (shared)	One PostgreSQL database for the team	Yes	Yes, around the clock	Project controls team

Desktop installers

Installers are built for every release: an NSIS installer for Windows, DMG files for Intel and Apple Silicon Macs, and AppImage/deb packages for Linux. They are produced by the project's GitHub Actions workflow and appear under *Actions → Desktop installers → latest run → Artifacts*.

The installers are not code-signed yet. On Windows, SmartScreen shows "Windows protected your PC": choose *More info → Run anyway*. On a Mac, right-click the app and choose *Open* the first time. The Windows installer lets you choose the install folder and adds Start-menu and desktop shortcuts. Double-clicking an `.xer` or `.ifc` file opens it in Stratum.

Stratum Server

See Stratum Server for teams for the Docker Compose setup, environment settings, users and roles.

3. First start

The first time Stratum opens, a short setup asks two questions. Both can be changed later in Settings.

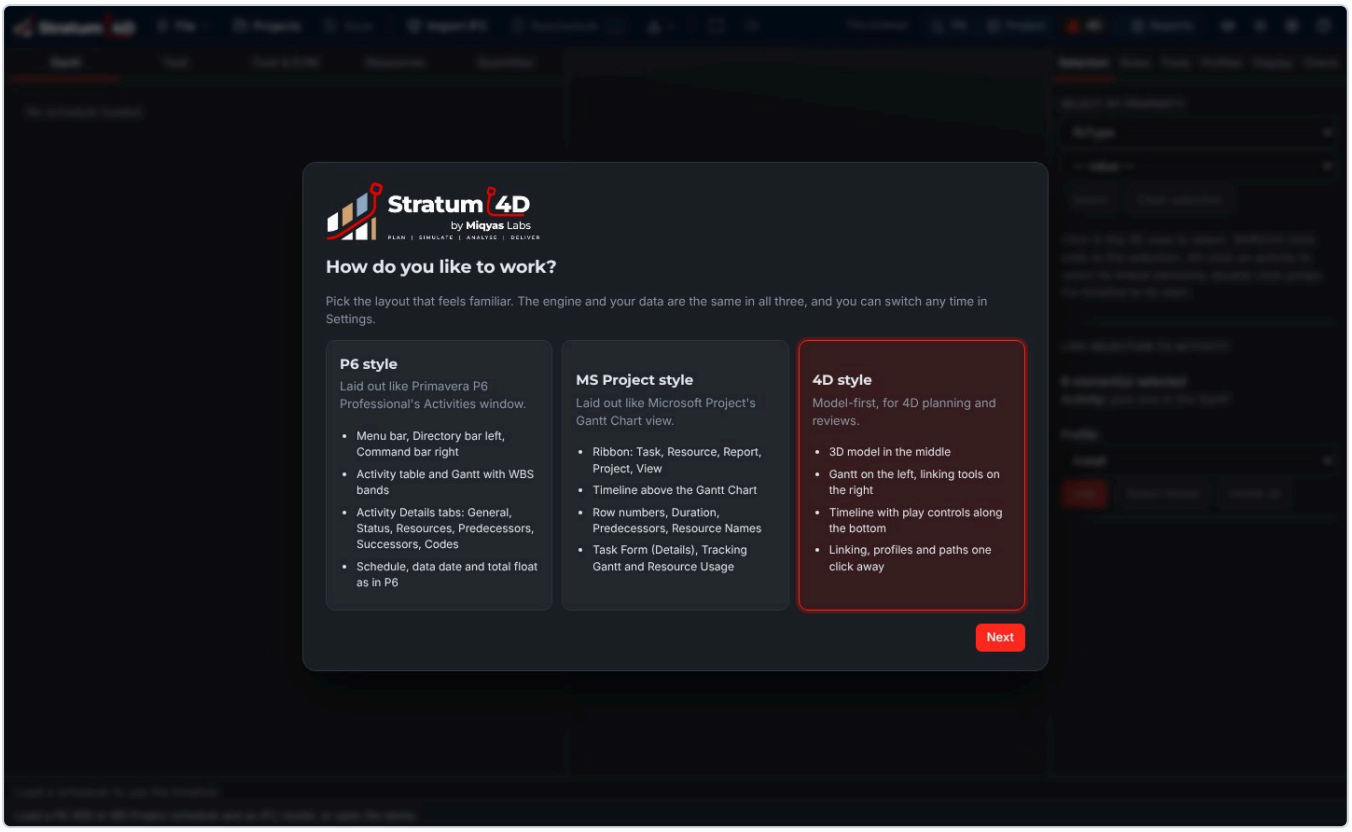


Figure 1. Step 1: choose the layout that feels familiar. The data and every feature are the same in all three.

Layout. *P6 style* is laid out like P6 Professional's Activities window, *MS Project style* like Microsoft Project's Gantt Chart view, and *4D style* puts the 3D model in the middle for 4D planning and reviews. Section 5 describes each layout.

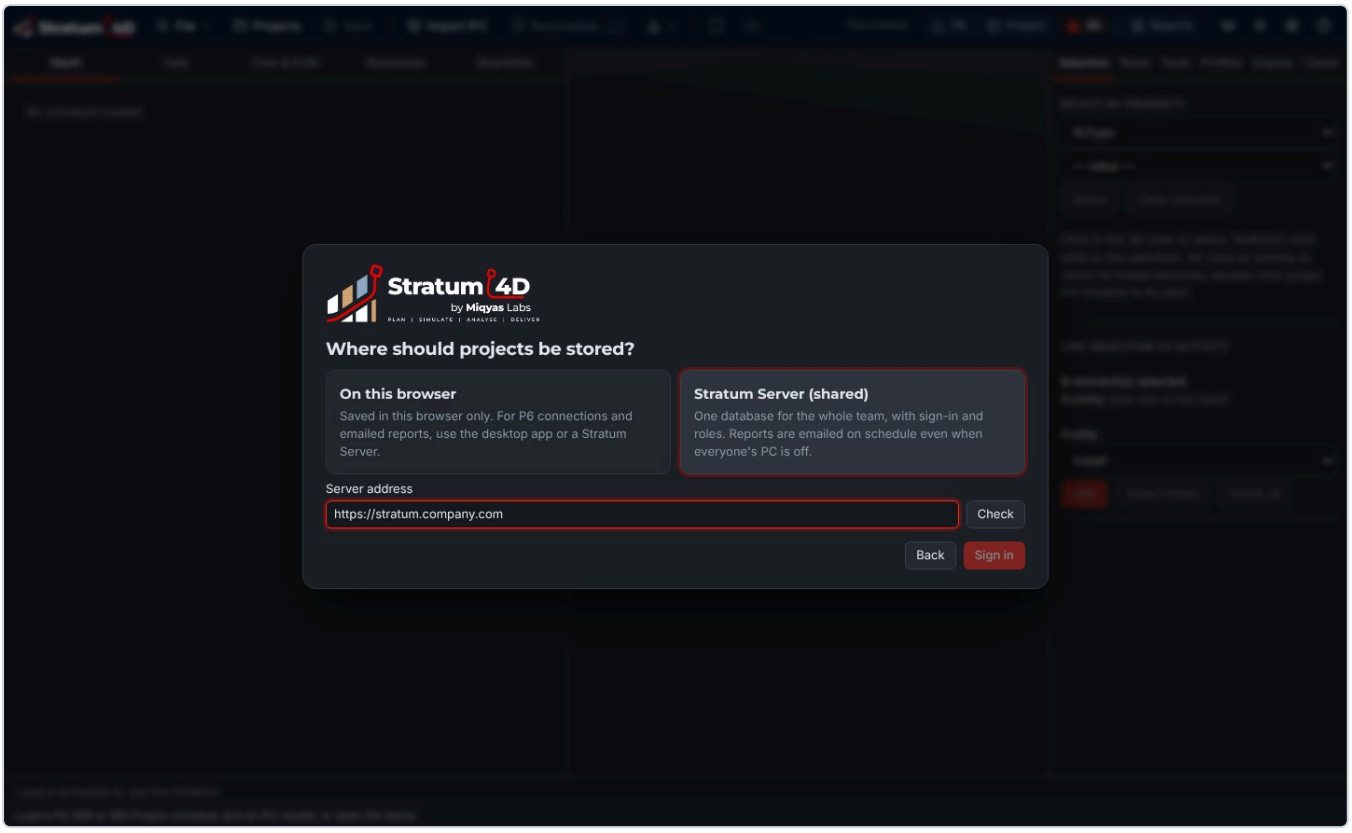


Figure 2. Step 2: where projects are stored. Choosing Stratum Server asks for the server address, then for an account; on a brand-new server the first account becomes the administrator.

Storage. In the browser the choice is between this browser and a Stratum Server. In the desktop app the first option is the local database on your computer. After setup Stratum offers to open the P6 and email settings, since both are usually configured once by an administrator.

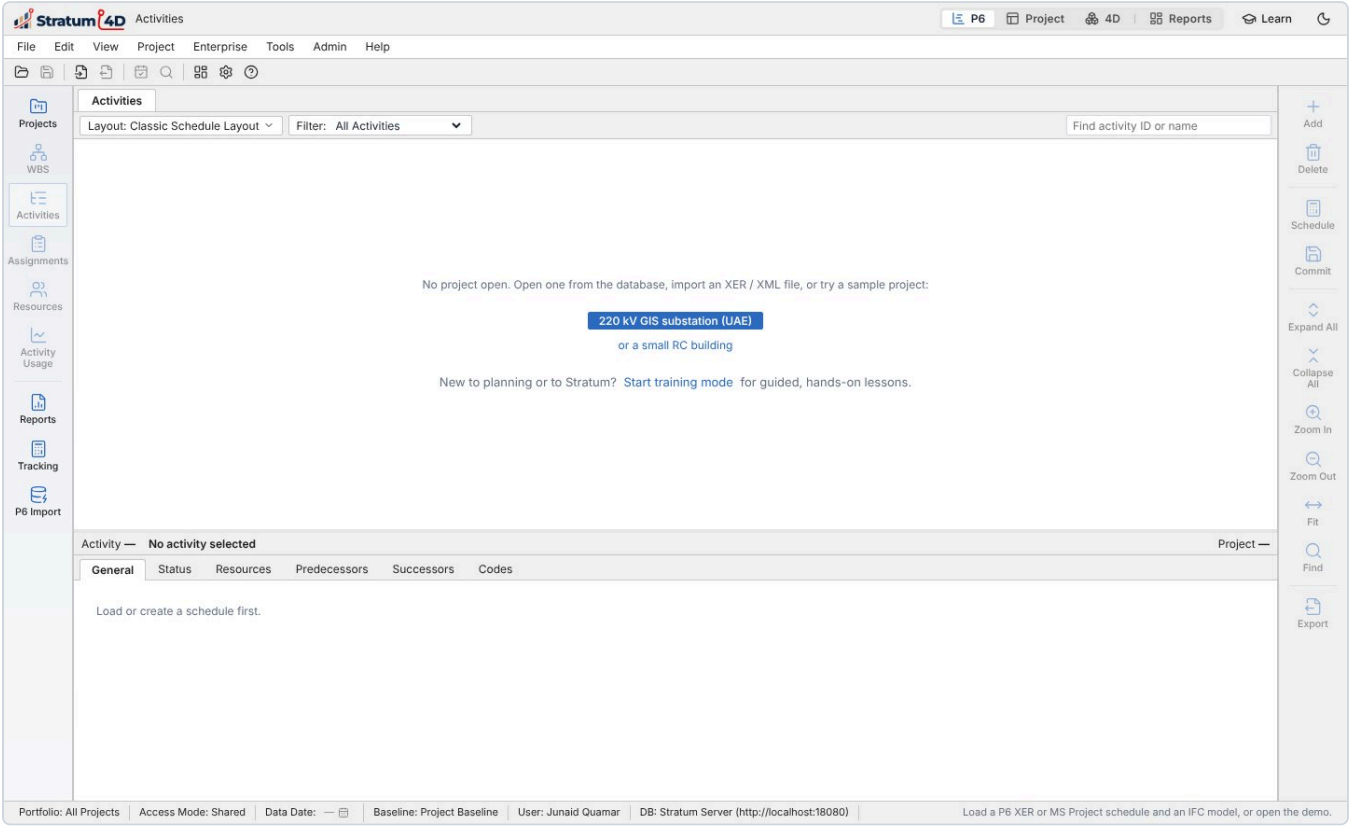


Figure 3. An empty P6 layout. The buttons in the middle load one of the two built-in sample projects.

4. Training mode

Training mode teaches planning and Stratum together, on the live application. A coach card walks through one idea at a time and a highlight shows where it lives on screen. Where a step asks you to do something, the card watches the application and ticks the step off by itself once it sees the change. It suits new planners, engineers moving into project controls, and P6 users learning Stratum.

Starting a lesson

Click **Learn** beside the layout switch (it is in every layout and in Reports), choose *Help → Training Mode*, or use the link on the empty start screen. The catalogue lists the lessons by course with their level, length and number of steps, and keeps your progress on this computer.

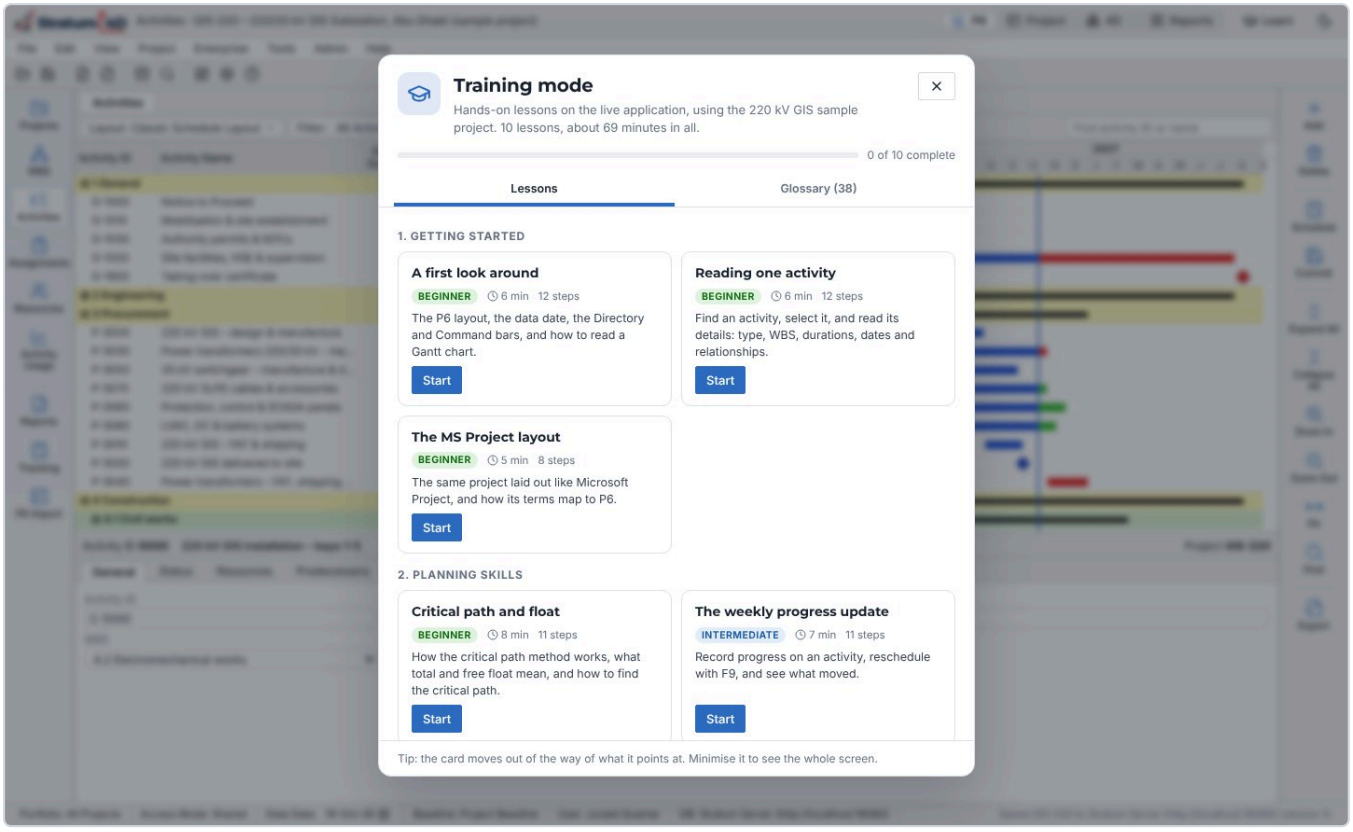


Figure 4. The training catalogue: ten lessons in four courses, about 69 minutes in all, with a progress bar and a glossary tab.

Course	Lesson	Level	Minutes	What you practise
Getting started	A first look around	Beginner	6	Layouts, data date, Directory and Command bars, reading the table and bars, WBS bands
	Reading one activity	Beginner	6	Find by ID, detail tabs, durations, predecessors and successors, FS/SS/FF/SF and lag
	The MS Project layout	Beginner	5	Ribbon, Timeline, Tracking Gantt, P6 ↔ MS Project terms
Planning skills	Critical path and float	Beginner	8	Forward and backward pass, total and free float, the critical path, negative float

Course	Lesson	Level	Minutes	What you practise
	The weekly progress update	Intermediate	7	The update cycle, % complete and remaining duration, F9, commit
	Adding an activity and its logic	Intermediate	8	Add, name, duration, a second predecessor with SS + lag, schedule, open ends, delete
Project controls	Baselines and variance	Intermediate	6	Baseline bars, finish variance, late-against-baseline filter
	Cost, resources and earned value	Intermediate	8	PV, EV, AC, SPI, CPI, EAC, resource usage
4D and reporting	4D simulation	Intermediate	7	Navigate the model, play and step, rules and profiles
	Dashboards and reports	Beginner	8	Templates, slicers, customise, add widgets, PDF, Excel and email

How a lesson works

Each lesson starts with a *Get ready* step. It switches to the layout the lesson needs and opens the 220 kV GIS sample, so the numbers on the card match the screen. Values such as durations, float and dates are read live from the project, not typed into the lesson.

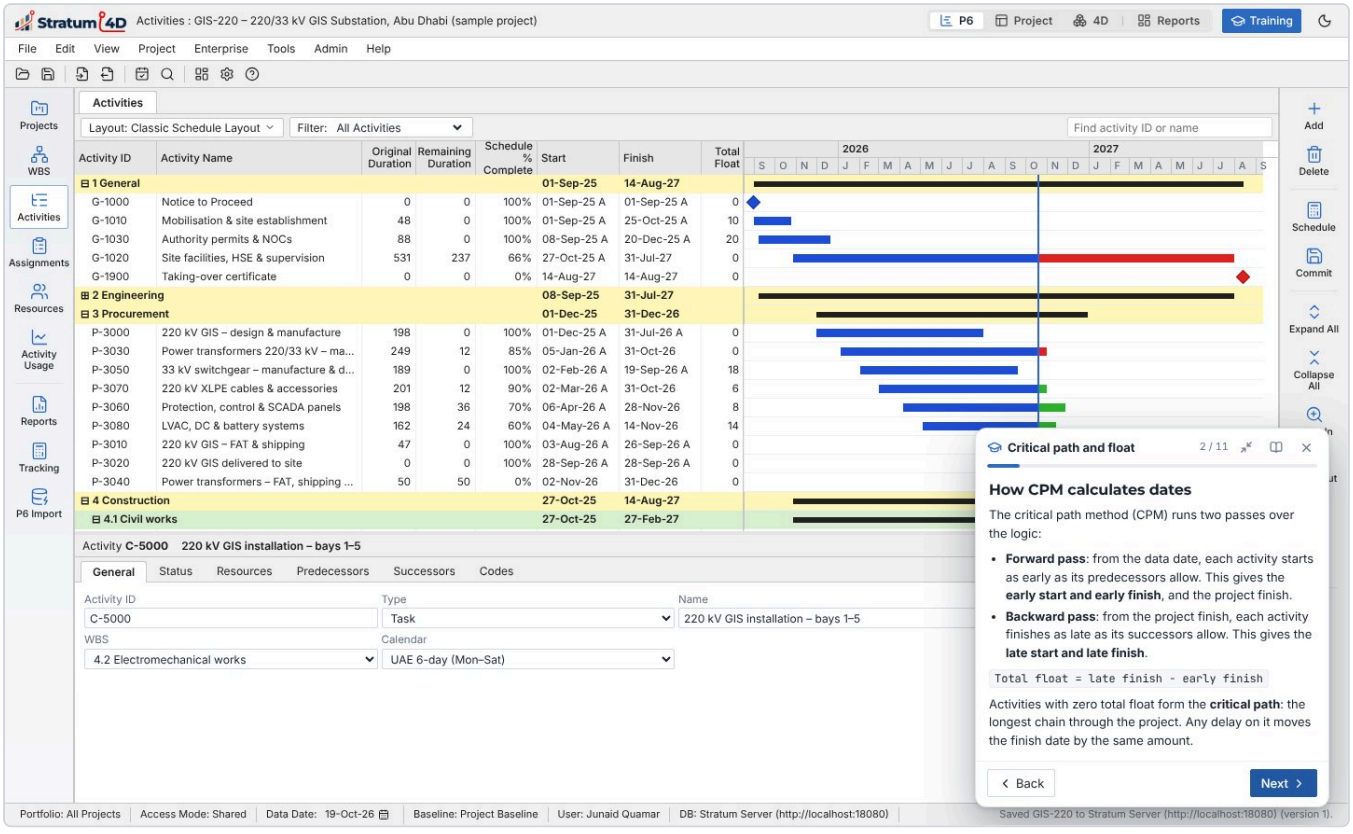


Figure 5. An explanation step. The card sits beside what it describes and never covers it; the progress bar shows how far through the lesson you are.

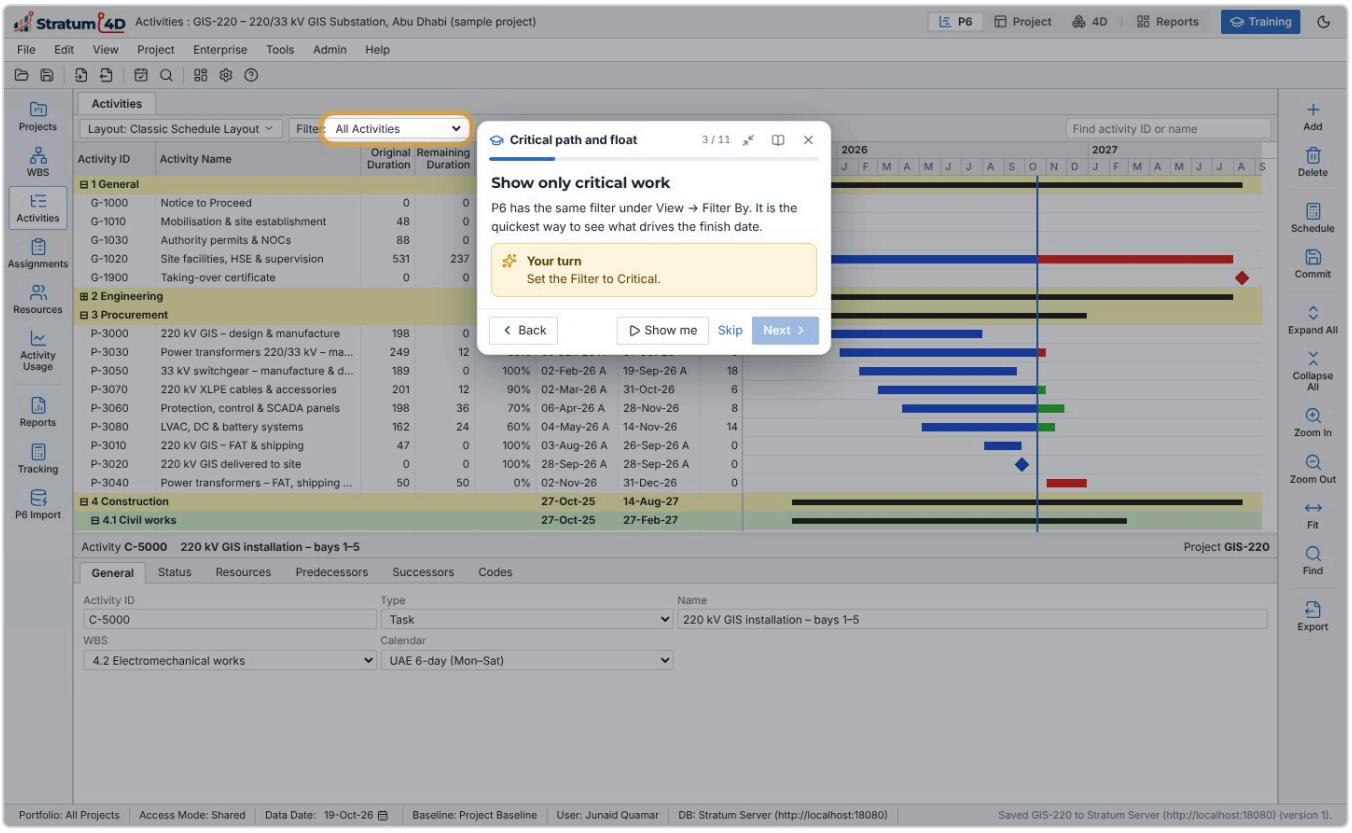


Figure 6. A task step. The highlighted filter is the control to use, and Your turn says what to do. Next stays off until it is done; Skip moves on anyway.

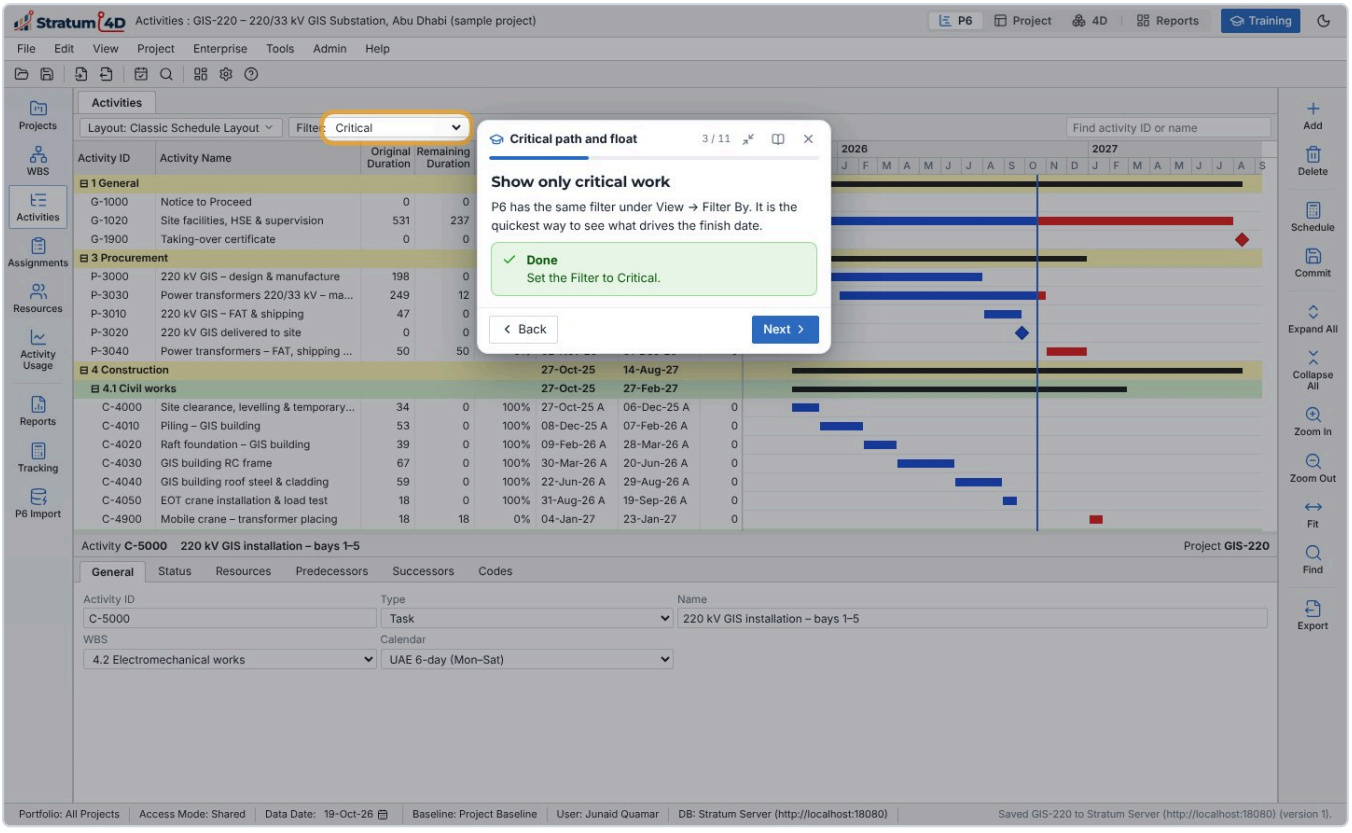


Figure 7. The moment the filter changes to Critical, the task turns green and the lesson moves on by itself a second later.

Show me performs the task for you: it types into the same boxes and clicks the same buttons you would, so you can watch where things are. **Back** returns to the previous step. The minimise button folds the

card into a small pill at the bottom right so you can look at the whole screen, and the book icon opens the catalogue.

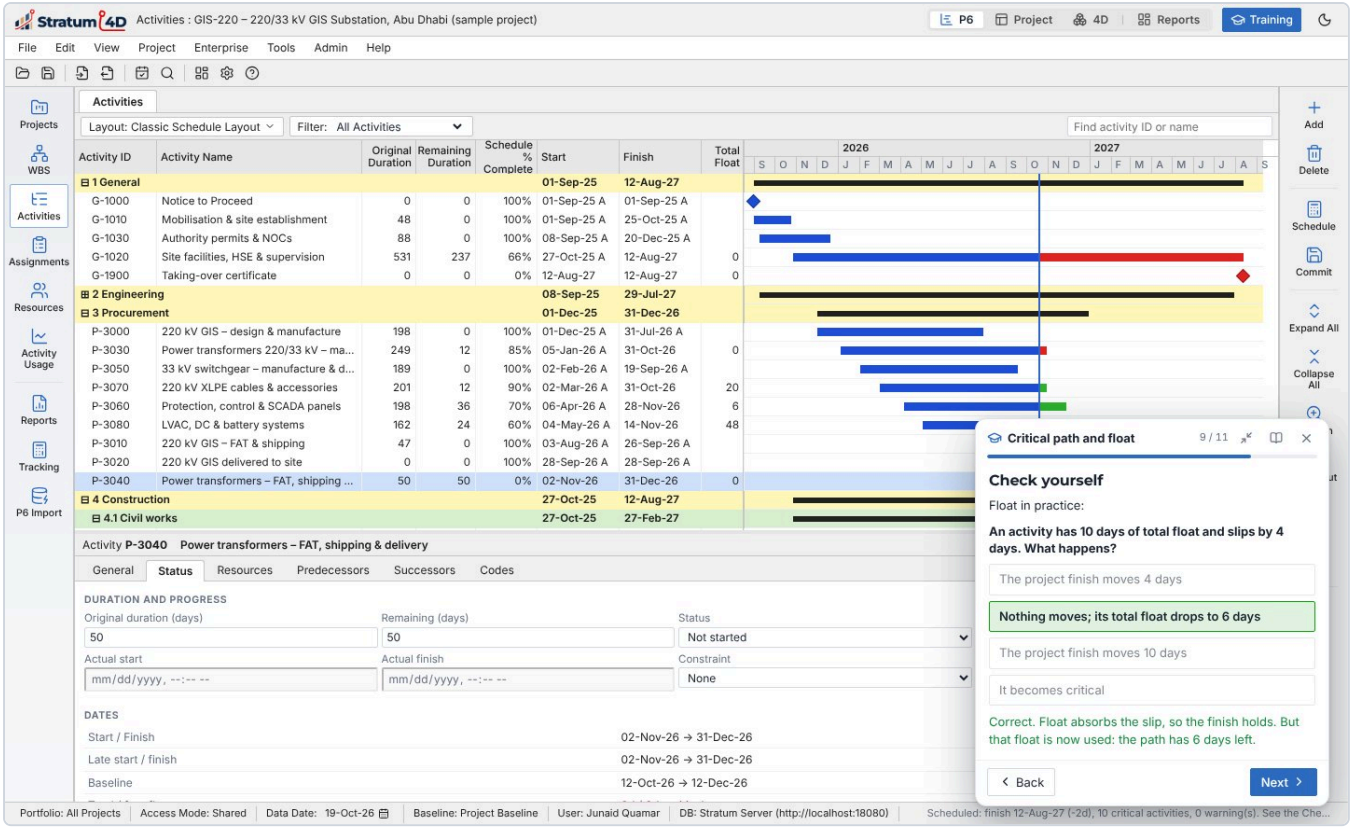


Figure 8. A check-yourself question. Choose an answer to see whether it is right and why.

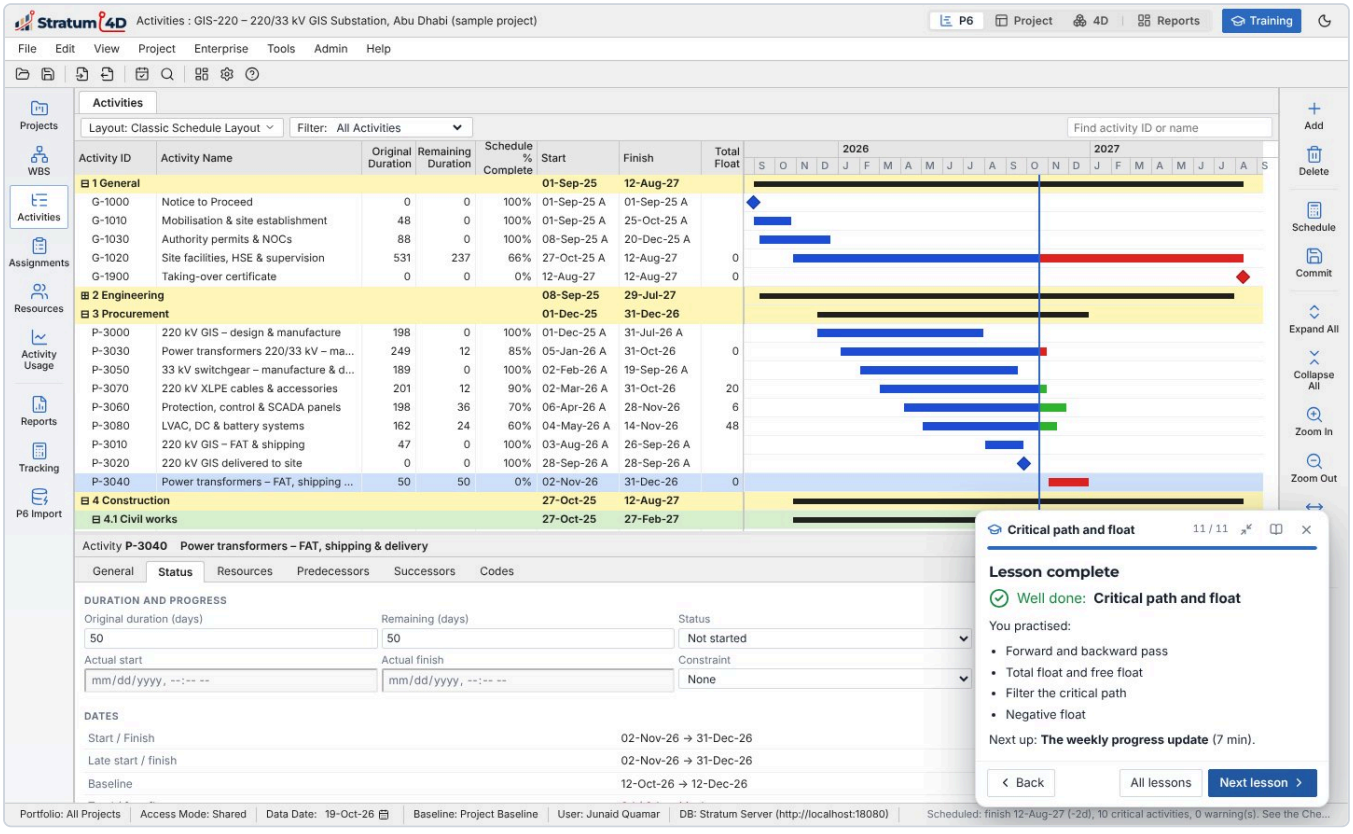


Figure 9. The end of a lesson: what you practised and the next lesson.

The glossary

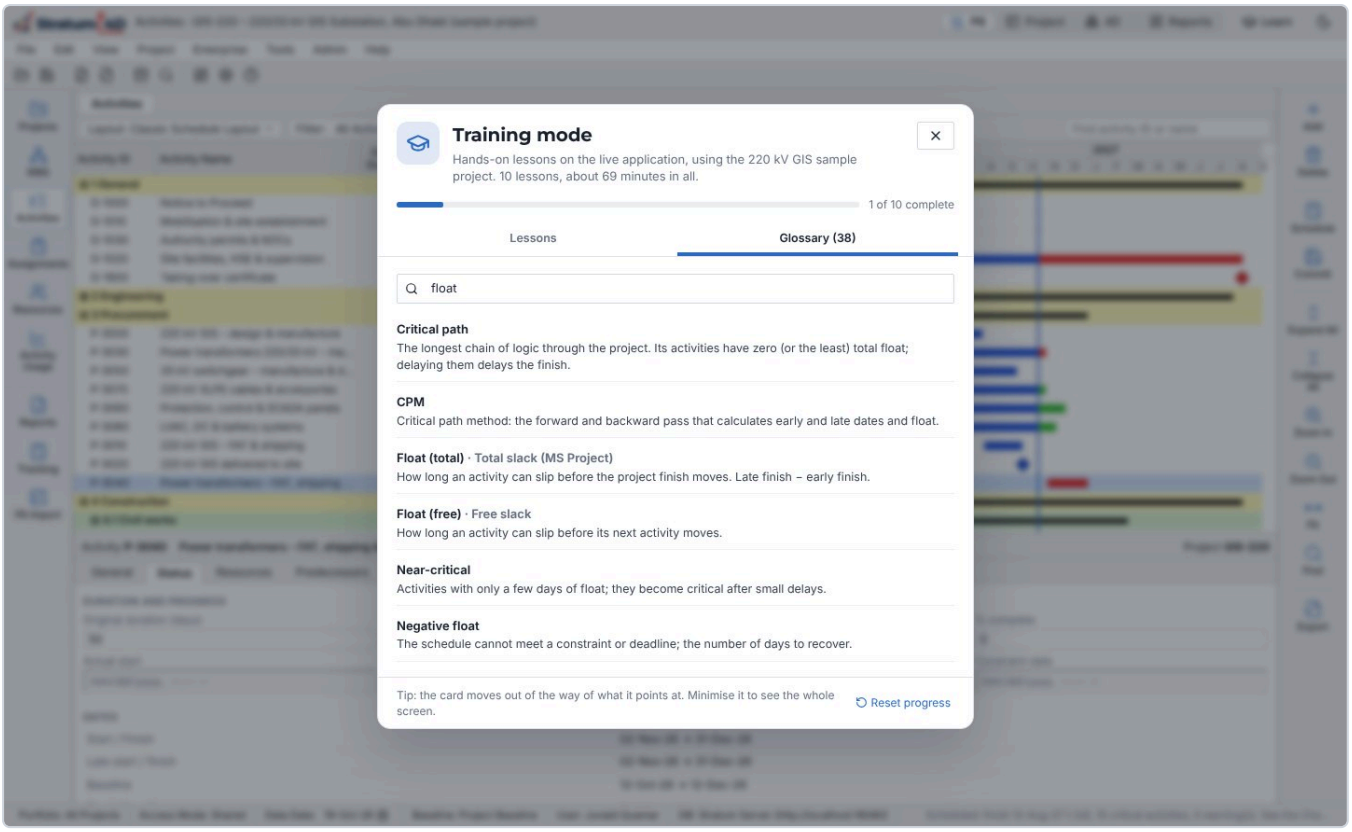


Figure 10. The glossary tab, searched for “float”. Each term shows its MS Project name where it differs. The full list is in Appendix C.

Notes for trainers

- Lessons change the sample project as a learner would (a new % complete, an added activity that is deleted again). Nothing is committed to the database unless the learner commits it. To start clean, open the sample again from the start screen.
- The Reports lesson creates a practice *Planning dashboard*. Delete it from *More* afterwards if you share a server.
- Progress is stored in the browser or desktop app on that computer. *Reset progress* at the bottom of the catalogue clears it.
- A sensible order for a new planner is the four courses in sequence, about 70 minutes, followed by the weekly update workflow in section 13 on a real project.

5. The three layouts

Stratum offers three layouts over the same engine and data. The **P6** and **MS Project** layouts are schedule-only, laid out like the applications planners already know, with no 3D view. The **4D** layout adds the model, linking tools and playback. The open project, the selection and the filters stay as they are when you switch.

Switching layouts

Click **P6**, **Project** or **4D** at the top right, or press `Ctrl`+`Alt`+`1`, `2` or `3`. The fourth button, **Reports** (`Ctrl`+`Alt`+`4`), opens the dashboard and report builder; the same switch sits in its top bar, so one click takes you back to any layout. The sun/moon button beside it switches between light and dark.

	P6 style	MS Project style	4D style
Laid out like	Primavera P6 Professional, Activities window	Microsoft Project, Gantt Chart view	A 4D planning tool
Top of the screen	Title bar, menu bar (File, Edit, View, Project, Enterprise, Tools, Admin, Help), icon toolbar	Green title bar with quick-access buttons, ribbon tabs (File, Task, Resource, Report, Project, View) and groups	Toolbar with icons and labels
Sides	Directory bar left, Command bar right	Vertical view label (Timeline, Gantt Chart, Task Form)	Gantt left, linking tools right
Main area	Layout and filter bar; activity table and Gantt with WBS bands	Timeline; Gantt Chart with row numbers, indicators, task mode, predecessors and resource names	3D model with the Gantt and timeline
Lower pane	Activity Details tabs, or a Resource / Activity Usage Profile	Task Form (Details), optional	Timeline playback
Gantt bars	Actual blue, remaining green, critical red; black WBS summary bars	Blue task bars with a progress line; black summary brackets; milestone dates; Tracking Gantt shows critical work and baselines	Blue bars, red critical, playback line
3D view	No	No	Yes

P6 style

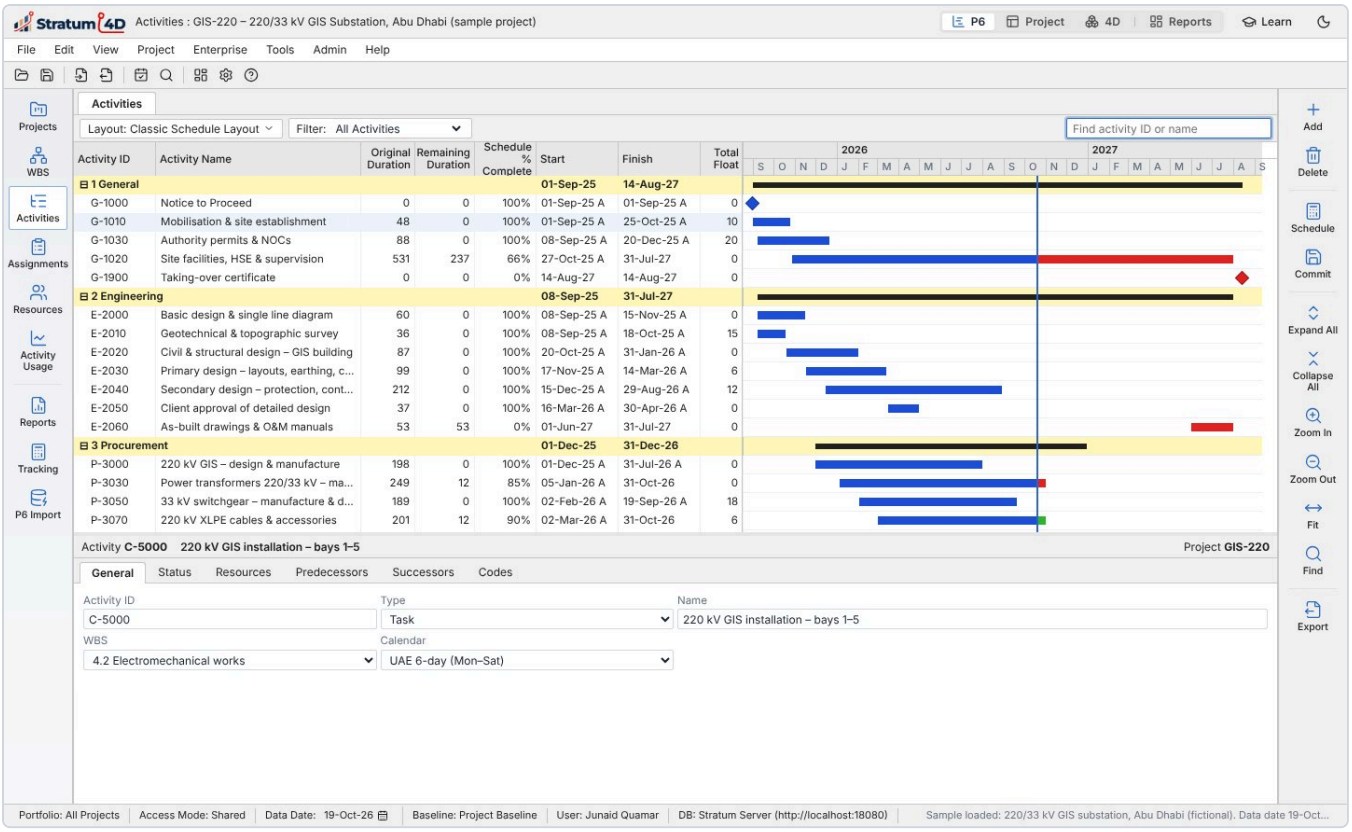


Figure 11. The P6 layout with the example project. C-5000 (220 kV GIS installation, bays 1–5) is selected and its General tab is open. WBS bands are coloured by level, actual dates carry an "A", and the vertical blue line is the data date, 19-Oct-26.

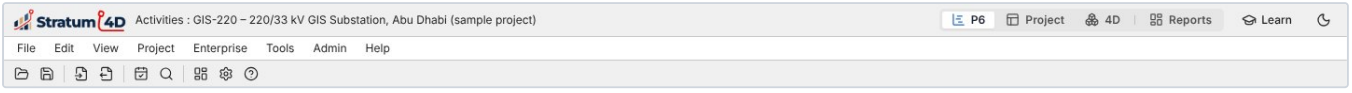


Figure 12. Title bar, menu bar and toolbar: the layout switch with Reports at the top right, then open, commit, import, export, schedule (F9), find, reports, preferences and help.

The **Directory bar** on the left decides what you look at: Projects, WBS, Activities, Assignments, Resources (Resource Usage Profile), Activity Usage (cost and earned value), Reports, Tracking and P6 Import. The **Command bar** on the right holds what you do: Add, Delete, Schedule, Commit, Expand All, Collapse All, Zoom In, Zoom Out, Fit, Find and Export.

Above the table, *Layout* switches between the Classic Schedule Layout and the same layout with baseline bars, and *Filter* shows all activities or only critical, not started, in progress, completed, late or milestone activities. The find box filters by activity ID or name. The table has P6's Classic Schedule Layout columns: Activity ID, Activity Name, Original Duration, Remaining Duration, Schedule % Complete, Start, Finish and Total Float.

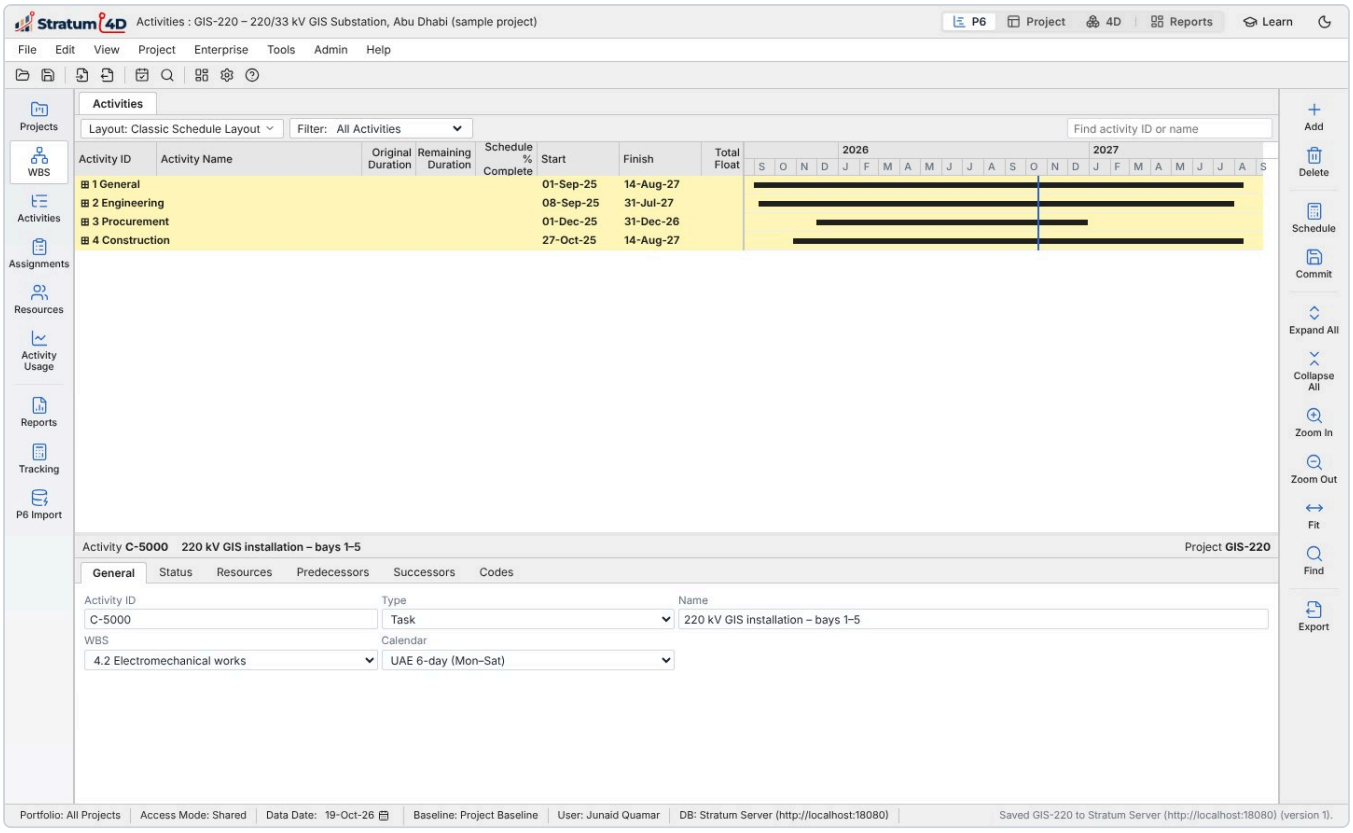


Figure 13. WBS on the Directory bar collapses the table to WBS bands: General, Engineering, Procurement and Construction with their dates. Activities expands it again.

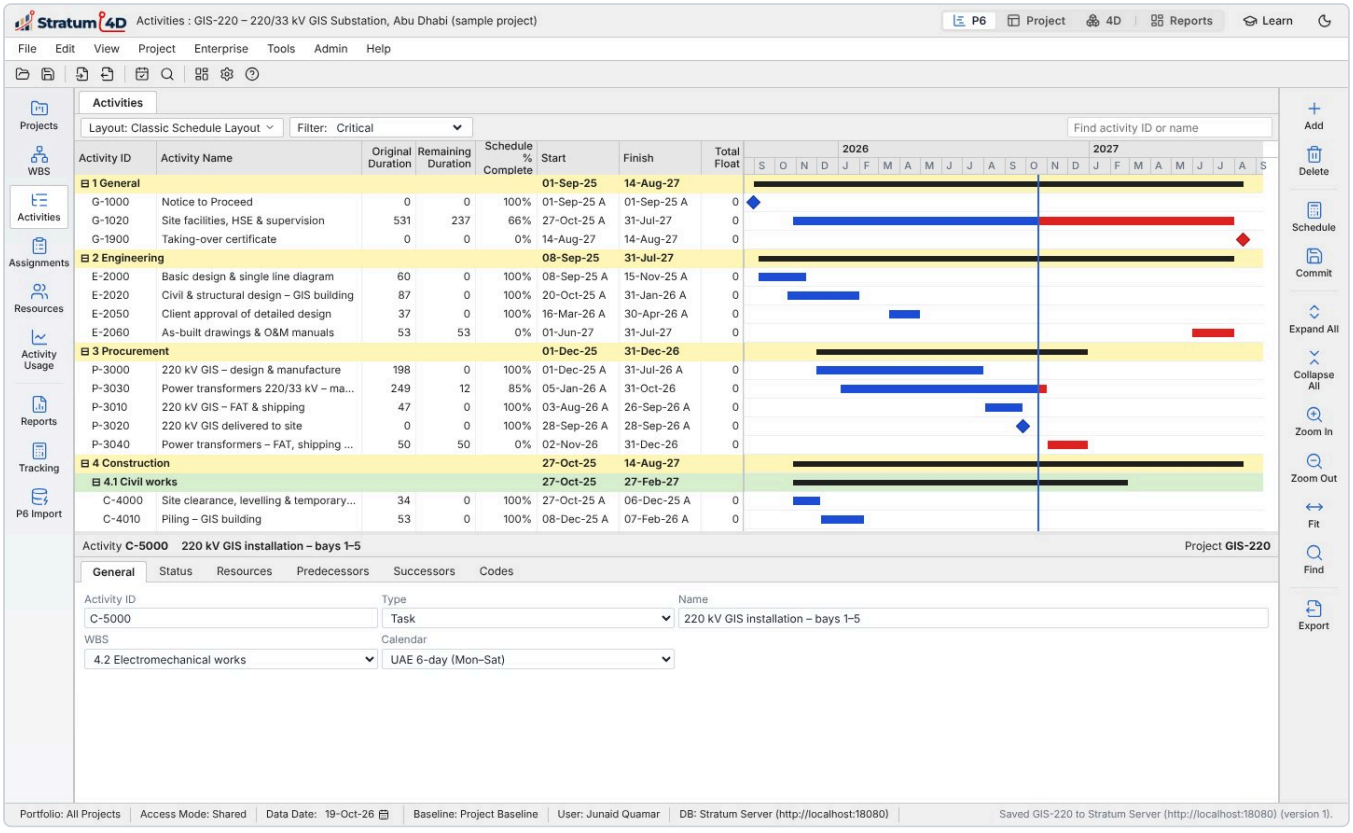


Figure 14. Filter: Critical keeps the critical activities with their WBS bands: transformer manufacture and delivery, transformer placing and erection, testing and energisation.

MS Project style

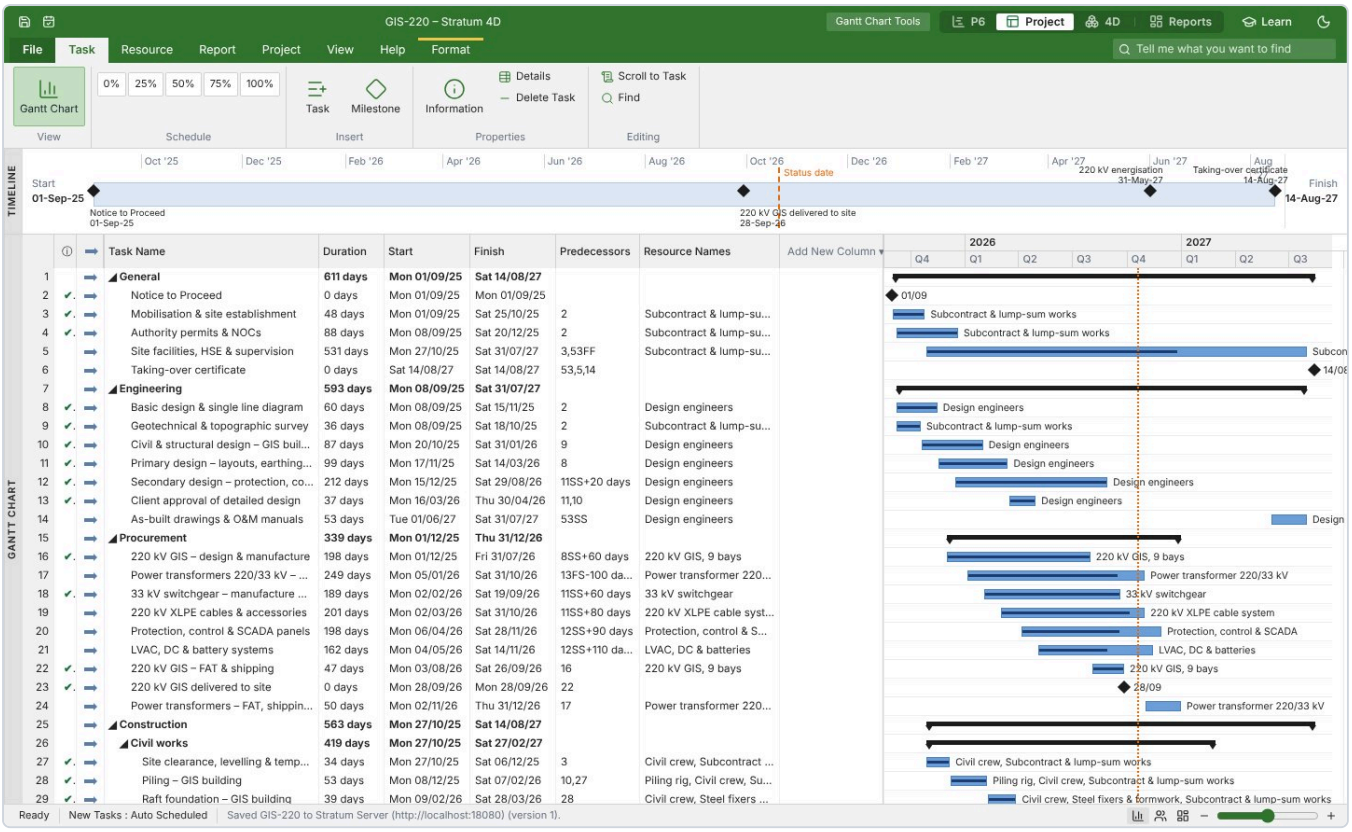


Figure 15. The MS Project layout: ribbon with the Task tab open (Gantt Chart, % complete buttons, insert Task or Milestone, Information, Details, Scroll to Task, Find), the Timeline with the status date and milestones, and the Gantt Chart with resource names beside the bars.

The Task tab's *0% – 100%* buttons set the selected task's progress, and *Calculate Project* on the Project tab reschedules (F9). *Details* opens the Task Form under the Gantt; *Tracking Gantt* on the View tab shows critical work in red with the baseline under each bar.

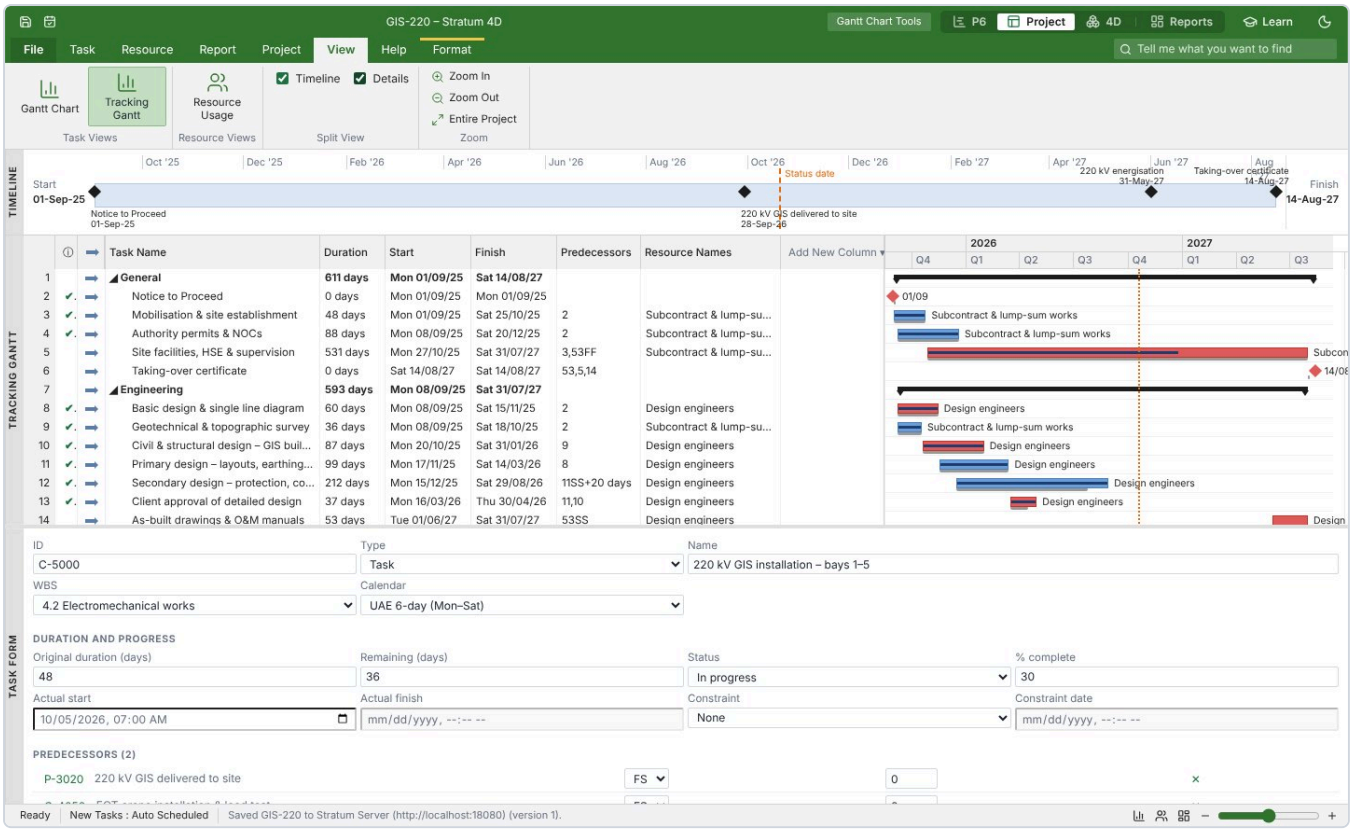


Figure 16. View → Tracking Gantt with Details ticked: critical tasks in red, baselines under the bars, and the Task Form below.

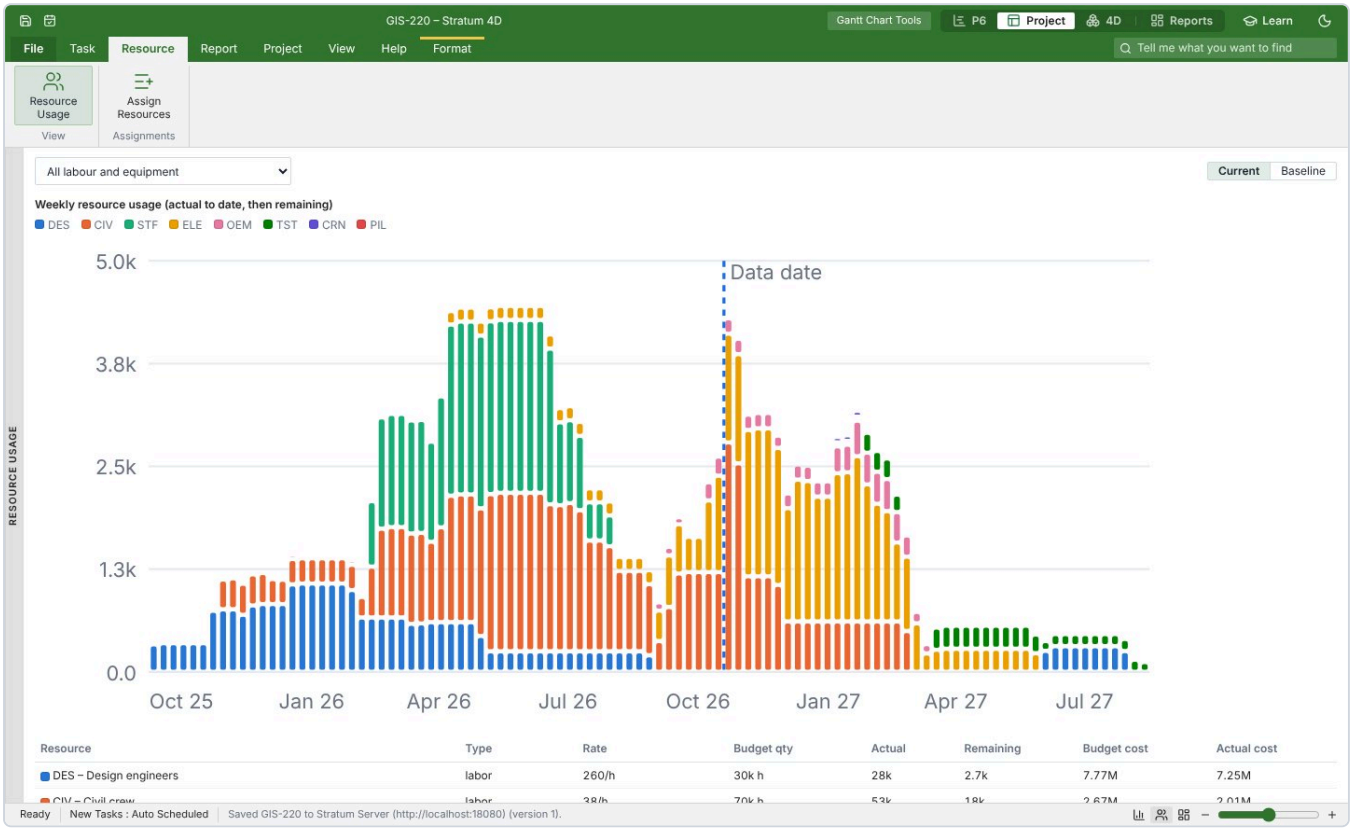


Figure 17. Resource → Resource Usage: weekly hours per resource, actual to date then remaining, with a table of budget, actual and remaining quantities and costs.

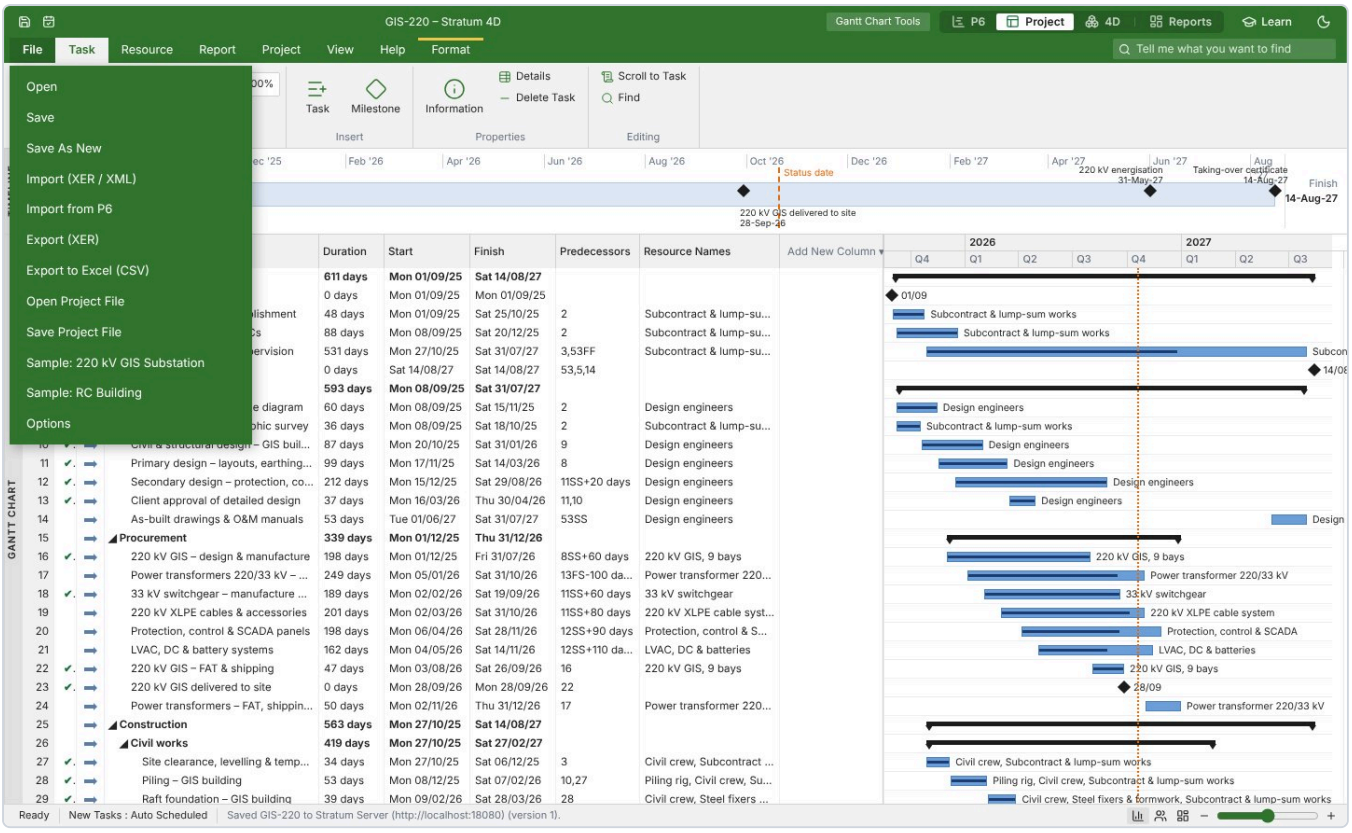


Figure 18. The File tab: open and save in the database, import and export, project files, the samples and options.

4D style

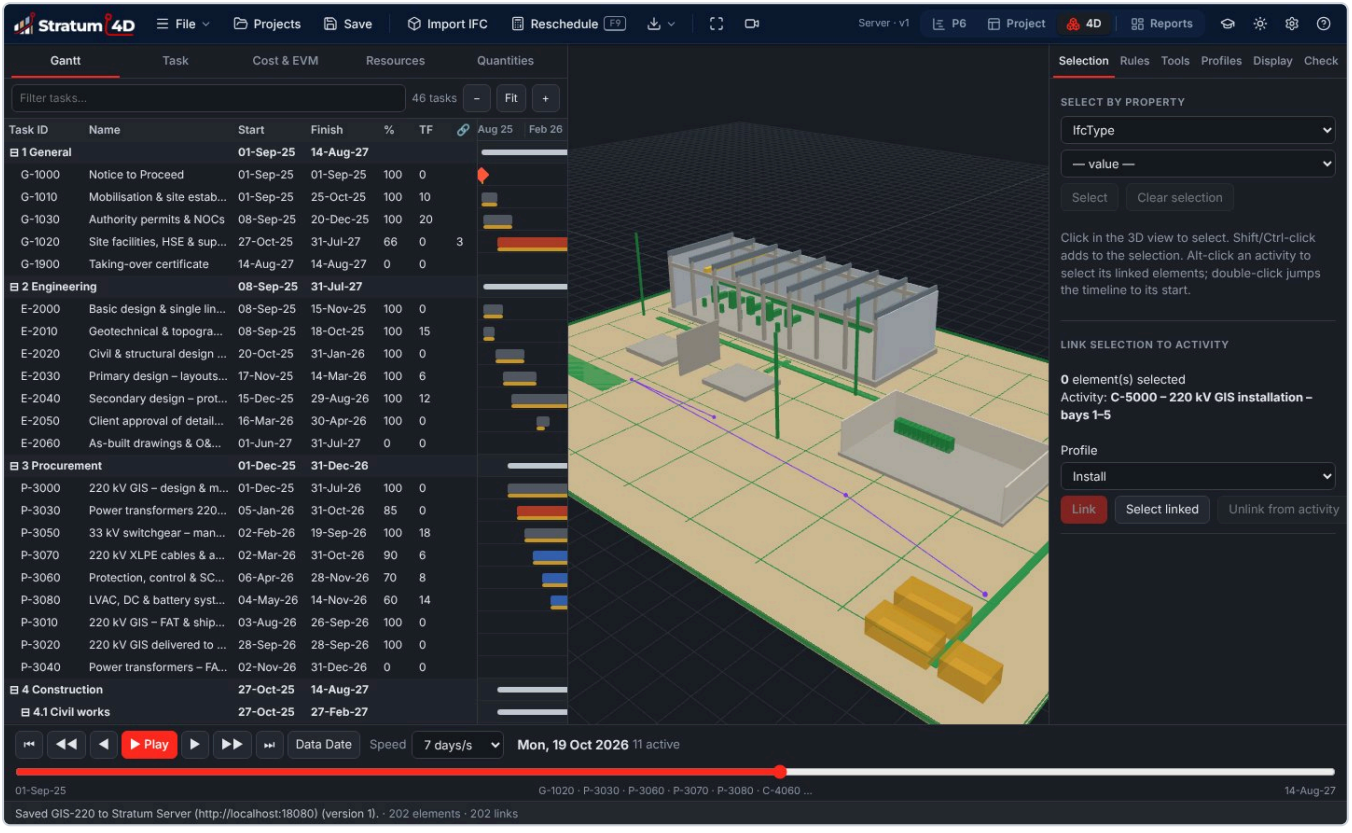


Figure 19. The 4D layout in its default dark theme. Green elements are being installed at the data date: GIS bays, cable trenches and 33 kV switchgear. The amber boxes are the site cabins, shown only while site facilities (G-1020) run.

Section 8 covers the 4D layout in detail.

Theme, font and text size

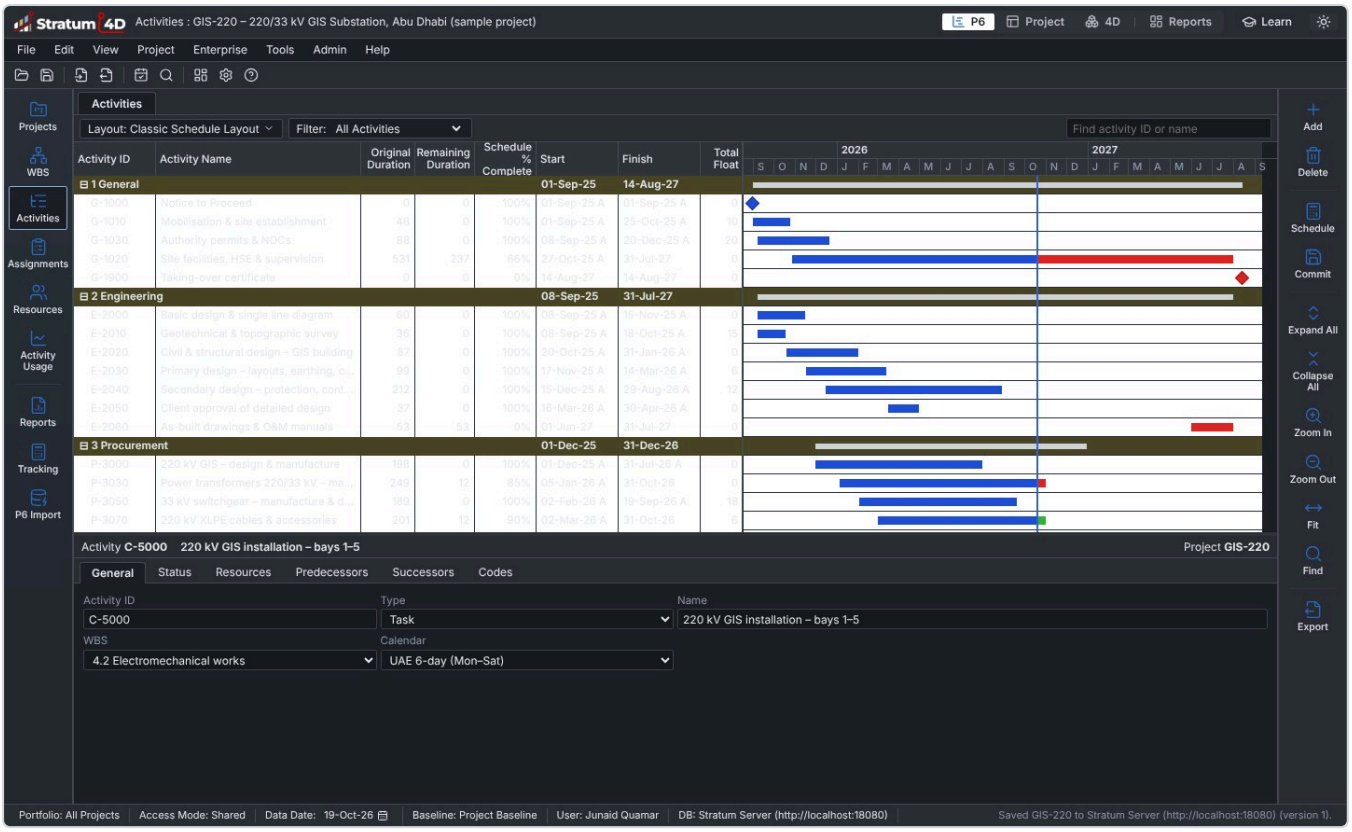


Figure 20. The P6 layout in the dark theme.

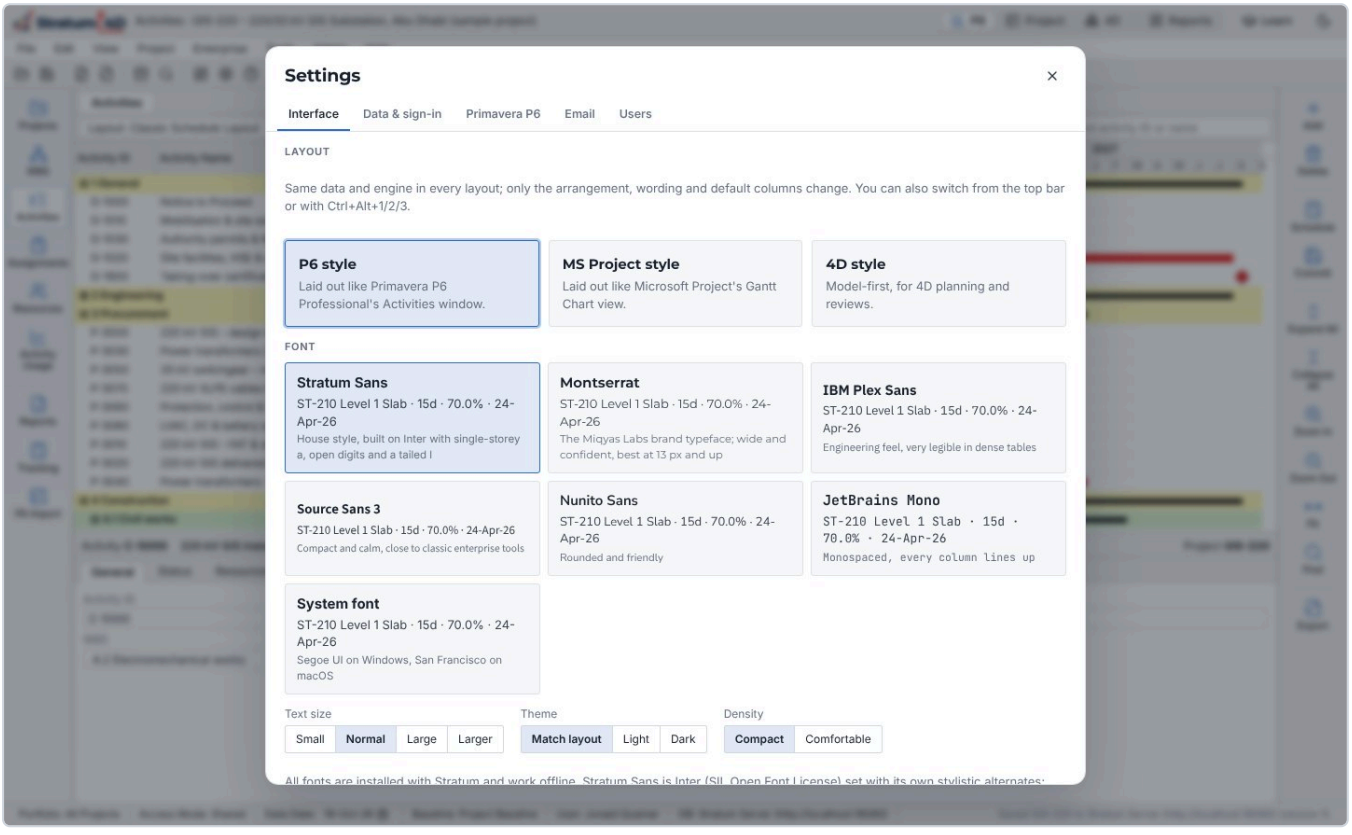


Figure 21. Settings → Interface (in the P6 layout: Admin → Preferences). Each font shows a sample line in its own face.

Fonts are installed with Stratum, so the desktop app looks the same offline. The default, **Stratum Sans**, is Inter set with its own stylistic alternates; headings, figures and the logo use **Montserrat**, the Miqyas

Labs brand typeface, which can also be chosen for the whole interface. IBM Plex Sans, Source Sans 3, Nunito Sans, JetBrains Mono and the system font are also offered, all under the SIL Open Font License. Table columns widen automatically for larger text or wider fonts.

6. Planning fundamentals

This chapter explains the ideas behind the screens: how CPM calculates dates, what float means, how progress is measured and how earned value works. Each part ends with what to look at in Stratum.

Activities, durations and calendars

An **activity** is a piece of work someone can be held to: it has a clear start and finish, a duration and a responsible party. Durations are in *working* days, converted to dates through the activity's calendar. The example project uses a UAE six-day calendar (Monday to Saturday, 8 hours a day), so a 6-day activity starting on a Monday finishes on the Saturday.

Two duration fields matter: **original duration**, the plan, which does not change during the job, and **remaining duration**, the current estimate of the time left. CPM uses remaining duration for every unfinished activity, so it is the remaining duration, not the % complete, that moves forecast dates.

Logic: the four relationship types

Type	Meaning	Typical use
FS · finish to start	B cannot start until A finishes	Most logic: foundations → steel erection
SS · start to start	B cannot start until A starts (plus lag)	Overlapping work: trenching started → cable pulling can start 10 days later
FF · finish to finish	B cannot finish until A finishes	Testing finishes after installation finishes
SF · start to finish	B cannot finish until A starts	Rare; hand-overs between shifts. Avoid in construction schedules

Lag adds time to a relationship (FS + 7 days for curing). **Lead** is a negative lag; most review standards reject leads, because they make the logic hard to follow. If a wait needs to be visible, model it as an activity.

A worked CPM example

Six activities, all finish-to-start. Days are counted from day 0 at the start of the job.

ID	Activity	Duration	Predecessors
A	Site survey	3	–
B	Foundations	5	A
C	Steel erection	4	B
D	Cable trench	4	A
E	Cable pulling	3	D
F	Energisation	1	C, E

Forward pass (early dates): each activity starts when its latest predecessor finishes. A runs 0–3. B runs 3–8 and C 8–12. D runs 3–7 and E 7–10. F waits for both C (12) and E (10), so it starts at 12 and the project

finishes on day **13**.

Backward pass (late dates): from day 13, each activity must finish by the earliest late start of its successors. F: 12–13. C: 8–12. E must finish by 12, so its late dates are 9–12. D must finish by E's late start of 9, giving 5–9. B: 3–8. A must finish by the earlier of B's and D's late starts, 3.

ID	Early start	Early finish	Late start	Late finish	Total float	Free float	Critical
A	0	3	0	3	0	0	•
B	3	8	3	8	0	0	•
C	8	12	8	12	0	0	•
D	3	7	5	9	2	0	
E	7	10	9	12	2	2	
F	12	13	12	13	0	0	•

Total float = late finish − early finish. A, B, C and F have none: they are the critical path, 13 days long. D and E share 2 days of total float. **Free float** = early start of the next activity − early finish. D has no free float, because any slip pushes E. E has 2 days, since F waits for C anyway. If D slips 2 days, the finish holds but E becomes critical: float belongs to the path, not to one activity.

In Stratum: press F9, then read early and late dates and float in the Status tab, or filter *Critical*. The lesson *Critical path and float* walks through the same ideas on the example project.

Constraints

A constraint imposes a date: *start on or after*, *finish on or before*, *mandatory start* and so on (Stratum supports P6's nine types). Soft constraints such as *start on or after* are normal for access dates or deliveries. Hard constraints such as *mandatory finish* override logic and hide the true critical path, so use them only for contract dates, and record why. A *finish on or before* that the logic cannot meet produces **negative float**: the number of days to recover.

The data date and updating

The **data date** (MS Project: status date) separates actual from forecast. At each update you record actual starts and finishes, % complete and remaining duration for work in hand, move the data date forward and reschedule. CPM then places all remaining work on or after the data date. An activity that should have started but has not is pushed to the data date, and its successors move with it. That is how a delay shows up.

Base % complete on something measurable: bays installed out of nine, metres of cable pulled, steel tonnage erected, drawings issued. Then check the remaining duration against the work left. An activity at 90% with a remaining duration still at 60% of the original is a warning sign.

Baselines and variance

The **baseline** is the approved plan, frozen. Variance compares the current schedule with it: `finish variance = forecast finish - baseline finish`. In the example the taking-over certificate G-1900 is forecast 14-Aug-27 against a baseline of 31-Jul-27, 14 days late, mainly because transformer manufacture P-3030 finishes three weeks after its baseline. Re-baselining hides history, so it is done only for approved changes (an extension of time, a variation), and the old baseline is kept.

Earned value

Earned value compares three cumulative numbers at the data date: **planned value** (the budget of the work scheduled by now), **earned value** (the budget of the work actually done) and **actual cost**. A worked example with a budget at completion (BAC) of AED 1,000,000:

Measure	Formula	Value	Reading
Planned value (PV)	budget scheduled by the data date	400,000	
Earned value (EV)	budget × % complete	360,000	
Actual cost (AC)	from cost reports	380,000	
Schedule variance (SV)	EV − PV	−40,000	behind schedule
Cost variance (CV)	EV − AC	−20,000	over budget
SPI	EV ÷ PV	0.90	10% less work earned than planned
CPI	EV ÷ AC	0.947	each dirham buys 0.95 of planned work
EAC	BAC ÷ CPI	1,055,556	forecast final cost if efficiency continues
ETC	EAC − AC	675,556	cost still to come
VAC	BAC − EAC	−55,556	forecast overrun
TCPI	(BAC − EV) ÷ (BAC − AC)	1.032	efficiency needed on the rest to finish on budget

The example project reports SPI 0.96 and CPI 0.99 at its data date (section 9). SPI drifts back to 1.0 as any project nears completion, whether or not it is late, so on long jobs read it together with the finish variance and the critical path.

Weighted physical progress

EPC contracts usually report progress as a weighted percentage: each discipline or package has a weight (by budget, hours or an agreed split), and overall progress is the weighted sum. Section 12 shows the example with cost weights and with the manual split 3 / 7 / 50 / 40.

S-curves

An S-curve plots cumulative progress or cost over time. Slow mobilisation, a steep middle and a slow finish give the S shape. The gap between the baseline curve and the actual curve at the data date is the progress shortfall; the forecast curve shows when the work will really finish.

7. Working with the schedule

Importing

Import reads Primavera P6 XER files and Microsoft Project XML (MSPDI). From XER, Stratum reads WBS, activities, relationships with lags, calendars, constraints, activity codes, user-defined fields, actual and baseline dates, resources, rates and assignments. Re-importing a newer XER over the open project updates it; links to the model are kept by Activity ID.

Activity details

In the P6 layout the lower pane holds the Activity Details tabs for the selected activity: General (ID, type, name, WBS, calendar), Status (durations, status, % complete, actual dates, constraint, early/late dates and float), Resources (assignments with budget, actual and remaining units and cost), Predecessors, Successors and Codes (user fields).

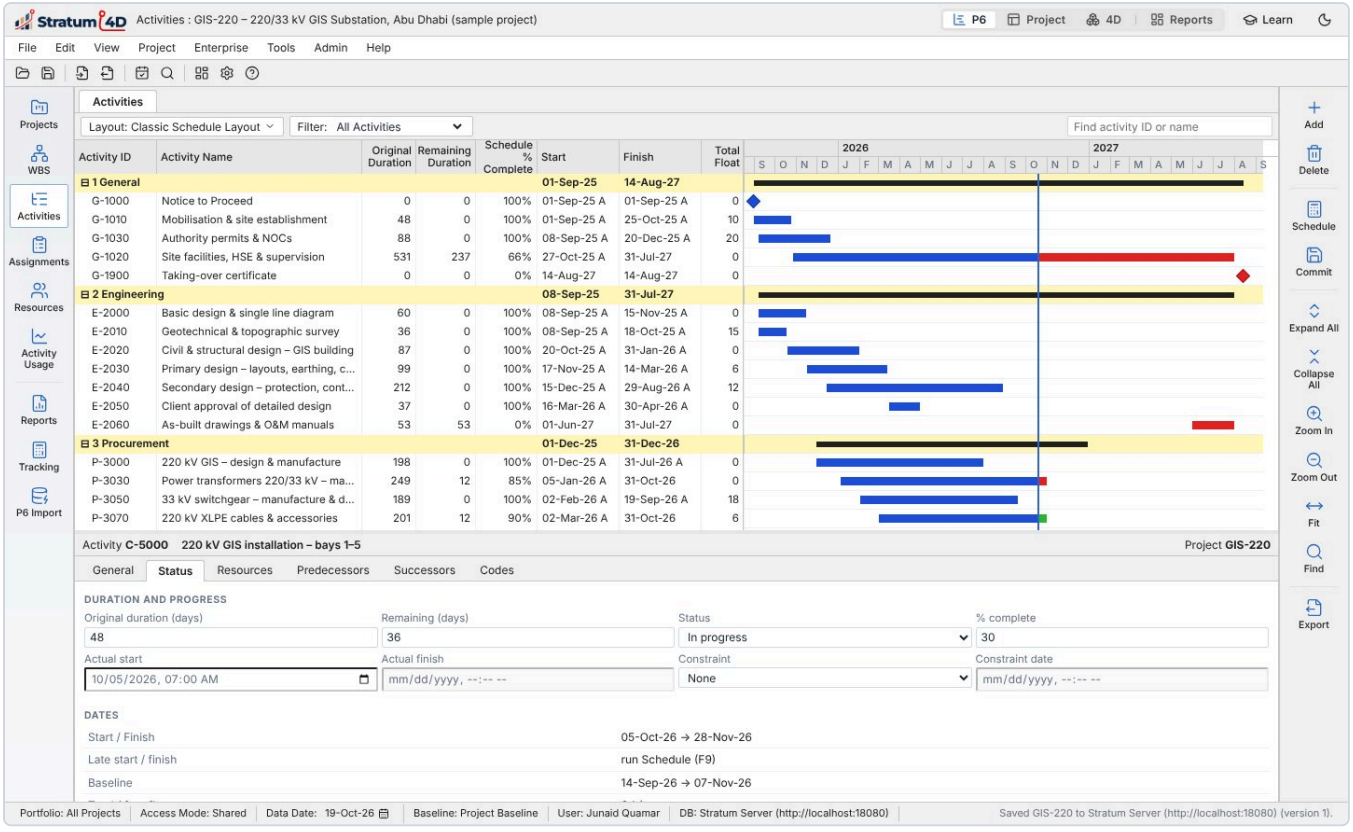


Figure 22. Status tab for C-5000: original 48 days, remaining 36, in progress at 30% with its actual start; the dates section below shows early and late dates and float after scheduling.

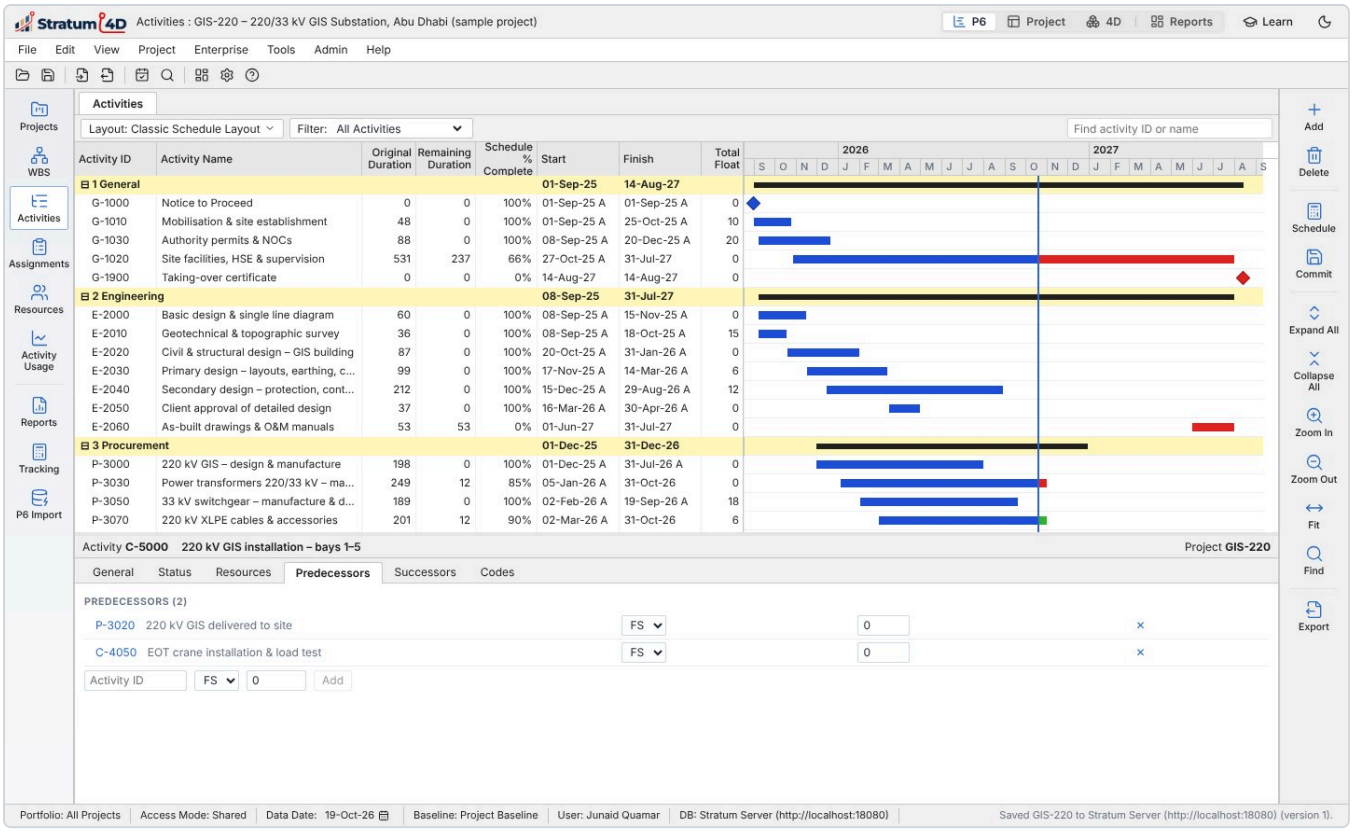


Figure 23. Predecessors tab: relationship type and lag for each predecessor, and a row to add one by activity ID.

Add (`Ins`) inserts an activity after the selected one with a 5-day duration and a finish-to-start link; *Delete* (`Del`) removes it with its relationships. Edits mark the schedule as changed and a banner asks you to reschedule.

Scheduling (F9)

Press `F9`, *Schedule* (P6) or *Calculate Project* (MS Project). The CPM engine follows P6's default options:

- FS, SS, FF and SF relationships with lags, on each activity's working calendar and holidays;
- nine constraint types;
- retained logic from the data date;
- total and free float, critical path, level-of-effort and WBS summary dates;
- loop detection and open-end warnings.

For the example project F9 finishes on 12-Aug-27, with the critical path running through transformer manufacture, delivery, erection and energisation.

Exporting

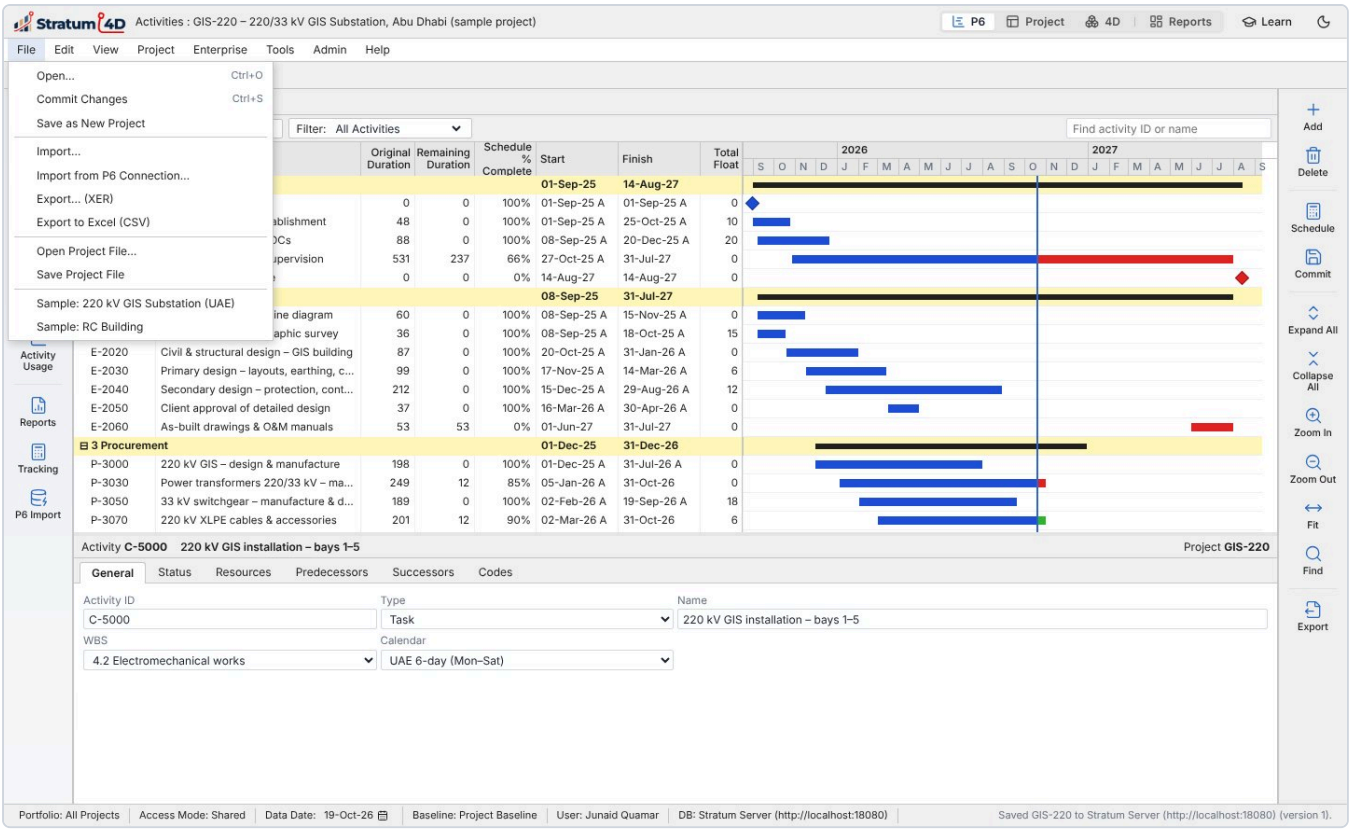


Figure 24. The P6 File menu: open and commit, import (file or P6 connection), export to XER or CSV, project files and the samples.

Export (XER) writes a file that updates the original project when imported in P6 with *Update Existing Project*; tables Stratum did not touch pass straight through. *Export to Excel (CSV)* writes the activity table.

8. 4D simulation

A 4D model links each model element to the activity that builds, removes or uses it, then plays the schedule against the model.

Linking the example model

The example model carries the activity ID in each element's property `Pset_Stratum.ActivityID` (for instance, every part of GIS bays 1–5 holds `C-5000`). One rule therefore links all 202 elements: *element property Pset_Stratum.ActivityID equals Activity ID*.

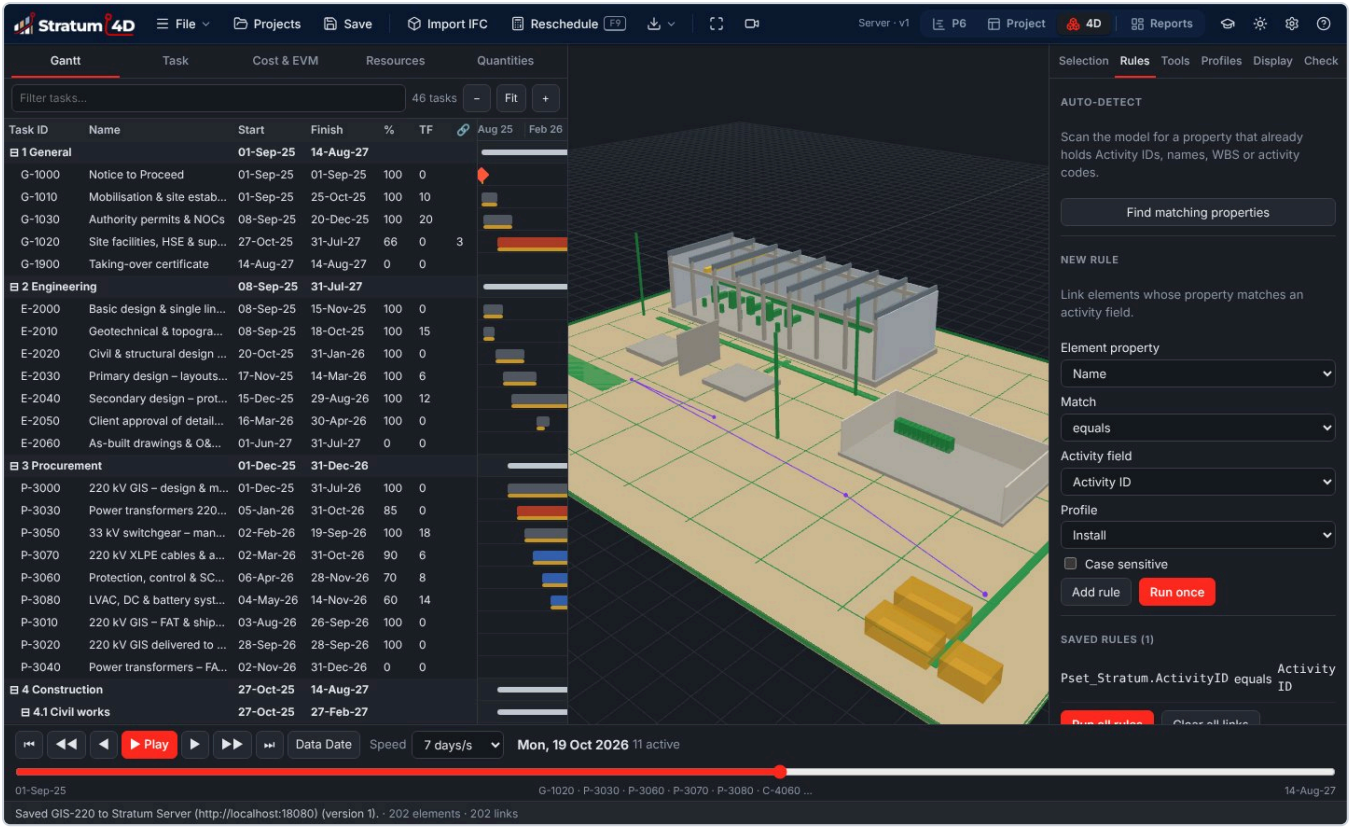


Figure 25. Rules tab. Find matching properties scans the model for a property that already holds activity IDs, names, WBS or codes. Below it, a rule is built from an element property, a match type, an activity field and a profile.

For your own models, the same approach works with a shared parameter from Revit, ArchiCAD or Tekla. Without one, select elements in 3D (or *Selection → Select by property*), pick the activity in the Gantt, choose a profile and press *Link*.

Appearance profiles

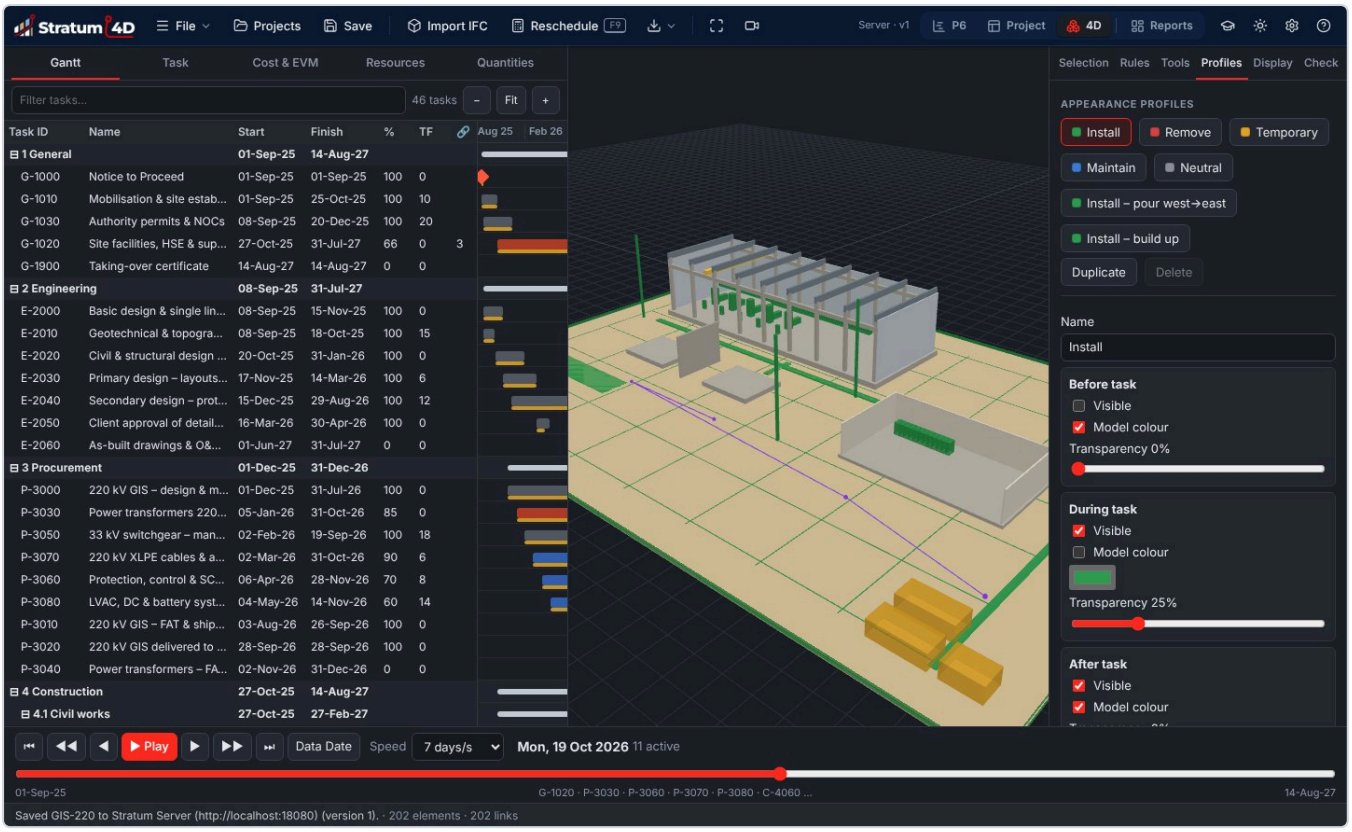


Figure 26. Profiles tab. Each profile sets visibility, colour and transparency before, during and after its activity.

The example uses *Install* for equipment, *Install – pour west→east* for rafts, slabs and trenches, *Install – build up* for walls, columns, piles and cladding, and *Temporary* for the site cabins and the 200 t mobile crane, which appear only while their activity runs.

Groups, pours and equipment paths

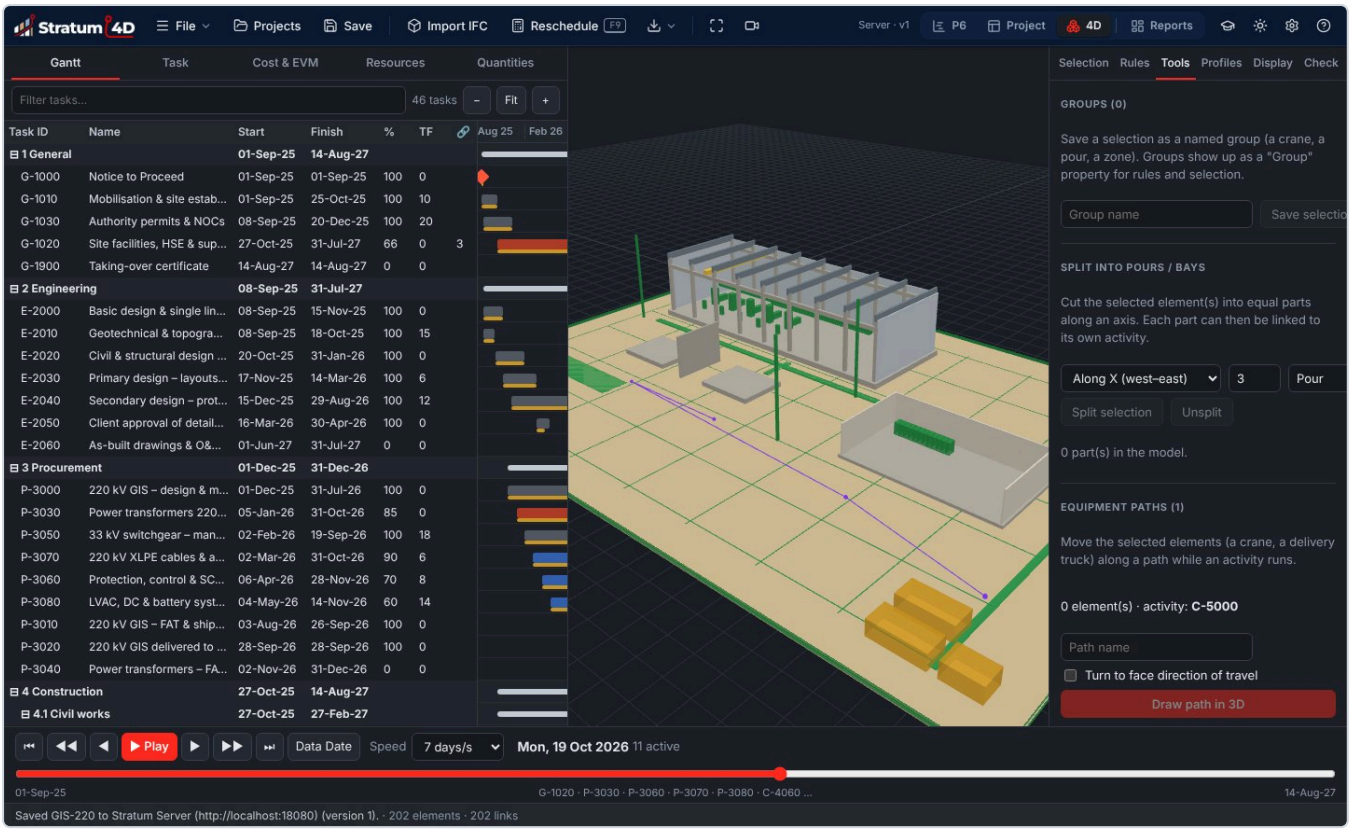


Figure 27. Tools tab: save selections as named groups, split elements into pours or bays, and move equipment along a path.

The example's crane path takes the 200 t mobile crane from the gate along the access road to transformer T1 and then T2 during activity C-4900 (transformer placing).

Display options and checks

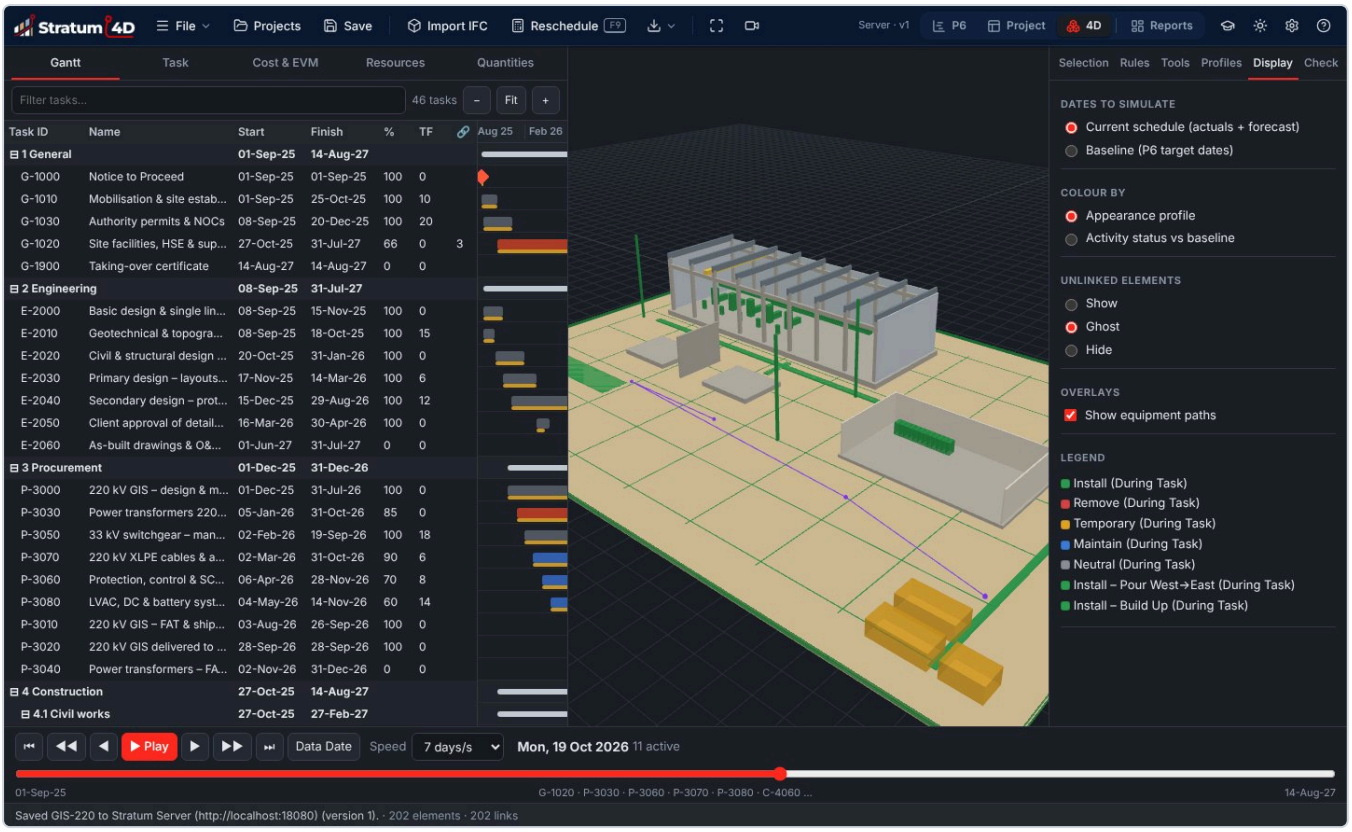


Figure 28. Display tab: simulate current dates or the P6 baseline, colour by profile or by status against baseline, ghost or hide unlinked elements, and show equipment paths.

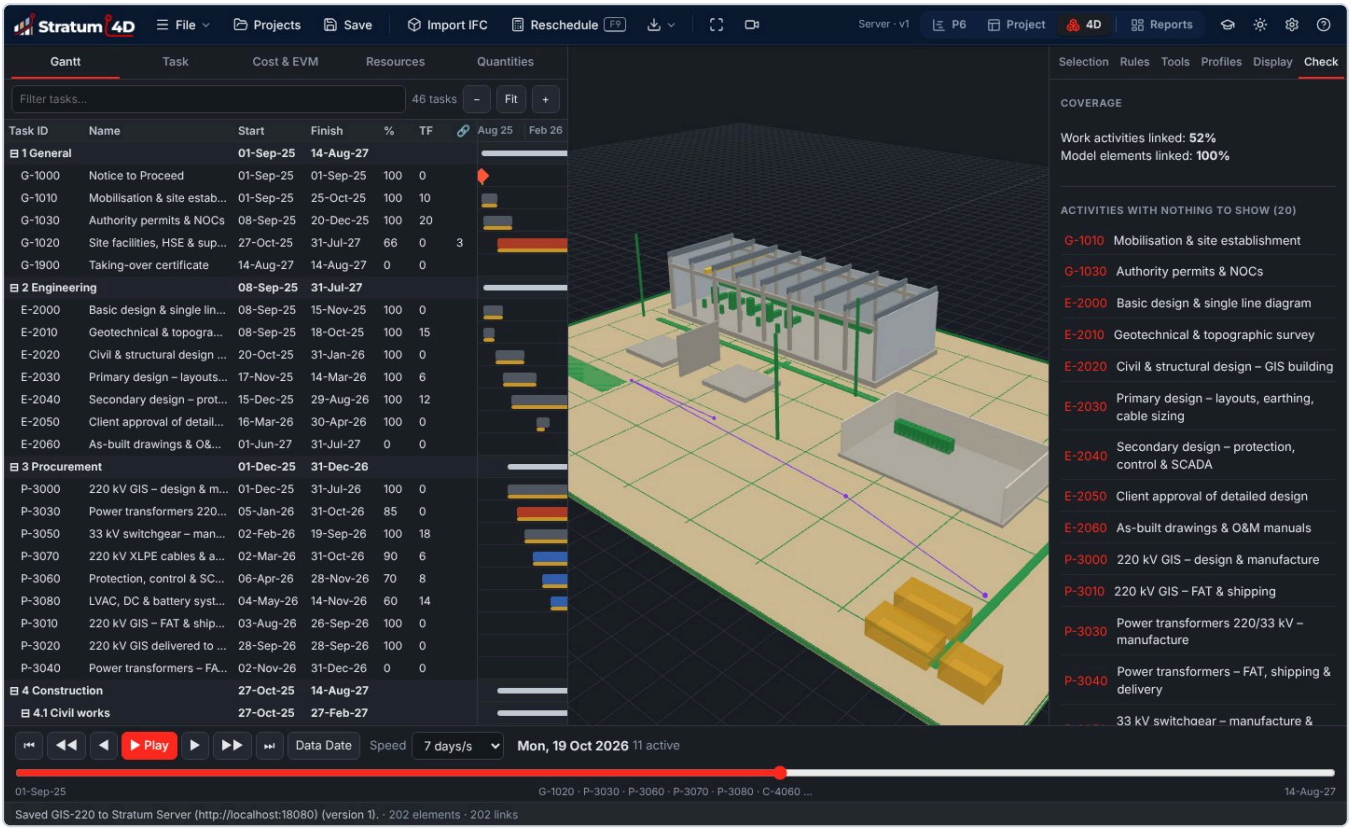


Figure 29. Check tab: link coverage, activities with nothing to show, elements never shown, and the change report after a schedule update.

Playing the sequence

Use the timeline at the bottom: Play/Pause (`Space`), step one day (`←` `→`) or one week (`Shift`+arrows), jump to the start, data date or finish, and set the speed. *Record* saves the playback as a WebM video.

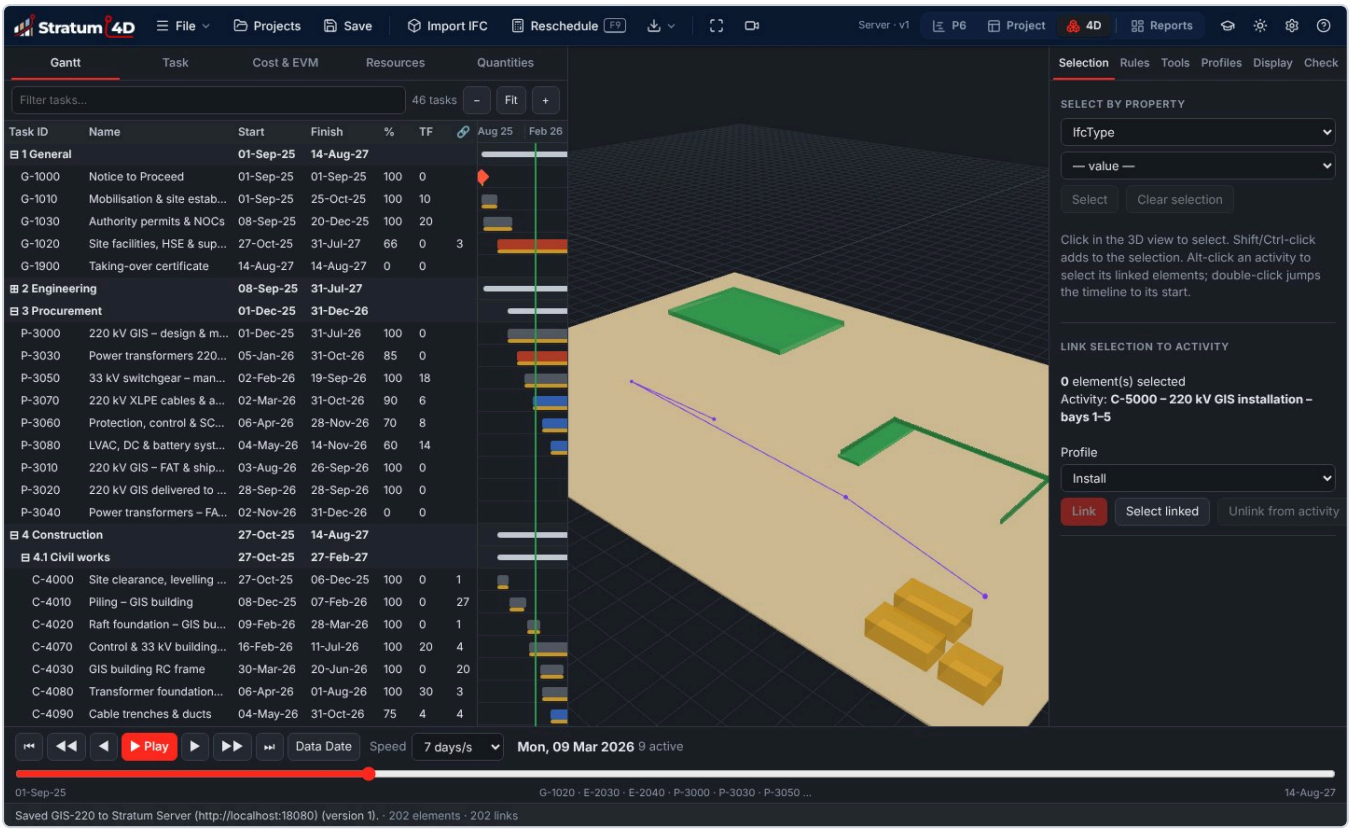


Figure 30. 9 March 2026: site levelled, piles done, the GIS building raft being poured (green) and the control building foundations started. The amber boxes are the site cabins.

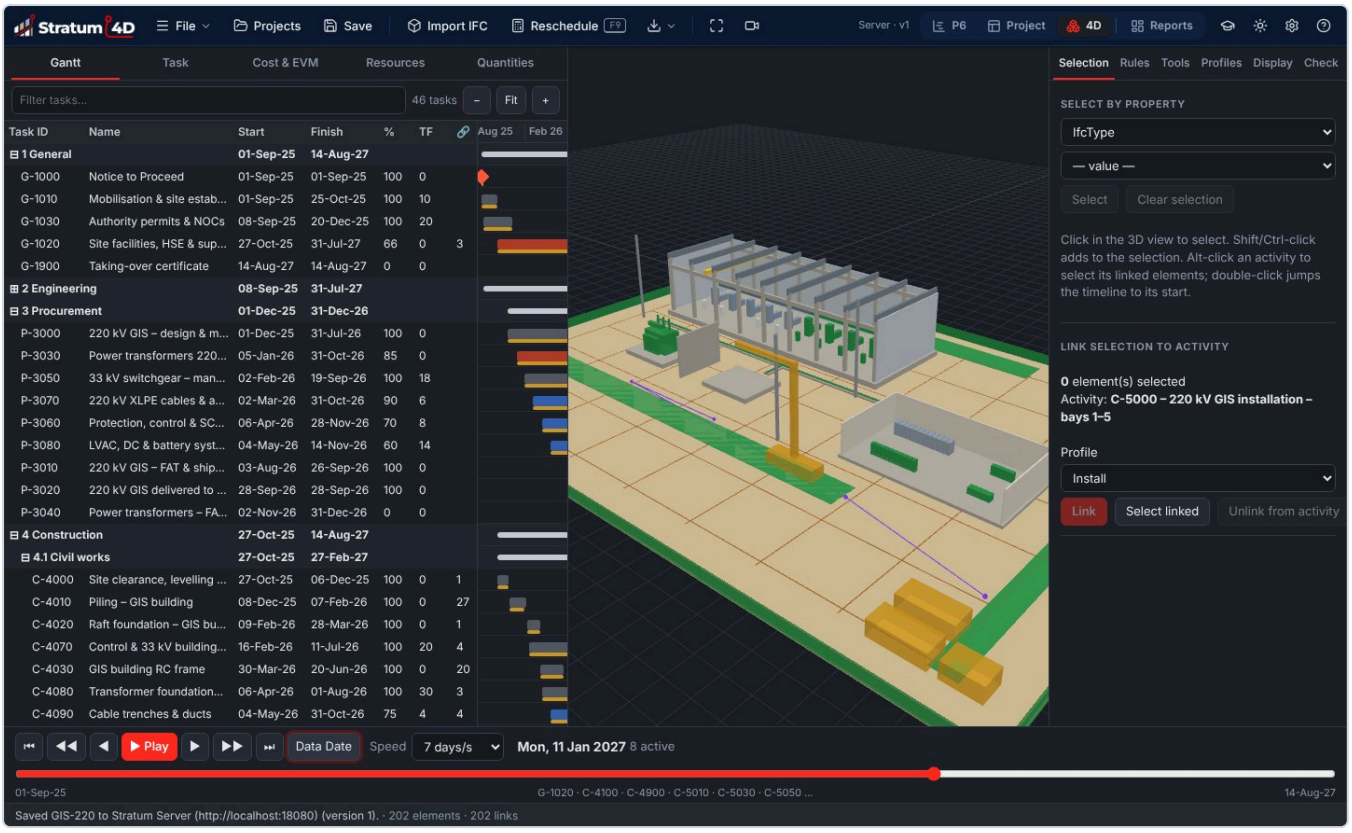


Figure 31. 11 January 2027: GIS hall complete with bays being installed, transformer T1 being erected, the crane (yellow) on its path, cables going into the trenches and the boundary wall under way.

9. Cost, resources and quantities

In the P6 layout, choose what the lower pane shows from the Directory bar or *View → Show on Bottom*: Activity Details, the **Resource Usage Profile** or the **Activity Usage Profile**. In the MS Project layout, Resource Usage is a view of its own (Resource or View tab). Quantities come from the 3D model, so they live in the 4D layout.

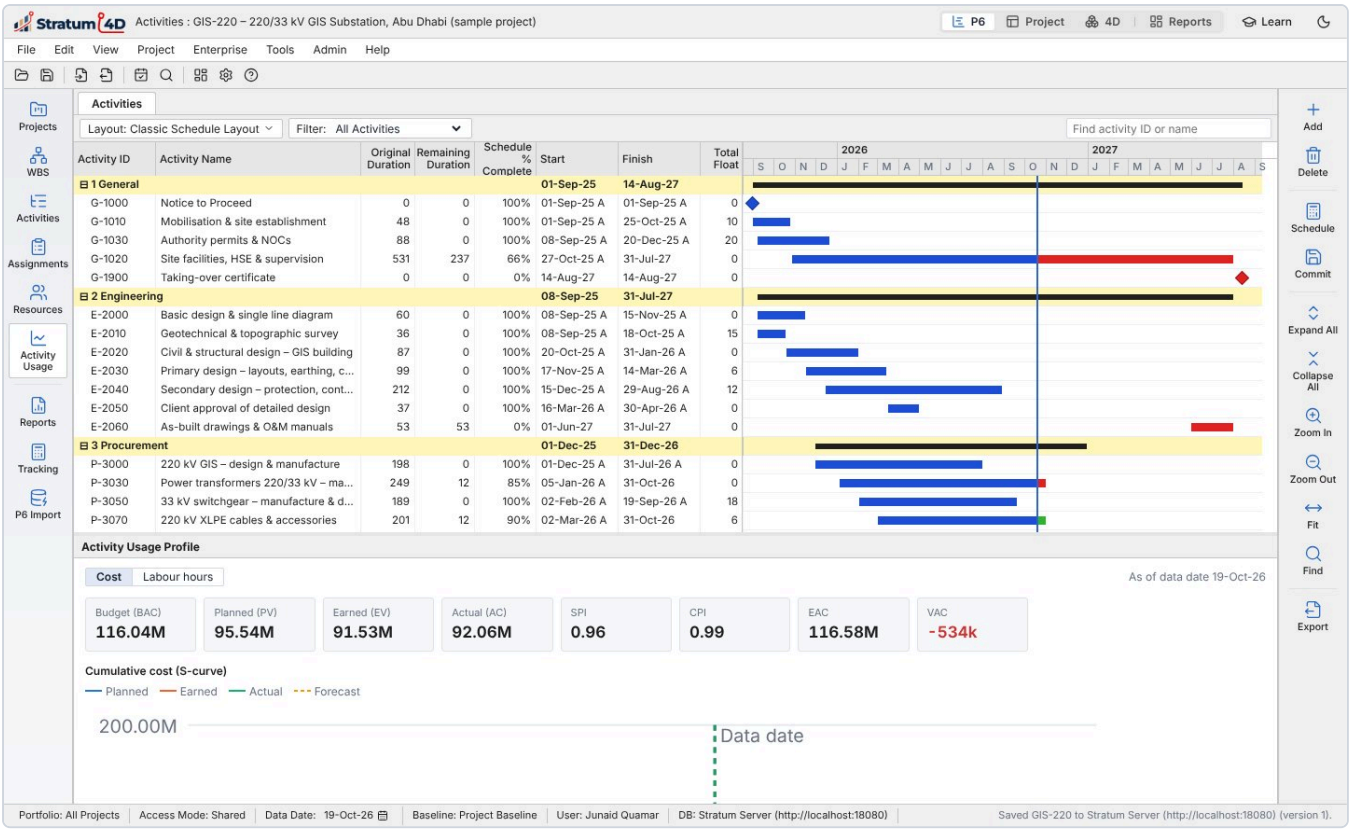


Figure 32. Activity Usage Profile for the example at the data date: budget at completion AED 116.04M, planned value 95.54M, earned value 91.53M, actual cost 92.06M, SPI 0.96, CPI 0.99, EAC 116.58M and VAC −534k, with the cumulative S-curve below.

Earned value is calculated from resource assignments, in cost or labour hours. In the example, SPI 0.96 reflects the late GIS delivery and transformer manufacture; the small cost overrun comes from piling, cladding and secondary design.

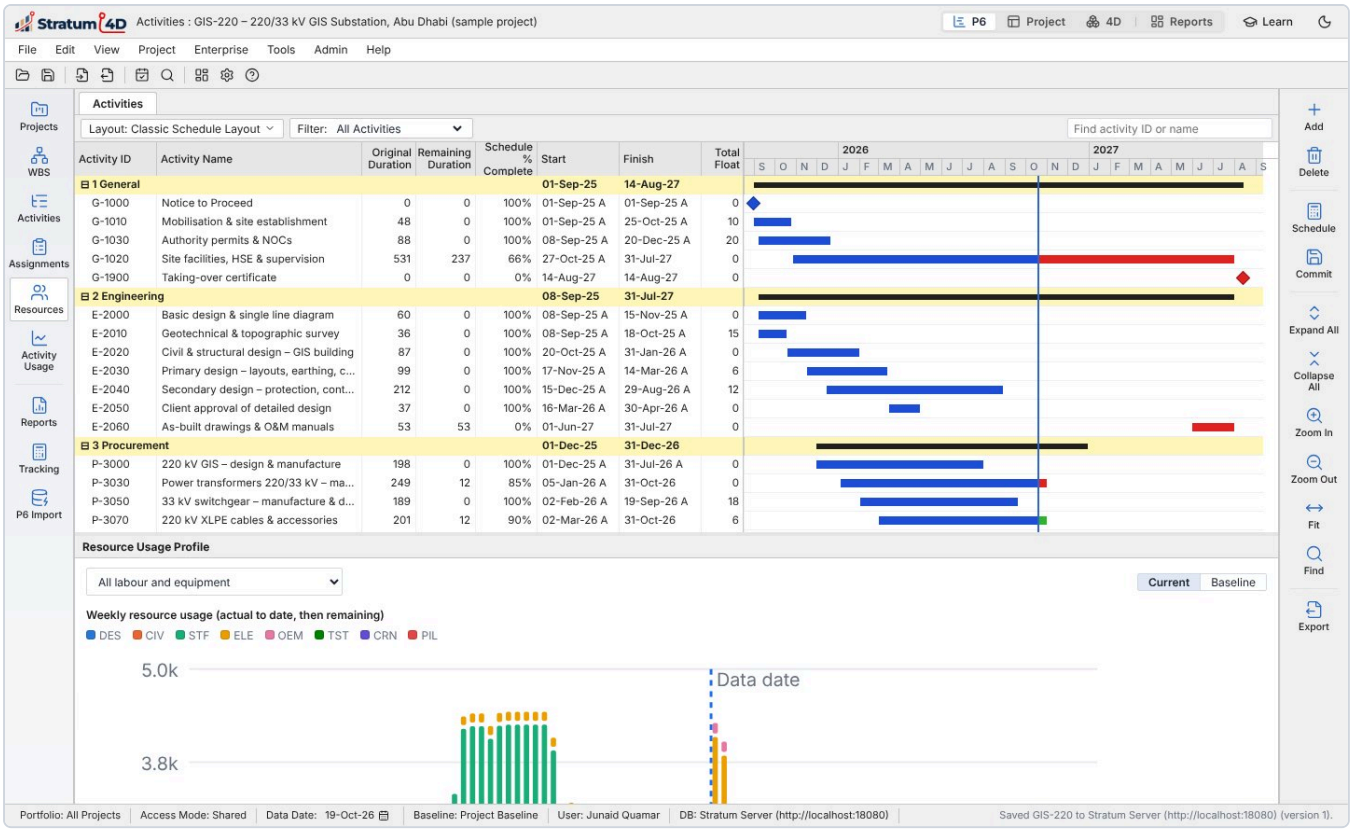


Figure 33. Resource Usage Profile: weekly usage per resource, actual to date then remaining (or the baseline), with a resource table underneath.

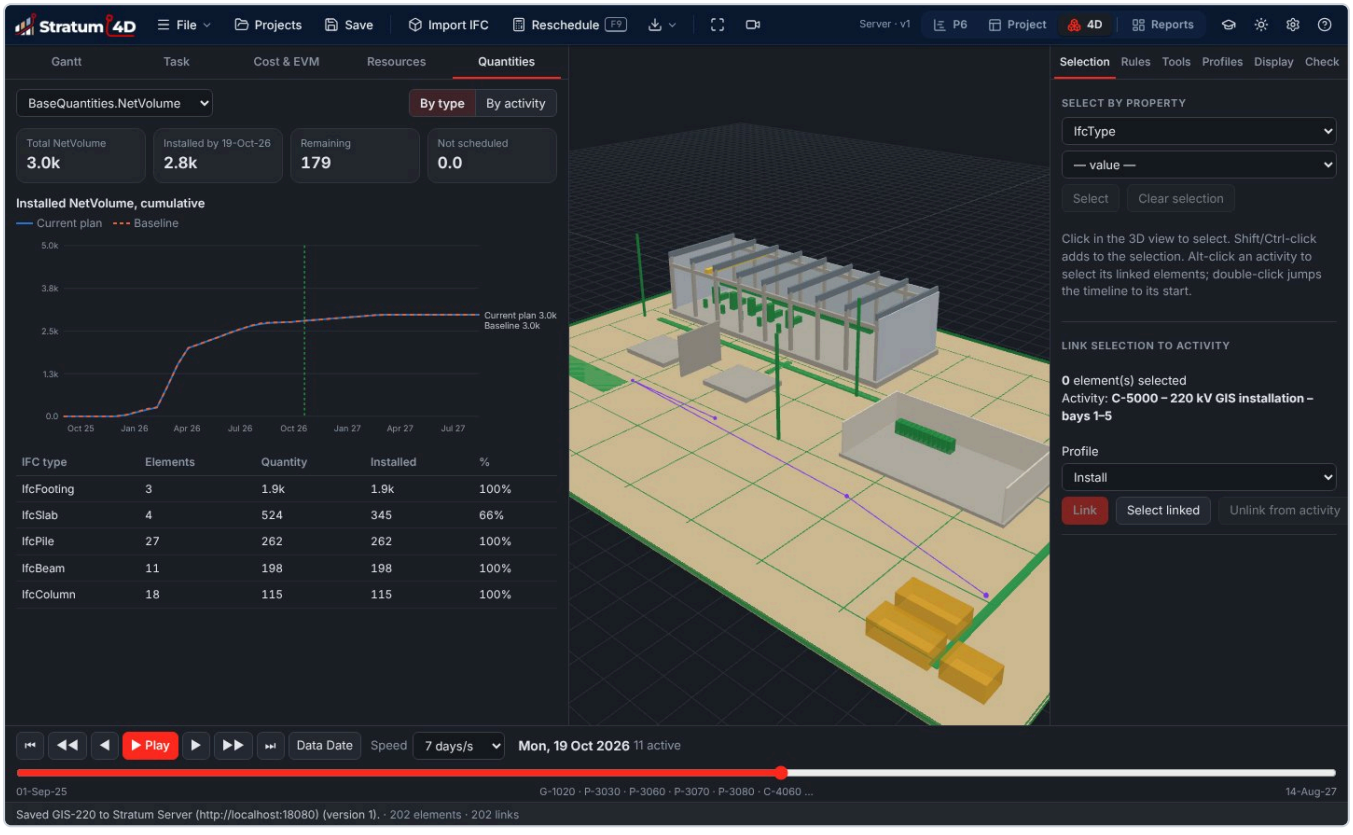


Figure 34. 4D layout → Quantities: any numeric IFC quantity (here NetVolume) installed by the playback date, as tiles and a cumulative curve against the baseline.

10. The project database

Save (*Commit Changes* in the P6 layout, `Ctrl` + `S`) stores the open project in the database: schedule, model links, rules, profiles, groups, pours and paths. Every save gets a version number, shown in the top bar. If someone else saved the project after you opened it, Stratum refuses to overwrite their work and offers *Save as new* instead.

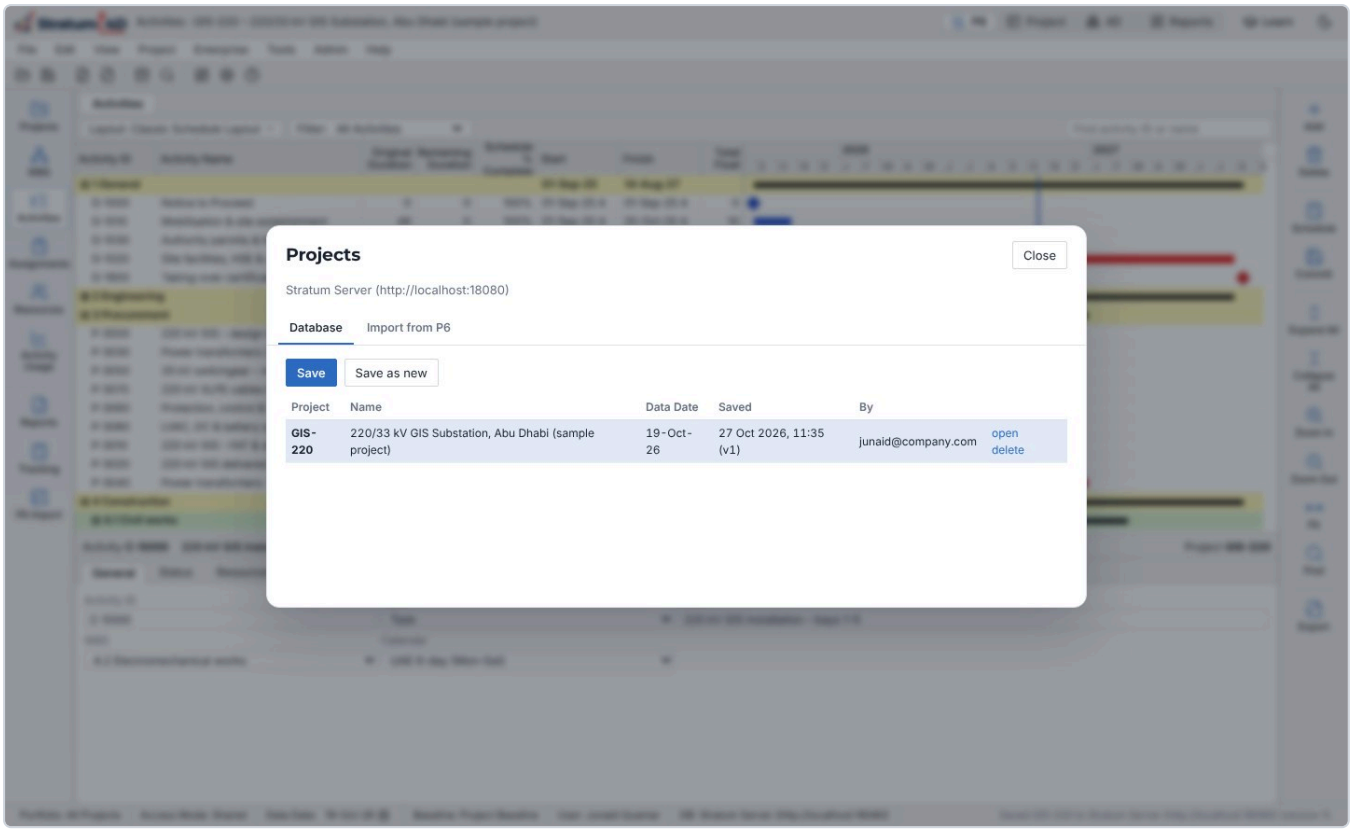


Figure 35. Projects → Database: open, save, save as new or delete. The current project is highlighted with its version.

The model geometry is not stored in the database; open the IFC again after opening a project and the links match by IFC GlobalId. `.stratum.json` project files are still available from the File menu for sharing outside the database.

11. Primavera P6 connections

The desktop app and Stratum Server can talk to P6 directly. The browser cannot, so use XER files there. Connections are set up once in *Settings → Primavera P6*, usually by an administrator.

Connection type	Reads	Writes back	Notes
P6 EPPM (REST API)	Projects, WBS, activities, relationships, calendars, codes, UDFs, resources, assignments	Actual start/finish, % complete, remaining duration	Address normally ends in /p6ws/restapi. Username/password or a bearer token (Oracle-hosted P6).
P6 Professional database (SQL Server, Oracle, SQLite)	The same tables as an XER export	No (read-only by design)	Use a read-only database login; send changes back by XER.

Checking a connection

Test connection checks each link in the chain and stops at the first one that breaks, with a plain-language fix. *Copy report for IT* puts the result on the clipboard.

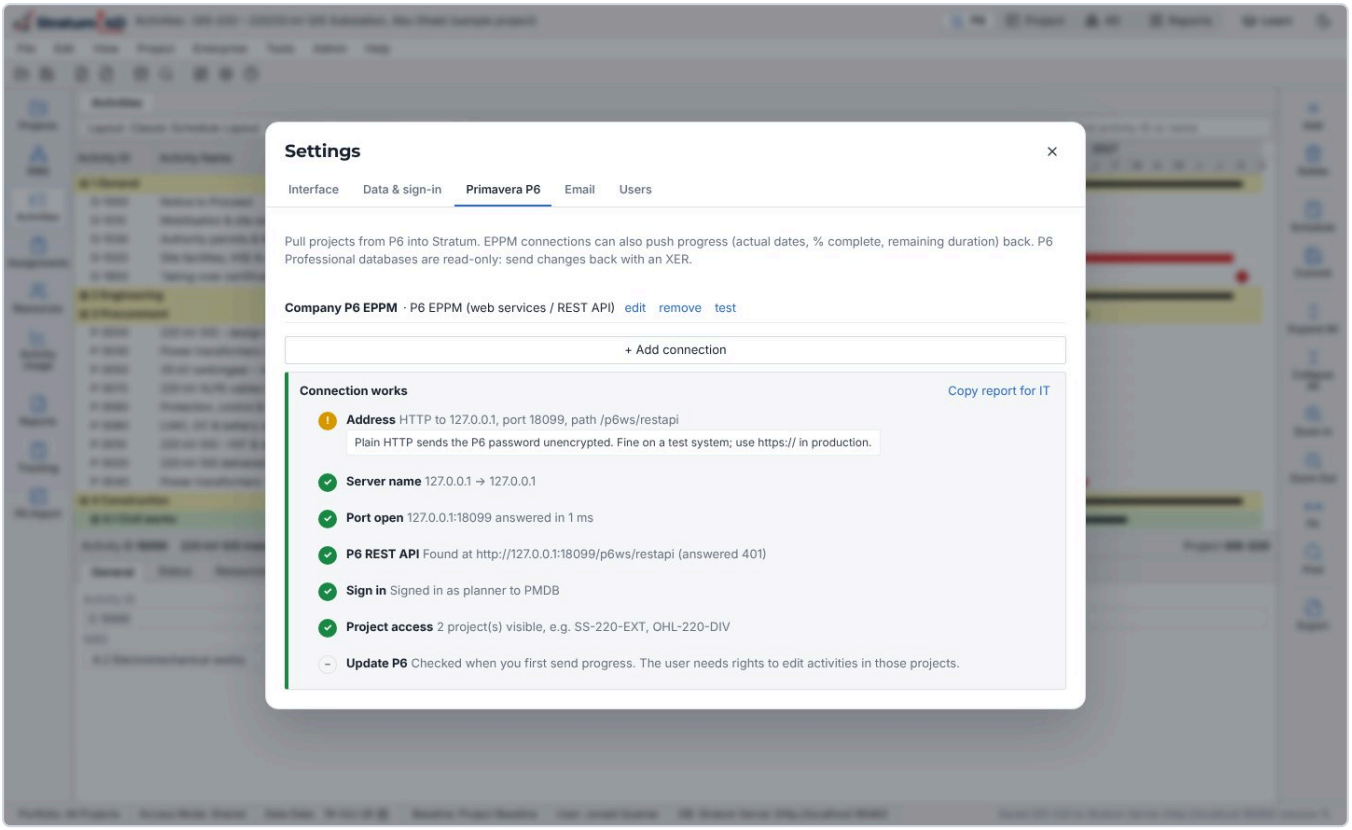


Figure 36. A successful check: address, server name, port, P6 REST API, sign-in and project access.

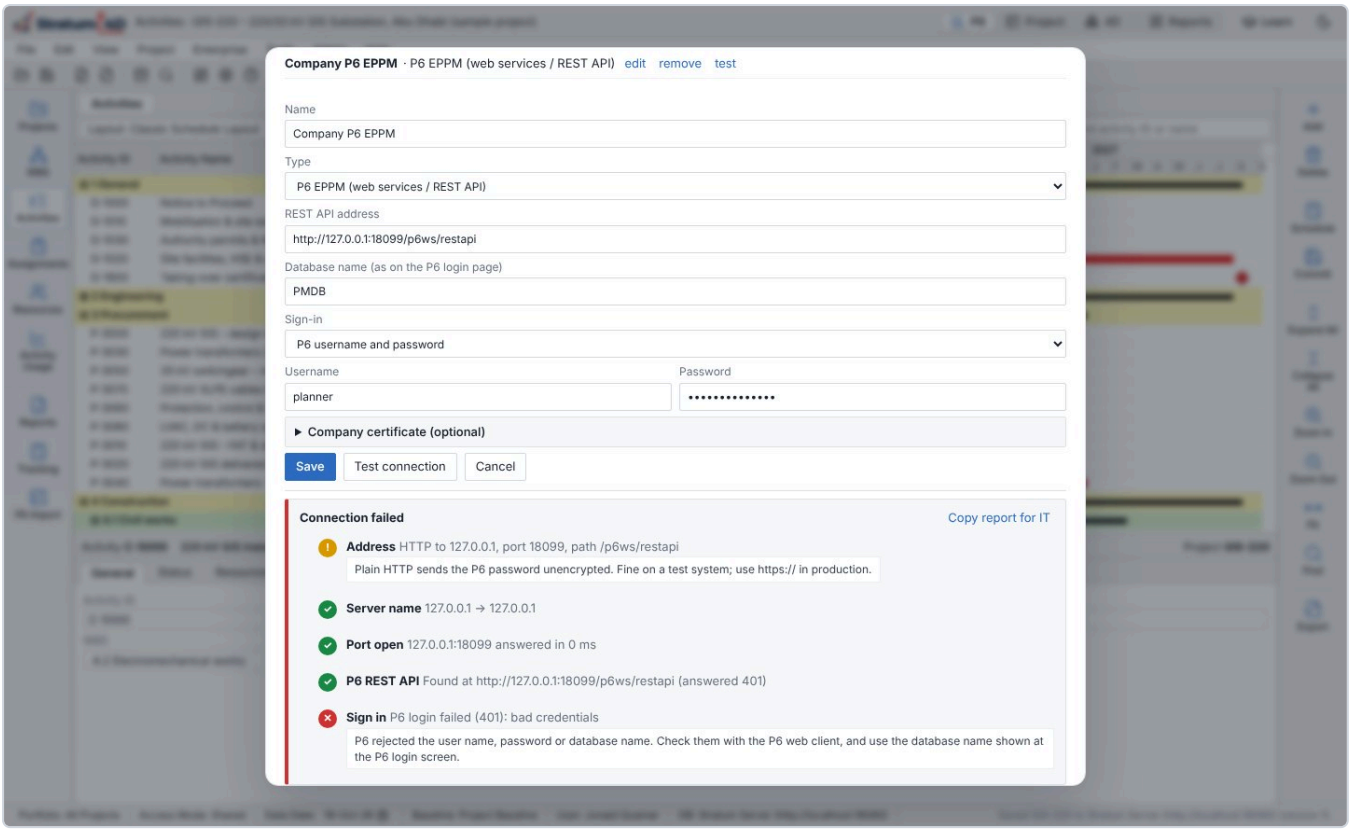


Figure 37. A failed check: sign-in rejected, with what to check.

Step	What it catches
Address	Not a web address; plain http:// (password sent unencrypted); path not ending in /restapi
Server name	Name not found: spelling, or not on the company network / VPN
Port open	Connection refused (P6 stopped, wrong port) or no answer (firewall, VPN)
Certificate	Expired, issued for a different name, or from a company authority this machine does not trust
P6 REST API	Nothing at this path (REST not deployed), or an application-server error
Sign in	Wrong user, password or database name; user without API access; single sign-on
Project access	Signed in but no projects visible: give the user OBS access

Network and certificates

Stratum only makes outgoing HTTPS calls to P6, so nothing needs port forwarding on the Stratum side. The machine running Stratum must reach the P6 server the same way a browser reaches the P6 web client. If Stratum Server runs in the cloud and P6 on company servers, connect them with a VPN or private link rather than exposing P6 to the internet.

- **Desktop app:** P6 calls use the operating system's network stack, so certificates installed by IT in Windows or the macOS keychain are trusted, and the system proxy is used.
- **Stratum Server:** set `NODE_EXTRA_CA_CERTS` to a file holding the company root certificate, and `HTTPS_PROXY` / `NO_PROXY` if a proxy is needed.
- **Either:** paste the root certificate into *Company certificate (optional)* on the connection.

Importing and updating P6

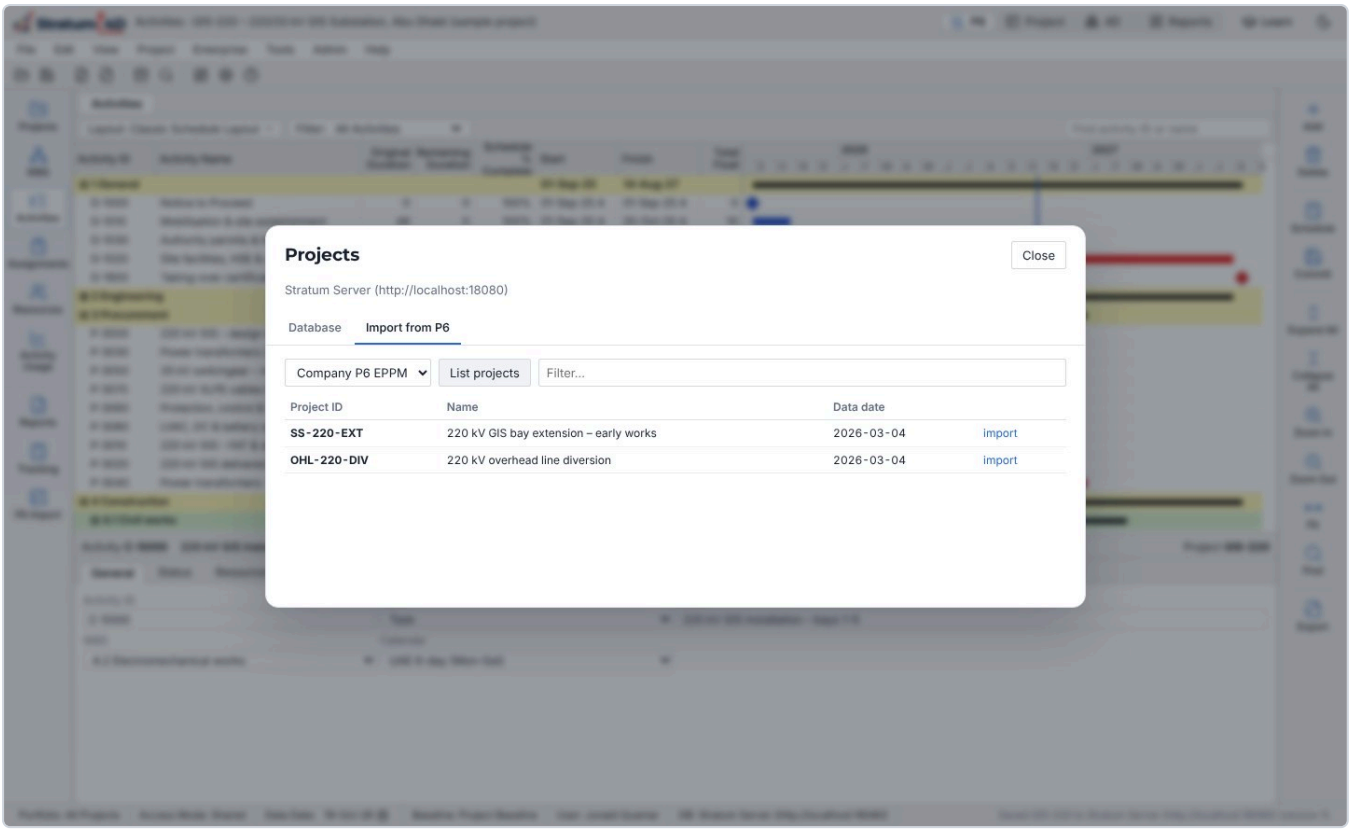


Figure 38. Projects → Import from P6: list the projects on a connection and import one.

An imported project behaves like any other: link it to the model, record progress and report on it. For EPPM projects the top bar shows *Update P6*, which sends actual dates, % complete and remaining duration back for activities that exist in P6. Activities added in Stratum are listed, not pushed. After the update, reschedule in P6 with F9.

12. Dashboards and reports

Reports in the top switch opens the report builder, in the spirit of Power BI. It works the same from the P6, MS Project and 4D layouts. On the left are your dashboards, your own templates and the starting points under *New*. Each dashboard shows the open project or a saved project chosen at the top.

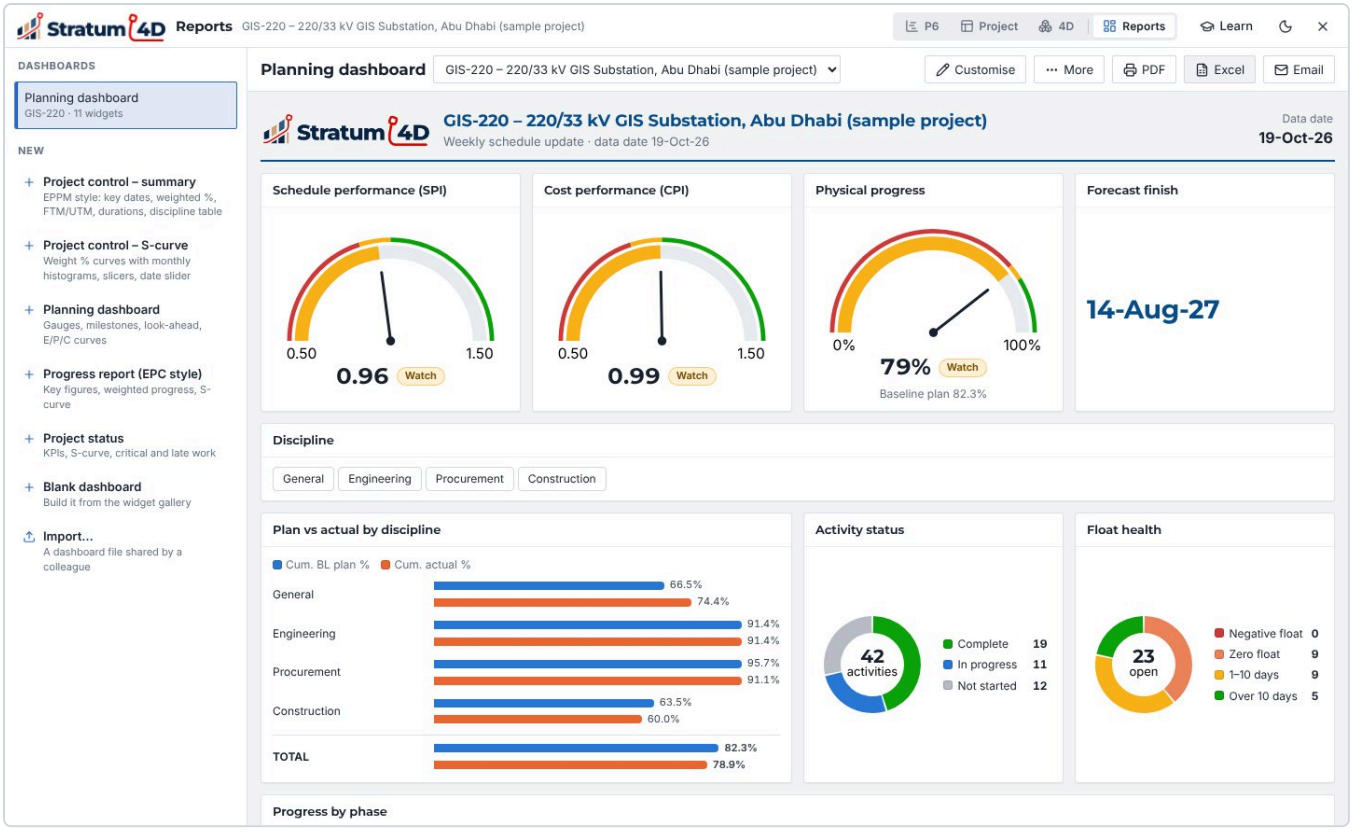


Figure 39. The Planning dashboard template on the example project: SPI and CPI gauges with red, amber and green limits, physical progress against the baseline plan, forecast finish, a Discipline slicer, plan against actual by discipline, activity status and float health.

The Planning dashboard collects the views planners most often post as their weekly P6 dashboards: SPI and CPI gauges, a milestone tracker, plan against actual by discipline, a separate S-curve for each of engineering, procurement and construction, and a look-ahead. In the example, SPI is 0.96 and CPI 0.99, both in the amber band between 0.90 and 1.00.

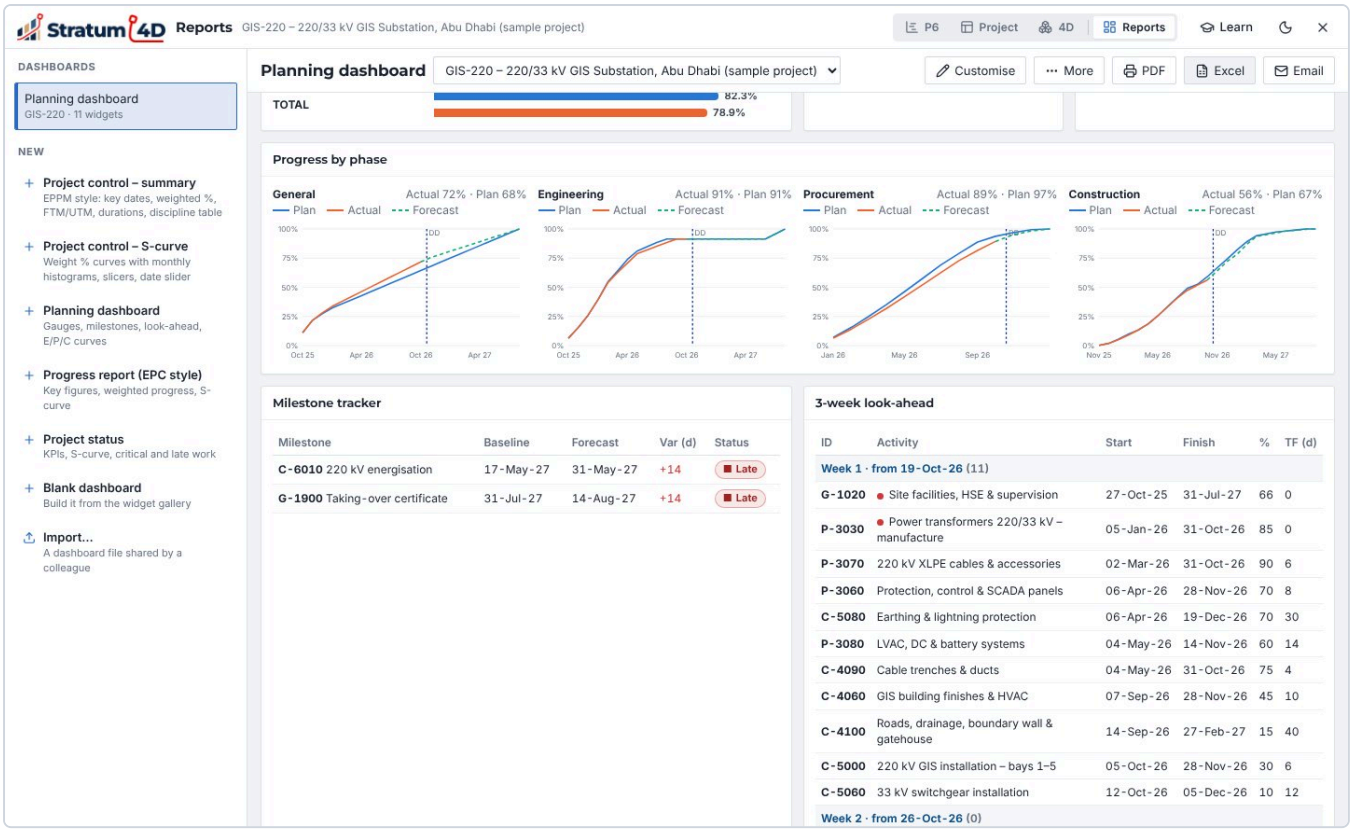


Figure 40. Lower half: one progress S-curve per WBS phase, the milestone tracker (both open milestones forecast 14 days late) and the 3-week look-ahead grouped by week from the data date.

The EPC progress report

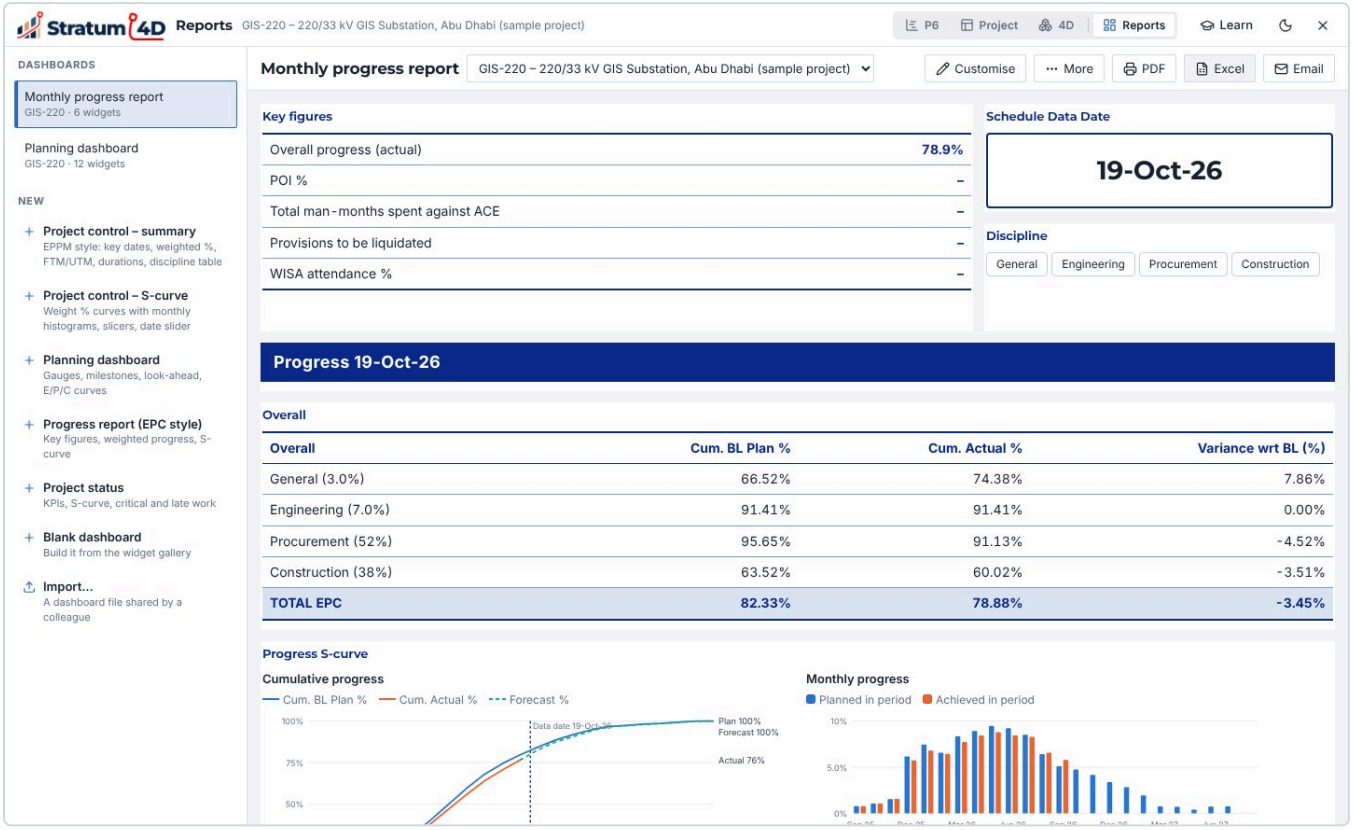


Figure 41. The Progress report (EPC style) template on the example project: key figures, the Schedule Data Date card, the Discipline slicer, the navy Progress banner and the weighted progress table with its TOTAL EPC row.

Progress is weighted by budgeted cost, labour hours, duration or equally, and grouped by WBS level 1 or 2, any activity code or any user field. With the example's cost loading, the weights come out at:

WBS level 1	Weight by cost	Cum. BL plan %	Cum. actual %	Variance
General	3.0%	66.52%	74.38%	7.86%
Engineering	7.0%	91.41%	91.41%	0.00%
Procurement	52.1%	95.65%	91.13%	-4.52%
Construction	37.8%	63.52%	60.02%	-3.51%
TOTAL	100%	**82.33%**	**78.88%**	**-3.45%**

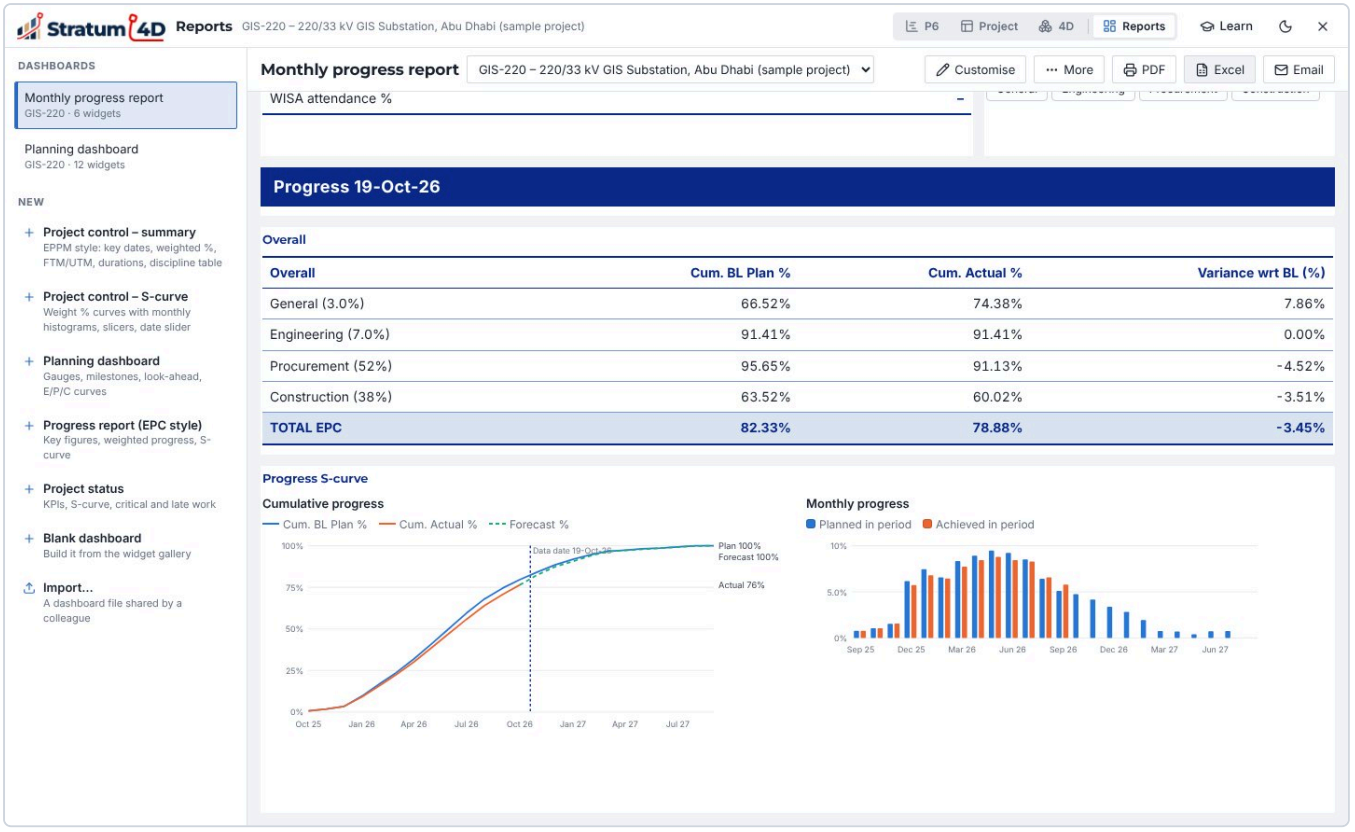


Figure 42. Lower half of the report: cumulative BL plan, actual and forecast S-curve, with monthly planned and achieved progress beside it.

Slicers

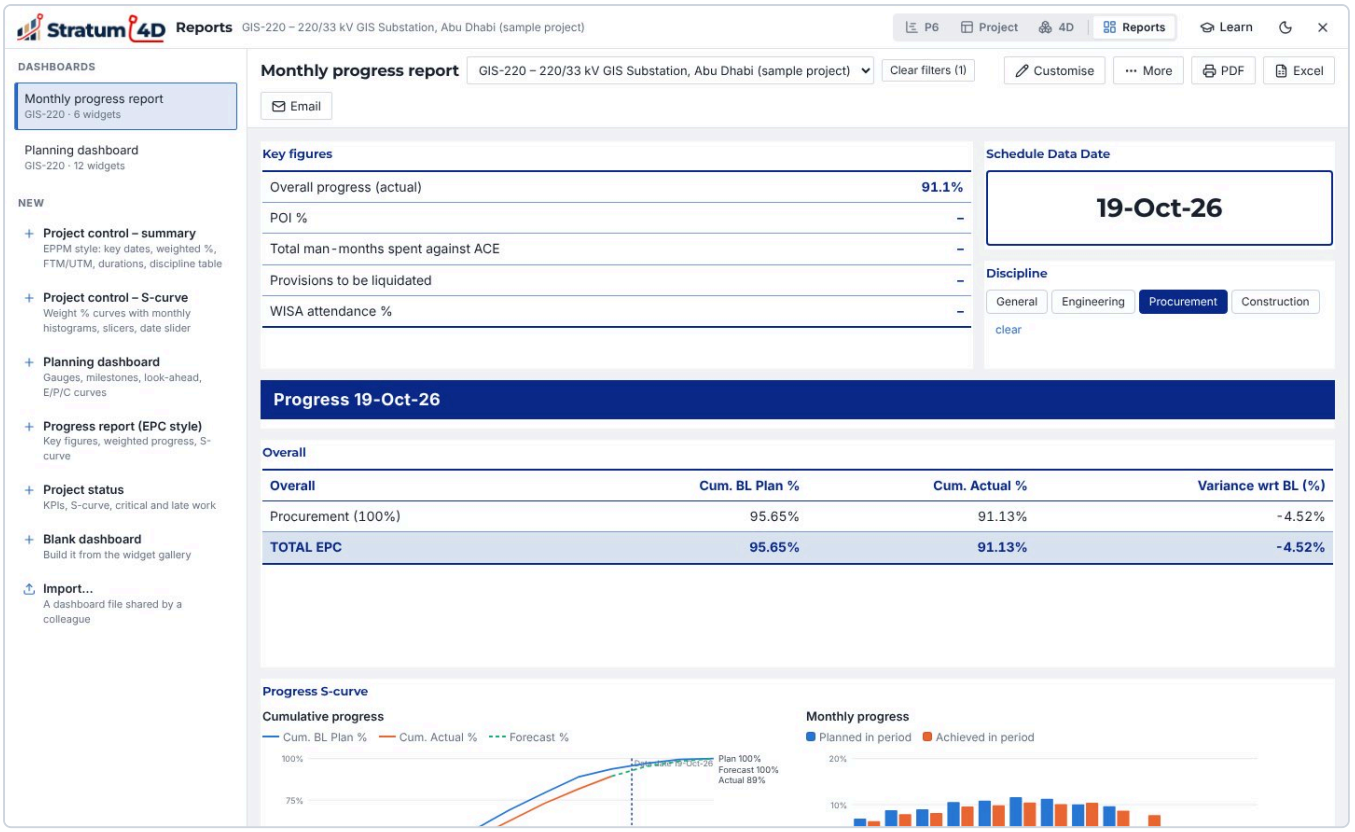


Figure 43. Clicking Procurement in the slicer filters every widget to procurement activities; the TOTAL row recomputes for the selection. Click again, or Clear filters, to remove it.

Customising a dashboard

Click *Customise*. The canvas stays live while you edit, and a panel opens on the right:

- **Move and resize.** Drag a card by its title bar to move it. Drag its bottom-right corner to resize it: widths run from 1/12 to the full page and heights in half-row steps. The size chip on the card shows the result, for example 6×2.
- **Data.** Title, widget type, project, measure, weighting basis, grouping, period and anything else the widget reads. KPI tiles and gauges take decimals, a prefix or suffix (such as `AED `), and the values at which they turn green and red.
- **Look.** Accent colour, card background, title bar (plain, underlined or filled), card style, text size, alignment, line or shaded-area charts, legend on or off, and which series to show.
- **Filter.** Limit one widget to part of the project, for example only Construction, and choose whether it follows the dashboard's slicer buttons.
- **Tables.** Choose columns (ID, name, WBS, status, %, current, baseline and actual dates, variance, total float, original and remaining duration, critical, any activity code), put them in order and set the default sort. Readers can click a heading to sort.

`Ctrl` + `Z` and `Ctrl` + `Y` undo and redo; `Del` removes the selected widget; the copy icon duplicates it.

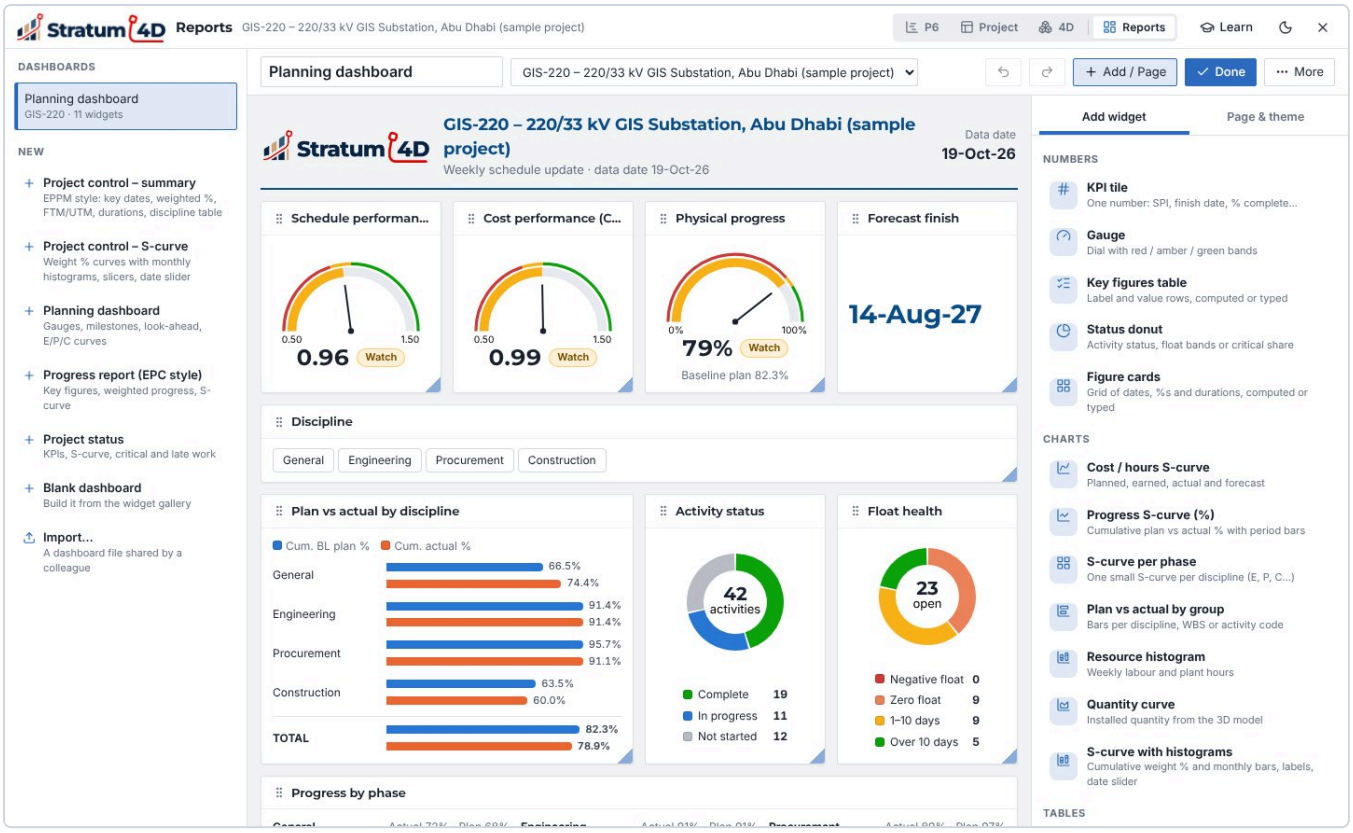


Figure 44. Customise with nothing selected: the widget gallery, grouped into numbers, charts, tables and layout. Click a widget to add it at the end of the page.

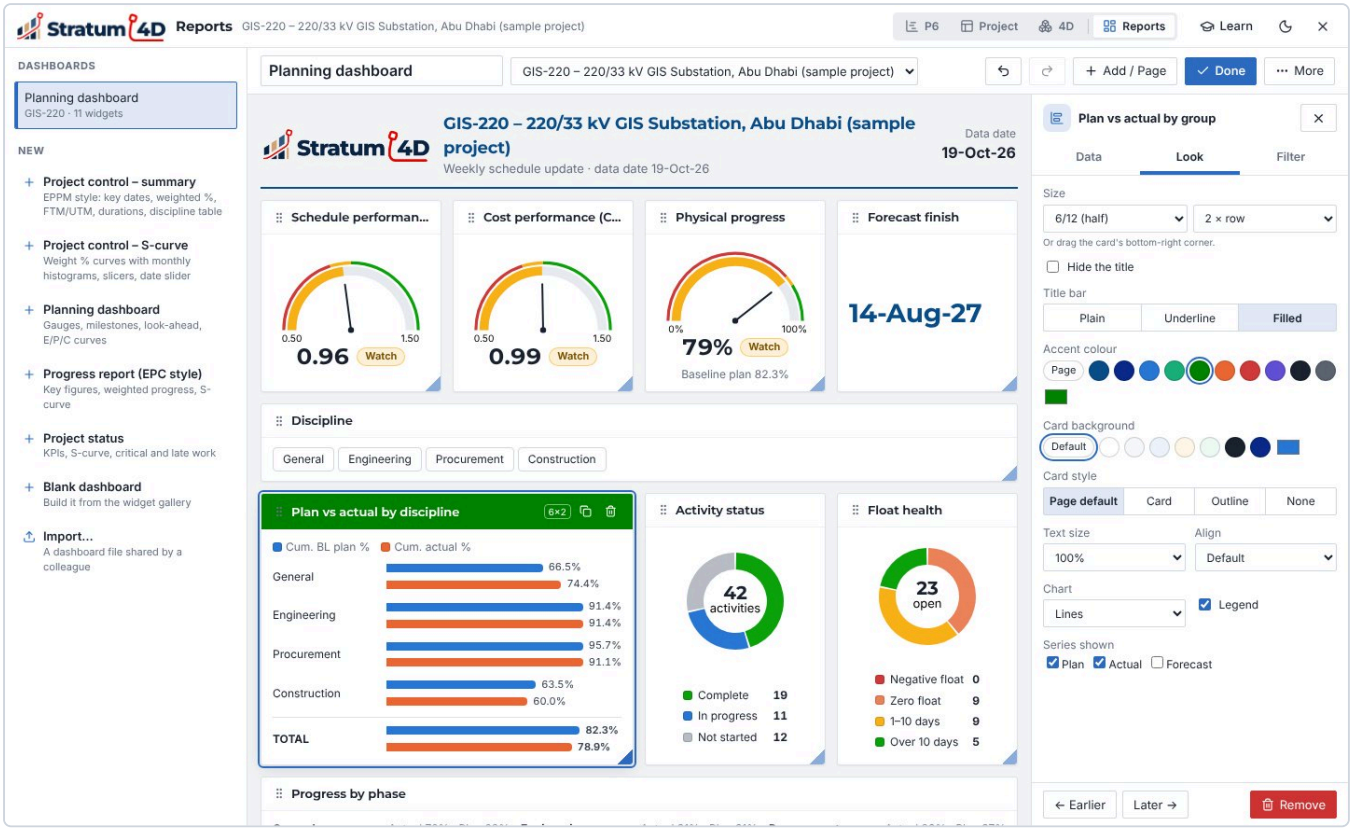


Figure 45. The Look tab for the plan-against-actual widget: a filled title bar in green, with the forecast series switched off.

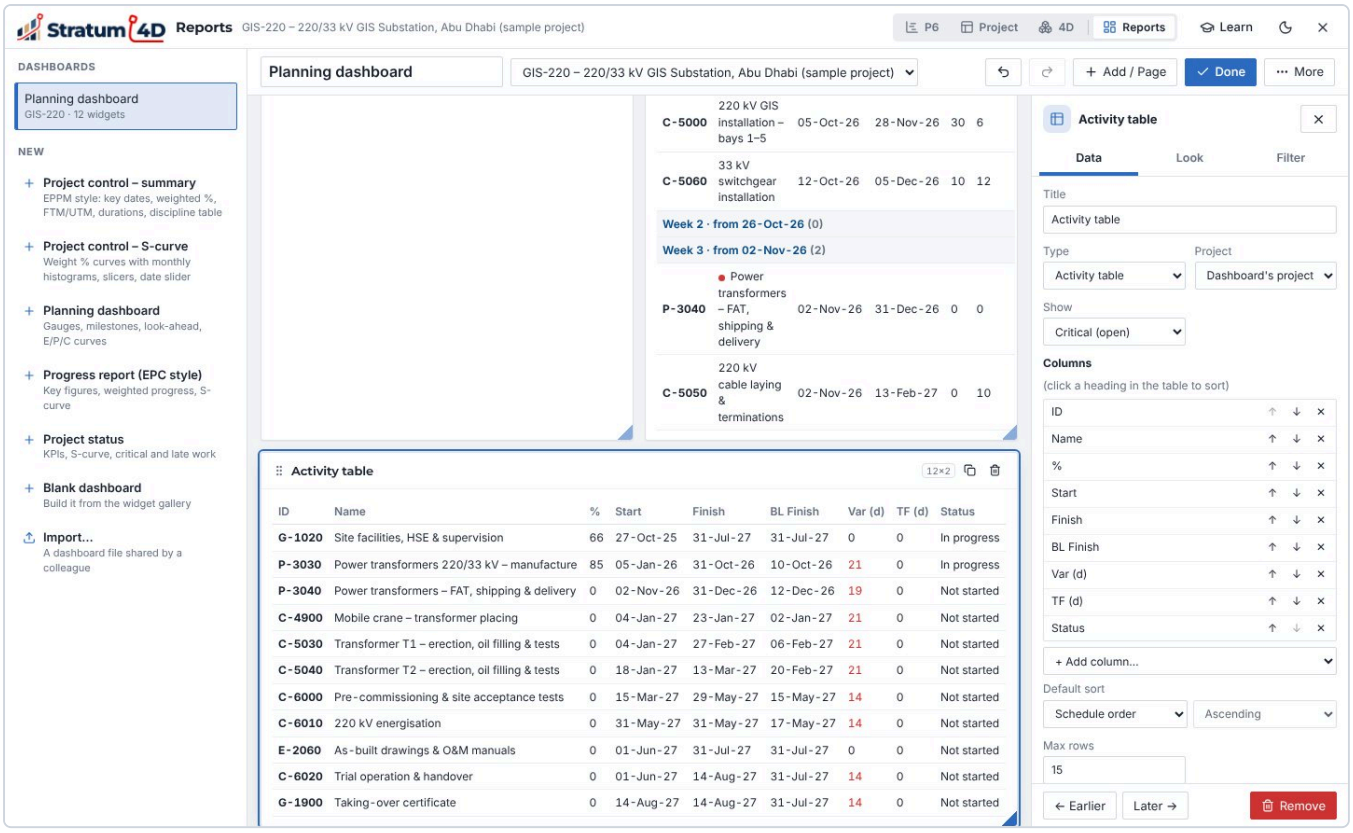


Figure 46. An activity table's Data tab: the column list with move and remove buttons, + Add column, default sort and maximum rows.

Page, theme and branding

With nothing selected, *Page & theme* sets the whole dashboard: theme (Stratum, EPC report or Executive dark), main colour, page background, card style, spacing, a header and subtitle with `{project}` and `{datadate}`, the data date at top right, a company logo, a footer, and A4 landscape or portrait for the PDF. Emailed PDFs use the same settings.

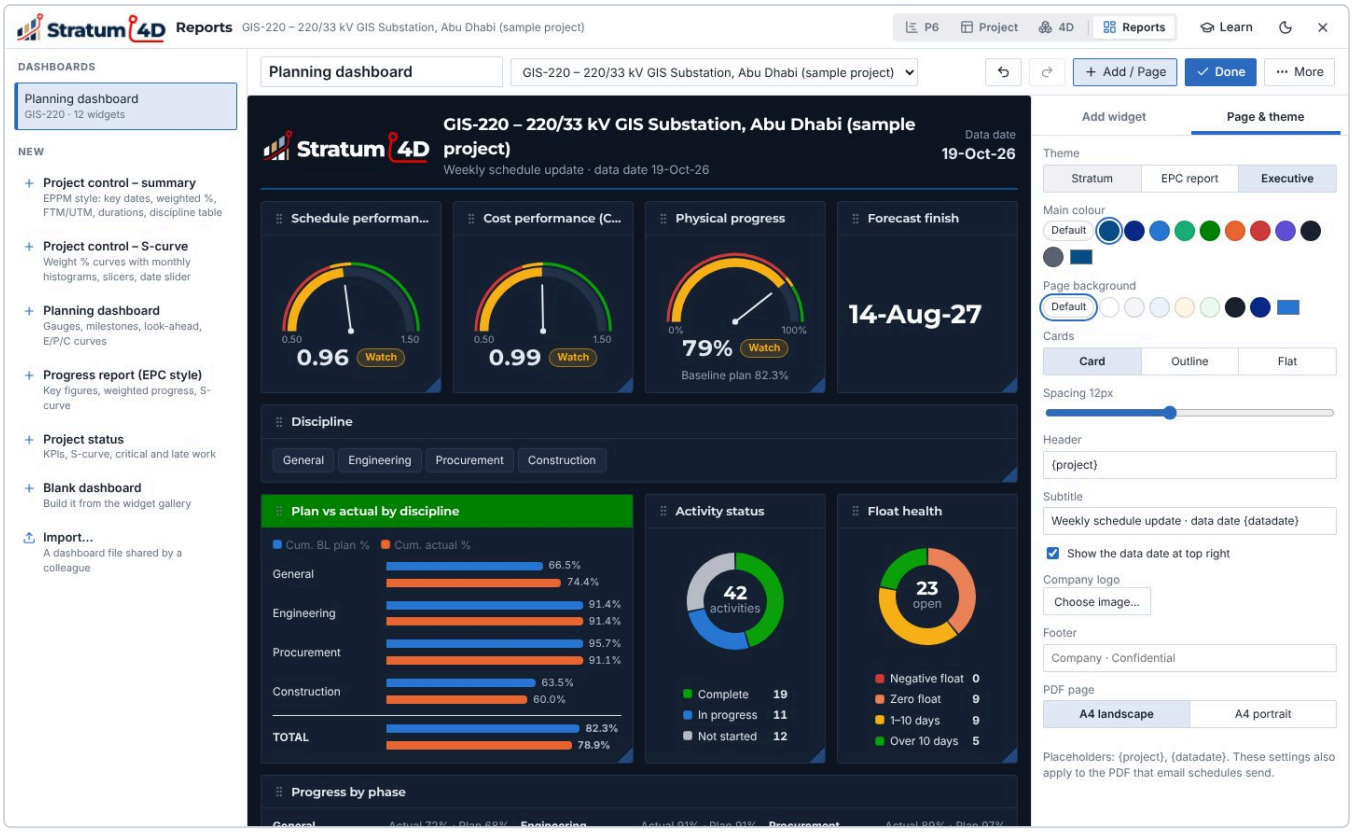


Figure 47. The same dashboard in the Executive theme, with the page settings on the right.

Manual weightage

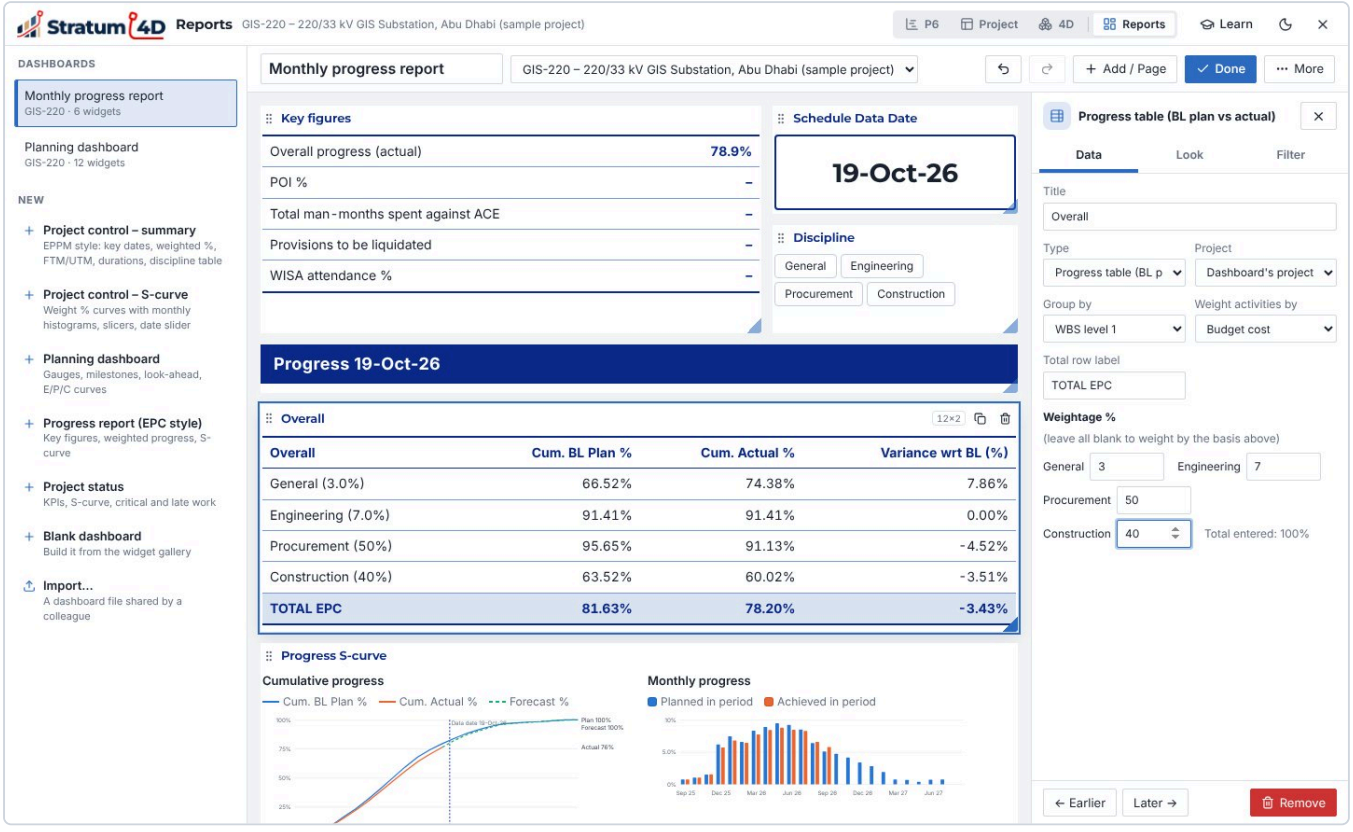


Figure 48. Selecting the progress table shows its weightage in the Data tab; here it is typed in as 3 / 7 / 50 / 40.

Typing a weightage per group overrides the calculated weights, so a report can follow the contract's agreed split. With 3 / 7 / 50 / 40 the example reports a cumulative plan of 81.63%, actual 78.20% and

variance -3.43%.

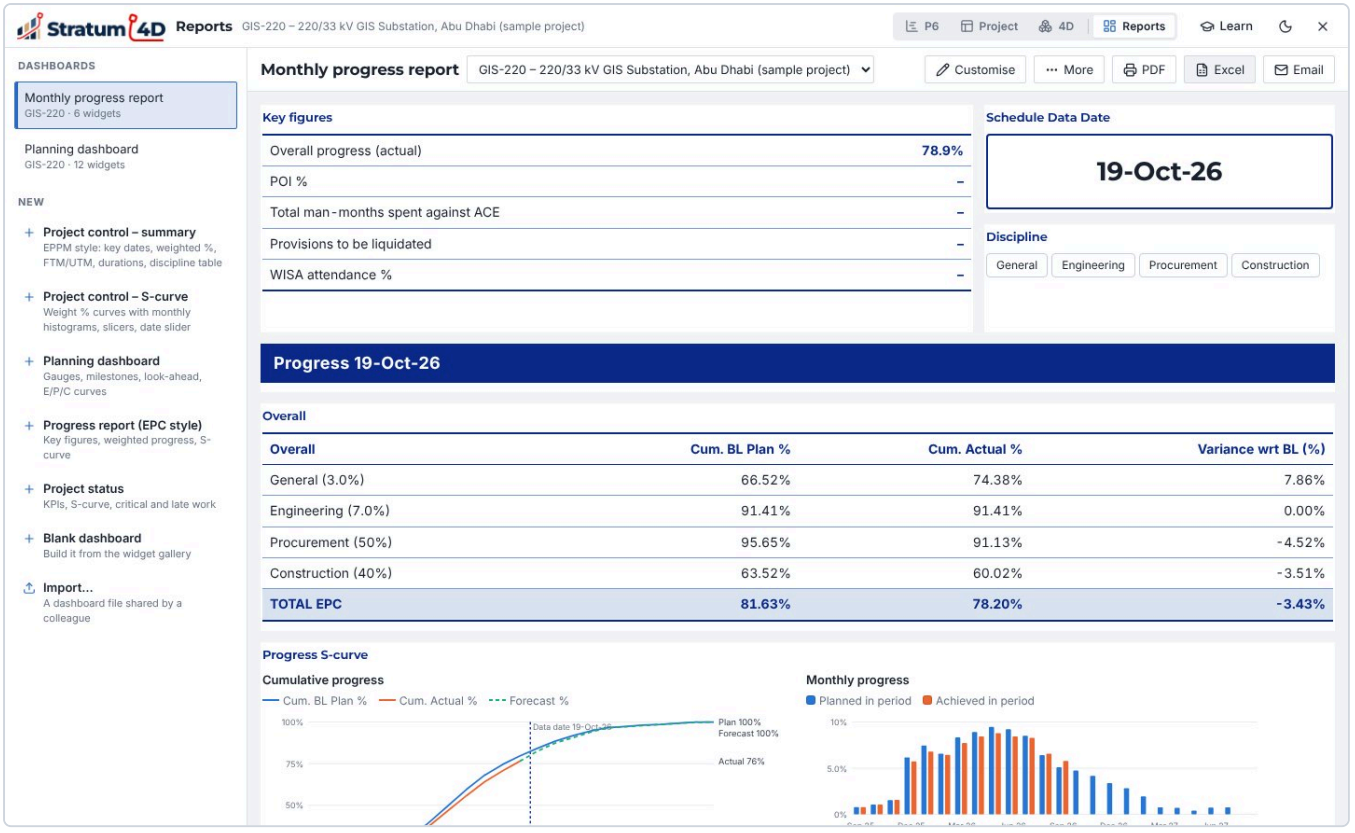


Figure 49. The report with the manual weightage: each row label shows its weight, and TOTAL EPC uses the typed split.

The key-figures table mixes calculated values (such as overall progress) with rows you type yourself, for figures that do not come from the schedule, for example POI %, man-months against ACE, provisions or WISA attendance.

Templates and sharing

More holds *Duplicate dashboard*, *Save as template* (it appears under *My templates* for everyone on the server), *Export dashboard file* (a .json file a colleague can bring in with *Import...*) and *Delete*.

Project status dashboard

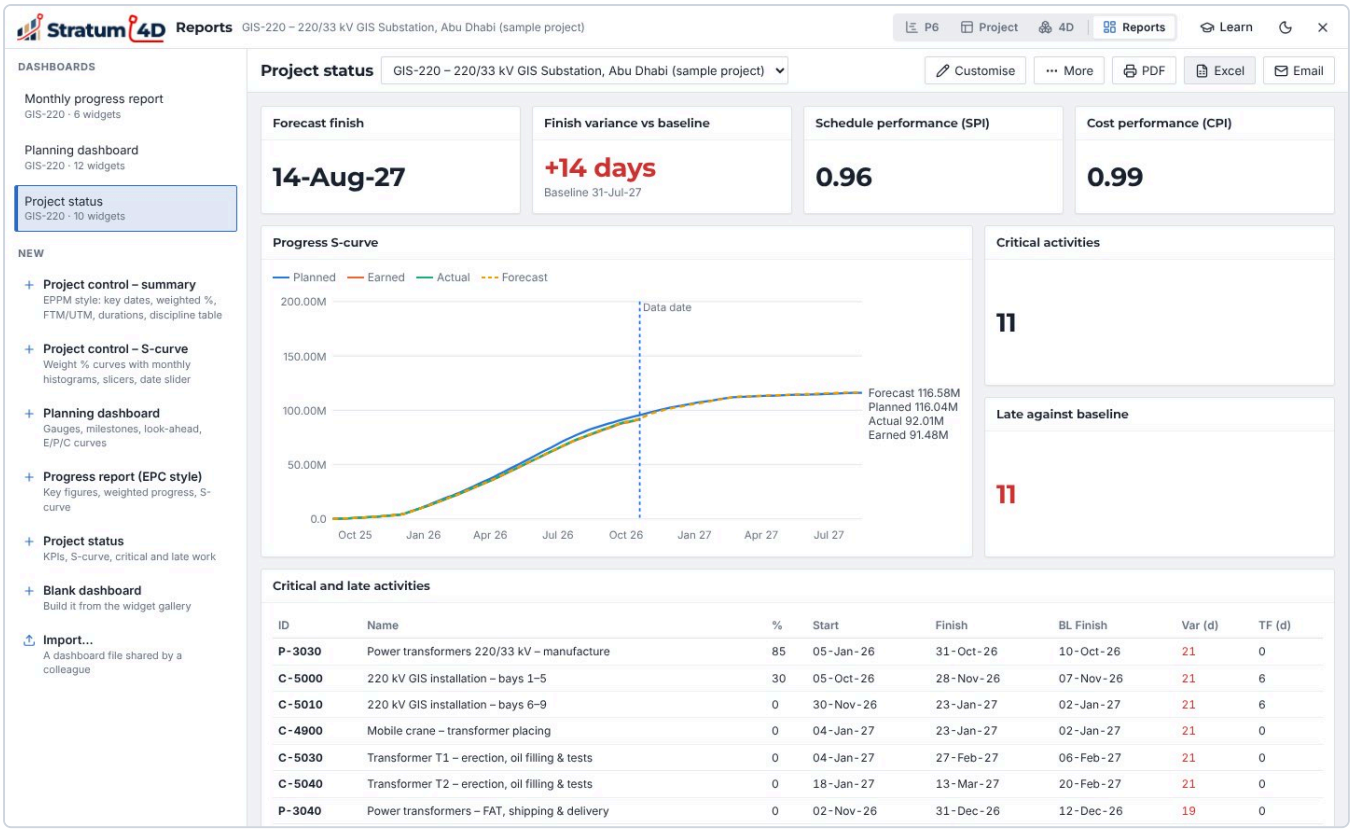


Figure 50. Project status template: forecast finish 14-Aug-27 (+14 days against the 31-Jul-27 baseline), SPI 0.96, CPI 0.99, the cost S-curve, critical and late counts, and the critical and late activities.

Group	Widgets
Numbers	KPI tile, gauge, key-figures table, status donut (progress status, float bands, critical share)
Charts	Cost or hours S-curve, progress S-curve with period bars, S-curve per phase, plan against actual by group, resource histogram, quantity curve
Tables	Activity table, milestone tracker, look-ahead (1–12 weeks), weighted progress table, portfolio of all projects
Layout	Section banner, text note, image or logo, slicer buttons

PDF and Excel

PDF prints the dashboard on A4 in the orientation set under *Page & theme*, with the page header and footer. *Excel* builds a workbook: summary, progress table, progress S-curve, late and critical, three-week look-ahead, all activities, S-curve data, weekly resources and portfolio.

Emailed reports

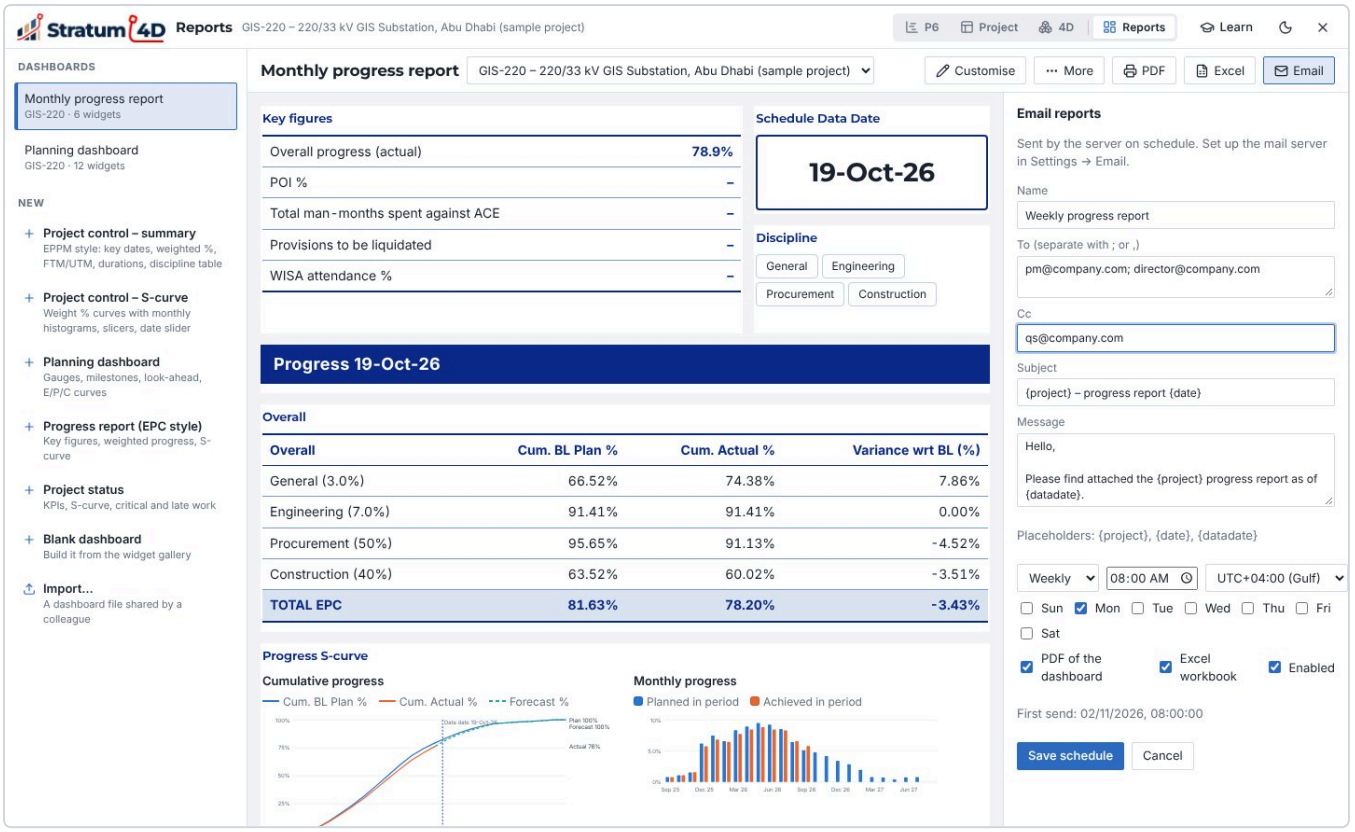


Figure 51. Email panel: a weekly schedule sending the report as PDF and Excel every Monday at 08:00 Gulf time to the project manager and director, copied to the QS.

Schedules run daily, weekly (on chosen days) or monthly (on a day of the month, or the last day). Subjects and messages accept `{project}`, `{date}` and `{datadate}`. Each run is logged with its result, and *Send now* sends immediately. On Stratum Server schedules run around the clock; in the desktop app they run while the app is open.

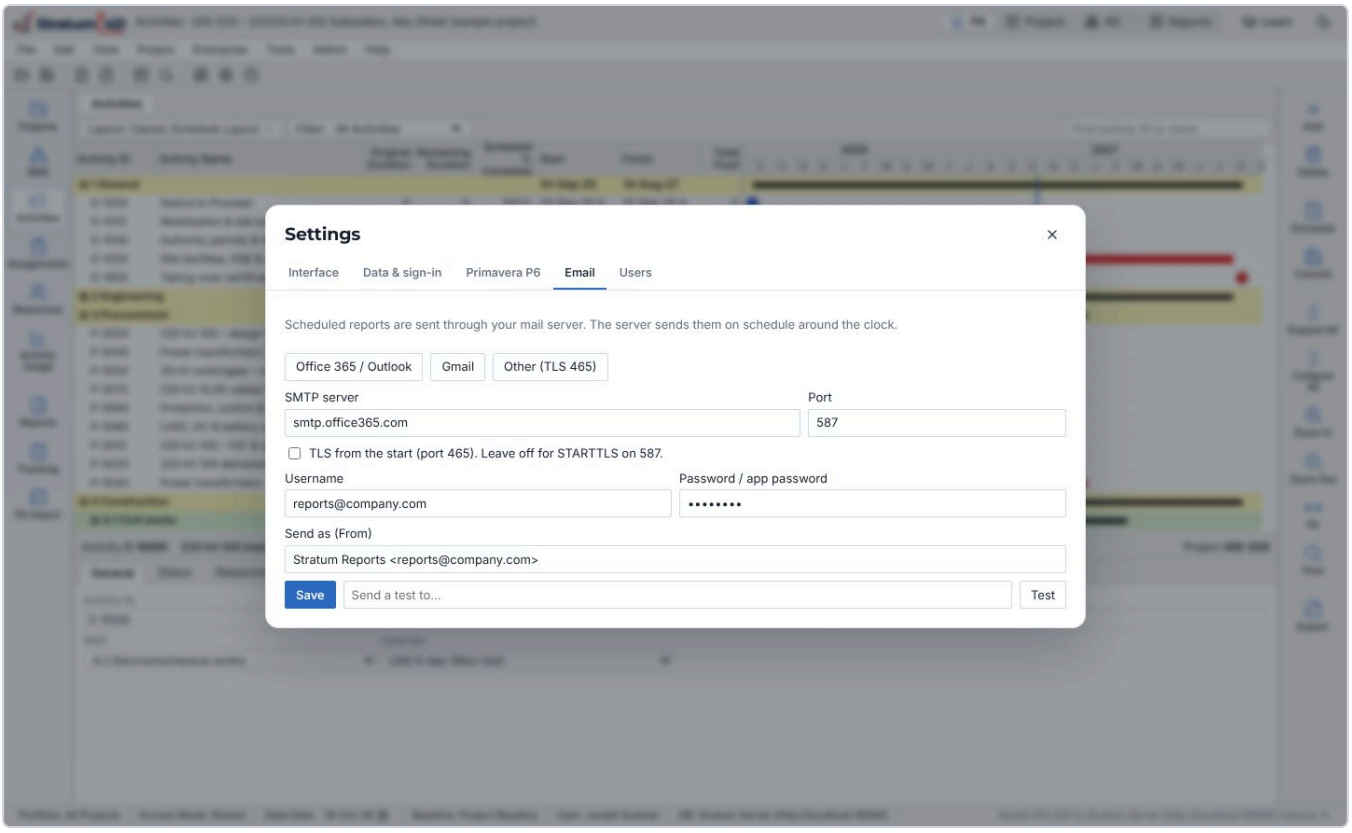


Figure 52. Settings → Email: the SMTP server (presets for Office 365 and Gmail), sender address and a test message.

13. Step-by-step workflows

These are the jobs planners do most often, start to finish. Each step names the screen to use.

Set up a project from a P6 export

1. In P6, export the project to XER (*File → Export → Primavera PM*), including the baseline if you report against one.
2. In Stratum, *File → Import* (P6 layout) or the Import button, and choose the XER. Or use *Projects → Import from P6* to read it straight from a P6 connection.
3. Check the status bar for the number of activities, relationships and resources read, and the data date.
4. Press `F9`. Compare the project finish with P6's. A difference usually means a P6 scheduling option Stratum does not use (see Known limits).
5. Filter *Critical* and confirm the critical path makes sense.
6. *Commit* (`Ctrl` + `S`) to store it in the database for the team.

The weekly update

1. Collect progress from site for the period: actual starts and finishes, % complete and the remaining time for work in hand, and any new or changed scope.
2. Open the project and move the data date: click *Data Date* in the P6 status bar (MS Project layout: *Project → Status Date*) and pick the new date. If P6 is the master schedule, do the update there and re-import instead.
3. Filter *In Progress*, then *Not Started* with starts before the data date. For each activity, enter actual dates, % complete and remaining duration on the Status tab.
4. Press `F9`. Read the result in the status bar: new finish date and its change.
5. Filter *Critical*. Has the critical path changed? Which activities caused the movement?
6. Filter *Late against Baseline* for the slippage list.
7. Open Reports. Check the Planning dashboard: gauges, milestone tracker and look-ahead.
8. Write the narrative: progress this period, the critical path, the main delays and their causes, and recovery actions.
9. *Commit*. Scheduled emails pick up the committed version.

The lesson *The weekly progress update* practises steps 3 and 4 on the example.

Prepare the monthly progress report

1. Open Reports and choose *Progress report (EPC style)*, or your company's template under *My templates*.
2. *Customise*: select the progress table and set the weightage to the contract's split, if it has one.
3. Fill the typed rows of the key-figures table (POI %, man-months against ACE, provisions, WISA attendance).
4. Under *Add / Page → Page & theme*, add the company logo, a footer such as "Confidential", and choose A4 landscape or portrait.
5. *Done*, then *PDF* for the report and *Excel* for the backing data.

6. To send it every month, *Email → New email schedule*, monthly on the last day, PDF and Excel.

Build a 4D model

1. Switch to the 4D layout and open the IFC model (exported from Revit, ArchiCAD or Tekla).
2. On the Rules tab, press *Find matching properties*. If the model carries activity IDs in a property, one rule links everything.
3. Otherwise, select elements in 3D, pick the activity in the Gantt, choose a profile and press *Link*. Save repeated selections as groups.
4. Use the Check tab: link coverage, activities with nothing to show, elements never shown.
5. Play the timeline. Look for sequence problems and space clashes; record a video for the review.
6. Commit. The model itself is not stored; reopen the IFC next time and the links match by GlobalId.

Send progress back to P6 (EPPM)

1. Import the project from the EPPM connection.
2. Record progress in Stratum and press F9 to check it.
3. Press *Update P6*. Stratum sends actual dates, % complete and remaining duration for activities that exist in P6, and lists any it did not send.
4. In P6, run *Schedule (F9)* so P6 recalculates with its own options.

Set up a team server

1. Install Docker on a company server and run the Compose file from the README; it starts Stratum Server with PostgreSQL.
2. Open the server address in a browser and create the first administrator.
3. Settings → Users: add planners and viewers. Settings → Primavera P6: add the connection and press Test. Settings → Email: the SMTP server, and send a test.
4. On each planner's computer, choose *Stratum Server* in the first-start setup (or Settings → Data) and enter the address.

14. Schedule quality checklist

Before a schedule is used for reporting or submitted to a client, check its quality. The widely used DCMA 14-point assessment gives a common set of checks. The thresholds below are the usual DCMA ones; your contract may set others.

#	Check	Usual threshold	How to look in Stratum
1	Missing logic: activities without a predecessor or successor	≤ 5% of activities	F9 lists open ends in its warnings; the lesson Adding an activity and its logic shows one
2	Leads (negative lags)	0	Predecessors / Successors tabs: lag column
3	Lags	≤ 5% of relationships	Predecessors / Successors tabs
4	Relationship types: share of FS	≥ 90% FS	Predecessors tabs; XER export to check in bulk
5	Hard constraints	≤ 5% of activities	Status tab: Constraint
6	High float (over 44 working days)	≤ 5% of activities	Activity table sorted by TF on a dashboard
7	Negative float	0	Filter, or a donut with Float health
8	High duration (over 44 working days)	≤ 5% of activities	Activity table with Original duration, sorted
9	Invalid dates: actuals after the data date, forecasts before it	0	Status tab; F9 warnings
10	Resources: activities without resources or cost	Project-specific	Resources tab; Activity Usage Profile
11	Missed tasks: finished later than baseline	≤ 5% of completed	Filter Late against Baseline
12	Critical path test: delay one critical activity and check the finish moves by the same amount	Pass	Add duration to a critical activity, F9, then undo by reloading
13	CPLI (critical path length index)	≥ 0.95	Calculated outside Stratum for now
14	BEI (baseline execution index)	≥ 0.95	Count completed against baseline due, from the Late filter

Beyond the numbers, read the critical path as a story: does it run through the work you would expect to drive the finish? In a substation that is usually long-lead equipment (transformers, GIS) and then testing and energisation, as in the example.

15. Stratum Server for teams

Stratum Server is the shared version: one PostgreSQL database, sign-in, roles, P6 connections, scheduled emails and server-side PDF rendering. It ships as a Docker image with a Compose file that adds PostgreSQL 16.

```
cp .env.example .env      # set STRATUM_SECRET and POSTGRES_PASSWORD
docker compose up -d      # PostgreSQL 16 + Stratum Server on port 8080
```

Open `http://<server>:8080` and create the first administrator. In production, put the server behind your usual HTTPS front end (IIS, nginx or a load balancer).

Setting	Purpose
`STRATUM_SECRET`	Signs sign-in sessions and encrypts stored SMTP and P6 passwords. Keep it secret and do not change it later.
`DATABASE_URL`	PostgreSQL connection. Without it the server uses a SQLite file (`STRATUM_DB_FILE`).
`PORT` , `PUBLIC_URL`	Listening port (default 8080) and the address users see.
`NODE_EXTRA_CA_CERTS`	Company root certificate for reaching P6 over HTTPS.
`HTTPS_PROXY` , `NO_PROXY`	Outbound proxy for reaching P6.

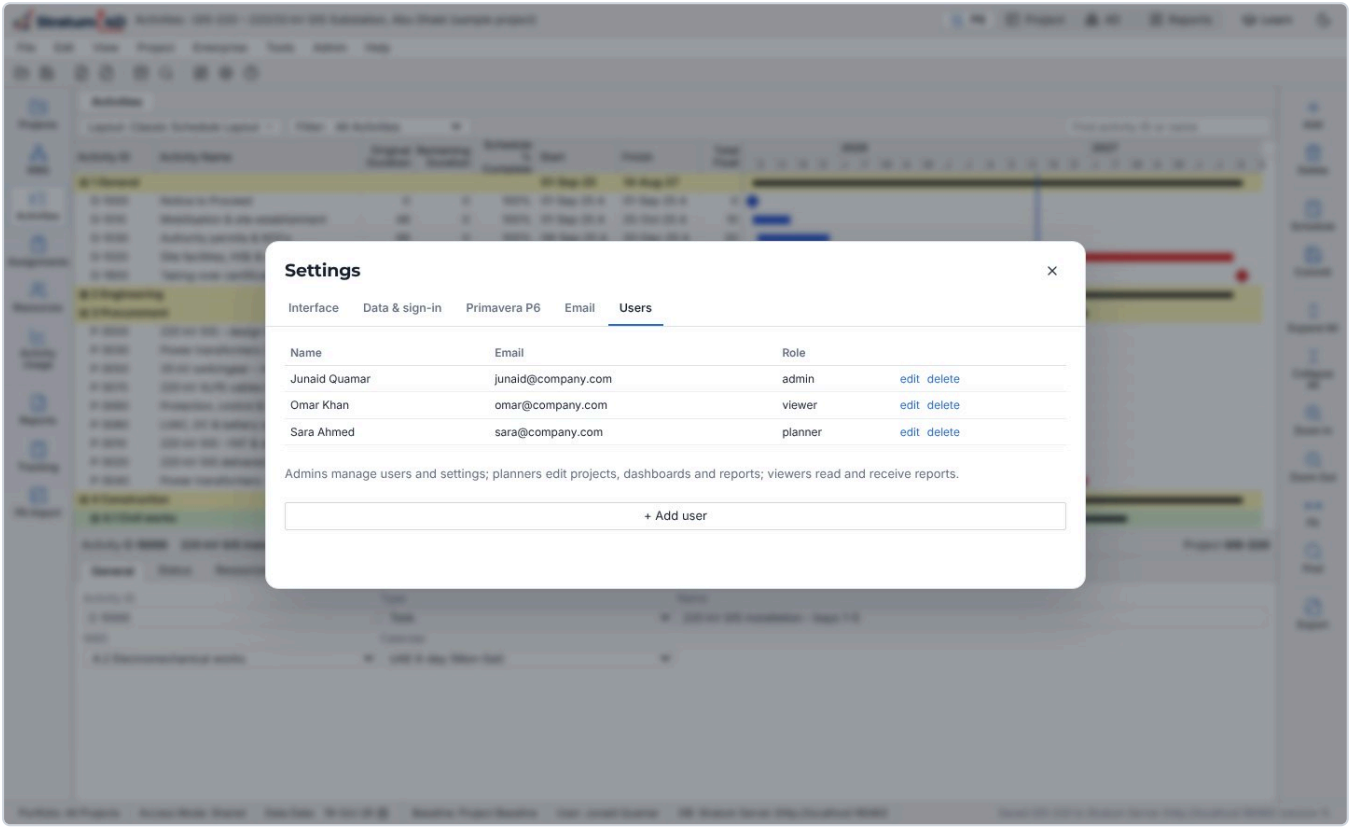


Figure 53. Settings → Users (administrators only): add users and give them a role.

Role	Can
Admin	Everything, including users, SMTP and P6 connection settings

Role	Can
Planner	Create and edit projects, dashboards and email schedules; import from P6
Viewer	Open projects and dashboards, and receive reports

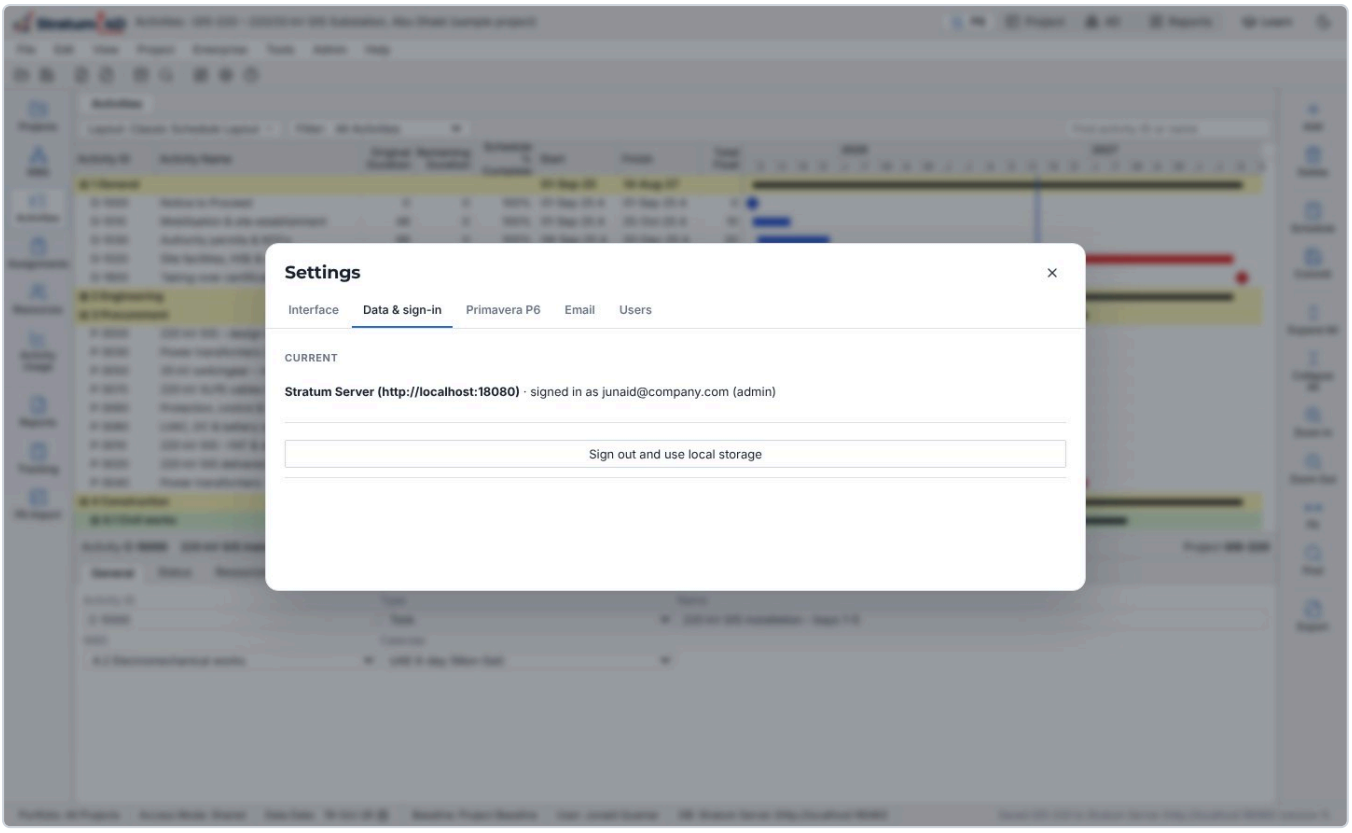


Figure 54. Settings → Data & sign-in: which database this client uses, and switching between local storage and a server.

16. Keyboard shortcuts

Keys	Action
`Space`	Play / pause
`←` `→`	Step one day
`Shift` + `←` `→`	Step one week
`Home` / `End`	Start / finish of the schedule
`F9`	Schedule (CPM)
`Ctrl` + `S`	Save (commit) to the database
`Ins` / `Del`	Add an activity after the selected one / delete the selected activity (P6 and MS Project layouts)
`Ctrl` + `F`	Find activities (P6 and MS Project layouts)
`Ctrl` + `Alt` + `1` / `2` / `3`	P6 / MS Project / 4D layout
`Ctrl` + `Alt` + `4`	Reports (dashboards)
`Ctrl` + `Z` / `Ctrl` + `Y`	Undo / redo while customising a dashboard
`Del` (dashboard)	Remove the selected widget while customising
Click, `Shift`-click in 3D	Select, add to selection
`Alt`-click an activity	Select its linked elements
Double-click an activity	Move the timeline to its start
`Esc`	Cancel path drawing, close dialogs
`?`	Getting-started help

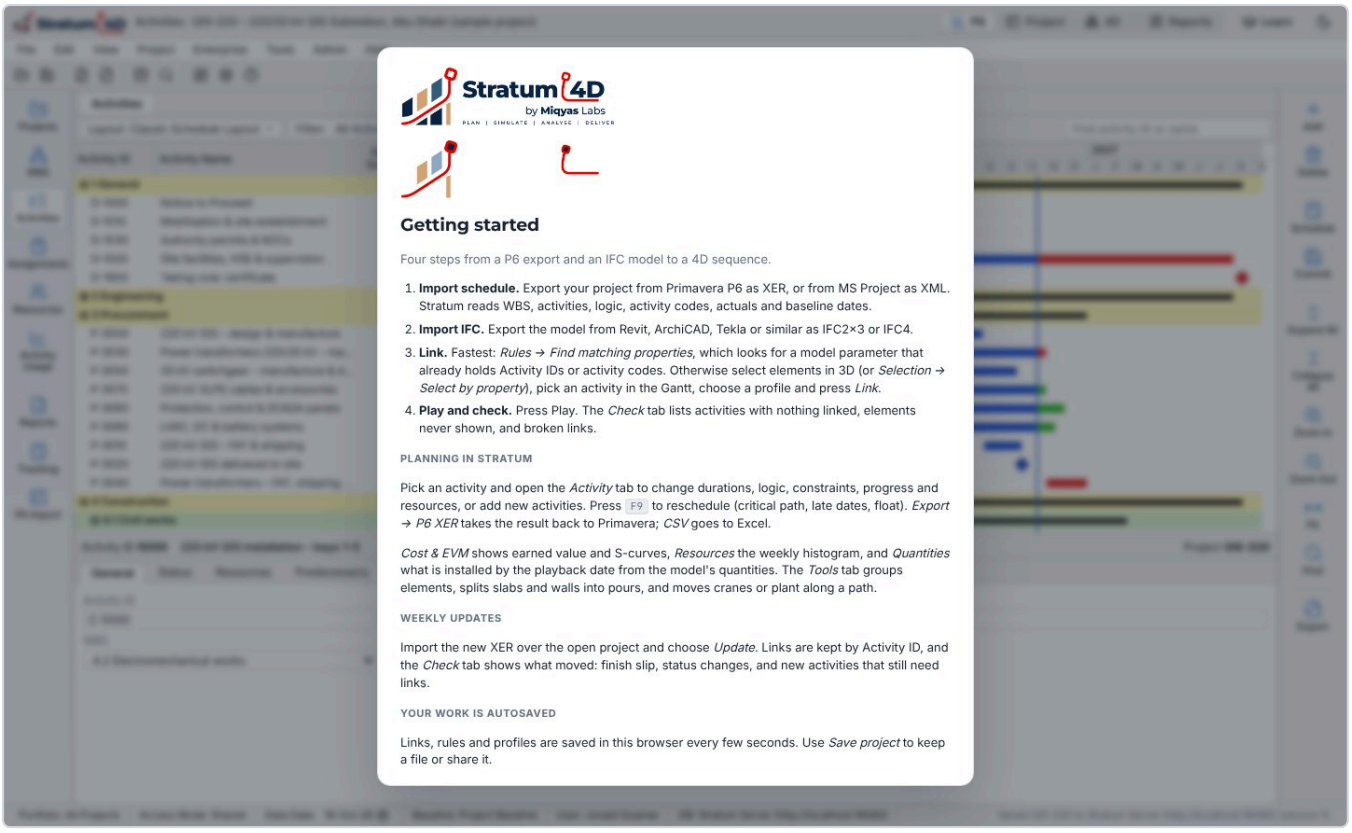


Figure 55. The getting-started help (?): four steps from a P6 export and an IFC model to a 4D sequence, plus shortcuts.

17. Troubleshooting

Symptom	Likely cause	What to do
P6 test stops at *Server name*	Not on the company network, or a typo	Connect to the VPN; check the address
P6 test stops at *Port open* with no answer	Firewall between this machine and P6	Ask IT to allow this computer (or the Stratum Server) to reach the P6 port
P6 test stops at *Certificate*	Company certificate authority not trusted	Install the root certificate in Windows/macOS, paste it into the connection, or set NODE_EXTRA_CA_CERTS on the server
P6 test stops at *P6 REST API* (404)	REST API not deployed at that path	Check the address ends in /p6ws/restapi; ask the P6 admin
Sign-in 401	Wrong user, password or database name	Check in the P6 web client; use the database name from the P6 login page
Sign-in 403	User has no API access	P6 Admin → Users → module access for web services / REST
No projects visible	No OBS access	Give the user project access in P6
“Not saved: someone else saved this project”	Another user saved a newer version	Use Save as new, or reopen the project and redo the change
Links lost after opening a project	The model is not stored in the database	Open the IFC again; links match by GlobalId
Emails not sent	SMTP settings, or the desktop app was closed	Send a test from Settings → Email; use Stratum Server for round-the-clock schedules
Windows protected your PC	Installer not code-signed yet	More info → Run anyway
Training card says a step is not done	The change was made somewhere the lesson does not watch (e.g. a different activity)	Press Show me, or Skip; Back returns to the step before
Training numbers differ from the guide	The sample was changed in an earlier lesson	Open the sample again from the start screen
A dashboard widget says “No project selected”	The dashboard shows the open project and none is open	Open a project, or choose a saved project in the bar above the dashboard
Slicer seems stuck on one discipline	Slicer choices are saved with the dashboard	Clear filters in the dashboard bar
PDF prints in the wrong orientation	Page set-up	Customise → Add / Page → Page & theme → PDF page

18. Known limits

- The CPM engine follows P6's default options. Progress override, resource levelling and multiple float paths are not implemented yet.
- Costs and quantities are spread linearly over each activity, so weekly figures can differ slightly from P6's calendar-weighted spread.
- XER export has been checked by re-importing into Stratum, not yet against a live P6 database. Use P6's import preview the first time.
- The P6 EPPM connector has been tested against a mock of the documented REST API; the P6 database reader against SQL Server 2022 and SQLite copies of the P6 schema. Oracle has not been run against a live instance yet.
- Very large IFC files load slowly; GPU instancing and level of detail are planned.
- Undo/redo, a column chooser and group and sort in the schedule views, and in-app baseline management, are on the roadmap (dashboards already have undo, redo and column choice). The P6 and MS Project layouts reproduce the arrangement of the originals; their menus contain only commands Stratum supports.
- Video export is WebM only.

Appendix A · Example project activities

All 46 activities of the example as imported (data date 19-Oct-26). OD = original duration in working days; TF = total float in days; ● = critical after F9. Baseline finish is shown only where it differs.

ID	Activity	WBS	OD	Start	Finish	BL finish	%	TF	Crit.
G-1000	Notice to Proceed	1	0	01-Sep-25	01-Sep-25		100	0	
G-1010	Mobilisation & site establishment	1	48	01-Sep-25	25-Oct-25		100	10	
G-1020	Site facilities, HSE & supervision	1	531	27-Oct-25	31-Jul-27		66	0	●
G-1030	Authority permits & NOCs	1	88	08-Sep-25	20-Dec-25		100	20	
G-1900	Taking-over certificate	1	1	14-Aug-27	14-Aug-27	31-Jul-27	0	0	●
E-2000	Basic design & single line diagram	2	60	08-Sep-25	15-Nov-25		100	0	
E-2010	Geotechnical & topographic survey	2	36	08-Sep-25	18-Oct-25		100	15	
E-2020	Civil & structural design – GIS building	2	87	20-Oct-25	31-Jan-26		100	0	
E-2030	Primary design – layouts, earthing, cable sizing	2	99	17-Nov-25	14-Mar-26		100	6	
E-2040	Secondary design – protection, control & SCADA	2	212	15-Dec-25	29-Aug-26	25-Jul-26	100	12	
E-2050	Client approval of detailed design	2	37	16-Mar-26	30-Apr-26	15-Apr-26	100	0	
E-2060	As-built drawings & O&M manuals	2	53	01-Jun-27	31-Jul-27		0	0	
P-3000	220 kV GIS – design & manufacture	3	198	01-Dec-25	31-Jul-26	11-Jul-26	100	0	
P-3010	220 kV GIS – FAT & shipping	3	47	03-Aug-26	26-Sep-26	05-Sep-26	100	0	
P-3020	220 kV GIS delivered to site	3	0	28-Sep-26	28-Sep-26	07-Sep-26	100	0	
P-3030	Power transformers 220/33 kV – manufacture	3	249	05-Jan-26	31-Oct-26	10-Oct-26	85	0	●
P-3040	Power transformers – FAT, shipping & delivery	3	50	02-Nov-26	31-Dec-26	12-Dec-26	0	0	●
P-3050	33 kV switchgear – manufacture & delivery	3	189	02-Feb-26	19-Sep-26		100	18	
P-3060	Protection, control & SCADA panels	3	198	06-Apr-26	28-Nov-26		70	8	
P-3070	220 kV XLPE cables & accessories	3	201	02-Mar-26	31-Oct-26		90	6	
P-3080	LVAC, DC & battery systems	3	162	04-May-26	14-Nov-26		60	14	
C-4000	Site clearance, levelling & temporary fencing	4.4.1	34	27-Oct-25	06-Dec-25		100	0	
C-4010	Piling – GIS building	4.4.1	53	08-Dec-25	07-Feb-26	31-Jan-26	100	0	
C-4020	Raft foundation – GIS building	4.4.1	39	09-Feb-26	28-Mar-26		100	0	
C-4030	GIS building RC frame	4.4.1	67	30-Mar-26	20-Jun-26		100	0	
C-4040	GIS building roof steel & cladding	4.4.1	59	22-Jun-26	29-Aug-26	15-Aug-26	100	0	
C-4050	EOT crane installation & load test	4.4.1	18	31-Aug-26	19-Sep-26		100	0	
C-4060	GIS building finishes & HVAC	4.4.1	72	07-Sep-26	28-Nov-26		45	10	
C-4070	Control & 33 kV building – foundations & structure	4.4.1	118	16-Feb-26	11-Jul-26		100	20	
C-4080	Transformer foundations, oil pits & firewall	4.4.1	97	06-Apr-26	01-Aug-26		100	30	
C-4090	Cable trenches & ducts	4.4.1	150	04-May-26	31-Oct-26		75	4	
C-4100	Roads, drainage, boundary wall & gatehouse	4.4.1	141	14-Sep-26	27-Feb-27		15	40	
C-4900	Mobile crane – transformer placing	4.4.1	18	04-Jan-27	23-Jan-27	02-Jan-27	0	0	●
C-5000	220 kV GIS installation – bays 1–5	4.4.2	48	05-Oct-26	28-Nov-26	07-Nov-26	30	6	
C-5010	220 kV GIS installation – bays 6–9	4.4.2	45	30-Nov-26	23-Jan-27	02-Jan-27	0	6	

ID	Activity	WBS	OD	Start	Finish	BL finish	%	TF	Crit.
C-5020	220 kV GIS HV testing	4.4.2	24	25-Jan-27	20-Feb-27		0	6	
C-5030	Transformer T1 – erection, oil filling & tests	4.4.2	48	04-Jan-27	27-Feb-27	06-Feb-27	0	0	●
C-5040	Transformer T2 – erection, oil filling & tests	4.4.2	45	18-Jan-27	13-Mar-27	20-Feb-27	0	0	●
C-5050	220 kV cable laying & terminations	4.4.2	87	02-Nov-26	13-Feb-27		0	10	
C-5060	33 kV switchgear installation	4.4.2	46	12-Oct-26	05-Dec-26		10	12	
C-5070	Protection, control & SCADA panels installation	4.4.2	77	07-Dec-26	06-Mar-27		0	4	
C-5080	Earthing & lightning protection	4.4.2	214	06-Apr-26	19-Dec-26		70	30	
C-5090	LVAC, DC & battery installation	4.4.2	47	07-Dec-26	30-Jan-27		0	14	
C-6000	Pre-commissioning & site acceptance tests	4.4.3	63	15-Mar-27	29-May-27	15-May-27	0	0	●
C-6010	220 kV energisation	4.4.3	0	31-May-27	31-May-27	17-May-27	0	0	●
C-6020	Trial operation & handover	4.4.3	65	01-Jun-27	14-Aug-27	31-Jul-27	0	0	●

Appendix B · Terms in each layout

Concept	P6 style	MS Project style	4D style
Activity	Activity	Task	Task
Activities	Activities	Tasks	Tasks
Activity Id	Activity ID	ID	Task ID
Activity Name	Activity Name	Task Name	Task Name
Wbs	WBS	Outline	WBS
Data Date	Data Date	Status Date	Data Date
Total Float	Total Float	Total Slack	Total Float
Free Float	Free Float	Free Slack	Free Float
Original Duration	Original Duration	Duration	Planned Duration
Remaining Duration	Remaining Duration	Remaining Duration	Remaining Duration
Percent Complete	Physical % Complete	% Complete	% Complete
Predecessors	Predecessors	Predecessors	Predecessors
Successors	Successors	Successors	Successors
Relationship	Relationship	Link	Dependency
Baseline	Baseline	Baseline	Baseline
Schedule	Schedule	Calculate	Reschedule
Resources	Resources	Resource Names	Resources
Constraint	Primary Constraint	Constraint Type	Constraint
Milestone	Milestone	Milestone	Milestone
Project	Project	Project	Project
Projects	Projects	Projects	Projects
Playback Date	Timescale date	Date	Focus date

Appendix C · Glossary

The same 38 terms as the glossary in training mode. Where MS Project uses another word, it is given in italics.

Term	Meaning
Activity *Task (MS Project)*	A unit of work with a duration, dates and logic, such as 'Install GIS bays 1–5'.
Actual cost (AC)	What the work done so far has really cost.
Actual start / finish	The dates the work really started or finished. Shown with an A after the date.
Baseline	The approved plan, frozen and used to measure variance. Re-baselining needs the client's agreement.
Budget at completion (BAC)	The total approved budget for the work.
Calendar	Working days, hours and holidays used to turn durations into dates.
Constraint	A date imposed on an activity (start on or after, finish on or before…). Use sparingly; logic should drive dates.
Cost performance index (CPI)	EV ÷ AC. Below 1.0: work is costing more than budgeted.
Critical path	The longest chain of logic through the project. Its activities have zero (or the least) total float; delaying them delays the finish.
CPM	Critical path method: the forward and backward pass that calculates early and late dates and float.
Data date *Status date (MS Project)*	The date progress is recorded up to. Past is actual; future is forecast.
Earned value (EV)	Budgeted cost of the work actually done: budget × % complete.
Estimate at completion (EAC)	The forecast final cost; simplest form BAC ÷ CPI.
Finish variance	Current finish minus baseline finish. Positive: late.
Float (total) *Total slack (MS Project)*	How long an activity can slip before the project finish moves. Late finish − early finish.
Float (free) *Free slack*	How long an activity can slip before its next activity moves.
Forecast	Dates and costs for remaining work, calculated by CPM from the data date.
4D	A 3D model linked to the schedule so it can be shown at any date.
Lag	A wait added to a relationship, e.g. FS + 7 days for curing.
Level of effort (LOE)	An activity that spans other work, such as site supervision or HSE, taking its dates from them.
Look-ahead	A short-term view (2–6 weeks) of the work starting or running, used to plan crews, materials and permits.

Term	Meaning
Milestone	A zero-duration event: contract award, energisation, taking-over.
Near-critical	Activities with only a few days of float; they become critical after small delays.
Negative float	The schedule cannot meet a constraint or deadline; the number of days to recover.
Open end	An activity with no predecessor or no successor. Schedule quality checks flag them.
Original duration	The planned duration in working days.
Planned value (PV)	Budgeted cost of the work scheduled by the data date on the baseline.
% complete	How far the work is done; base it on measurable quantities.
Relationship (FS, SS, FF, SF) *Dependency / link*	Logic between activities: finish-to-start, start-to-start, finish-to-finish, start-to-finish.
Remaining duration	Working days still needed; drives the forecast finish.
Resource	Labour, plant or material assigned to activities with quantities and rates.
S-curve	Cumulative progress or cost over time; plan, actual and forecast compared.
Schedule performance index (SPI)	EV ÷ PV. Below 1.0: less work done than planned.
Slicer	Dashboard buttons that filter every widget, e.g. by discipline.
Update	The periodic cycle of recording progress, rescheduling and reporting.
WBS *Summary tasks (MS Project)*	Work breakdown structure: the hierarchy of deliverables the activities sit under.
Weightage	Each group's share of overall progress in an EPC progress report, by cost, hours, duration or agreed split.
XER	Primavera P6's exchange file format.

Stratum 4D
by Miqyas Labs
PLAN | SIMULATE | ANALYSE | DELIVER

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ENGINEERING INTELLIGENCE

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The worked example (220/33 kV GIS Substation, Abu Dhabi) is fictional; names, quantities and values are illustrative. Fonts: Montserrat, Inter, IBM Plex Sans, Source Sans 3, Nunito Sans and JetBrains Mono under the SIL Open Font License.